

Energy Access and Climate Mitigation: Impacts on Technology Choice, Emissions, and Costs in East Africa

Submitted in partial fulfillment of the requirements for

the degree of

Doctor of Philosophy

in

Engineering and Public Policy

Joan R Nkiriki

B.S., Electrical Engineering, Louisiana Tech University

M.S., Electrical and Computer Engineering, Carnegie Mellon University

Carnegie Mellon University

Pittsburgh, PA

May 2023

© Joan Nkiriki, 2023

All Rights Reserved

I dedicate this dissertation to my heavenly Father – all of this is for your glory.

I also dedicate this feat to my earthly mother, Hon. Jovah Kamateeka;, because of your unshakeable faith in God and your sacrifices as a single mother, I have the opportunity to live a life beyond anything I could ever have imagined – I am eternally grateful.

ACKNOWLEDGEMENTS

This research was possible through the support of Carnegie Mellon's Engineering and Public Policy department and Carnegie Mellon's Steinbrenner Institute (2019 – 2020 Steinbrenner Fellow). I extend my deepest gratitude to my advising team, Prof. Paulina Jaramillo, Dr. Nathan Williams, and Dr. Aranya Venkatesh, for their invaluable support, guidance, and encouragement throughout this doctoral journey. Their dedication, patience, and commitment were instrumental in enabling me to achieve this milestone. I am grateful for their belief in my potential, even during the slowest and toughest seasons of my doctoral progress. They challenged me to grow and ensured that I was well-prepared to face the challenges ahead.

I would like to acknowledge Dr. Valerie Karplus, the other member of my committee, for her continued support and challenging me to think critically about the implications of missing sectors in my modeling results. I am grateful that the data scarcity explanation while relevant was never sufficient, this pushed me to articulate my contribution and explain any shortcomings more clearly. To her research group, the Laboratory for Energy and Organizations (LEO) team, I highly appreciate your feedback and comradery.

I owe a debt of gratitude to Dr. Michael Dioha and Dr. Destenie Nock for their invaluable contributions to my academic journey. Dr. Dioha graciously made time to help me think through the implications and communication of my dissertation work, and his guidance and laughter-filled Zoom calls were instrumental in helping me get through the final stretch of my program. Dr. Nock opened up the Society, Policy, Infrastructure, Climate, and Energy group gatherings to me and created safe spaces for me to share freely as a young black female researcher. Her support and friendship meant the world to me.

I also extend my gratitude to my exceptional mentors, including but not limited to, Mr. Michael Niyitegeka, Mrs. Crystal Rugege-Ntore, Mrs. Pamela Katooro, Dr. Rose Mutiso, Dr. Rebekah Shirley, Prof. Jared Cohon, Mr. Nathan Kagiri, and Dr. Twinemanzi Tumbweine, for their role in shaping my academic and career trajectory.

I am grateful to the faculty and administrative staff of the Engineering and Public Policy (EPP) department for their encouragement and support. I appreciate Vicki Finney, Debbie Kuntz, Adam Loucks, and many others who cheered me on and lifted my spirits. I also acknowledge the students who stood by me throughout my doctoral program, including Dini Maghfirra, Ana Caceres, Charles Van-Hein Sackey, Katrina Ramirez-Meyers, Nyla Khan, Kester Wade, Hannah Morin, Christine Chao, and the Green Design Institute

research team. The GROW+ community and the African Graduate Students at CMU (AfroGSA) family provided me with a space to feel at home and gain confidence in highlighting the need for diversity in my field of study.

I want to express my appreciation to the CMU Africa community, where I obtained my master's degree and met my advisor. The enabling environment and intentionality of every student, professor, instructor, and administrative staff member will forever leave a lasting positive impact on my life.

I would like to express my heartfelt appreciation to the other exceptional people outside Carnegie Mellon University who contributed to the success of this dissertation. I am grateful to the Open Energy Outlook (OEO) initiative and e-Guide team for their unwavering support, encouragement, and friendship. I extend my deepest gratitude to Dr. Joe DeCarolis, who alongside others, developed the Temoa model and provided consistent support through various communication channels including Zoom, Google forums, and Slack. I am eternally grateful to my e-Guide family, including Prof. Jay Taneja, Aggrey Muhebwa, Dr. June Lukuyu, Dr. Simone Fobi, Joel Mugenyi, Bob Muhwezi, Lefu Magelepo, and Fhazhil Wamalwa, for their invaluable contributions and treasured friendship.

I am blessed to have an amazing support system of fiercely loyal friends who have stood by me through thick and thin. To my housemate, Lynn Kirabo, you are one of the greatest gifts in my life. I am grateful that for the past five years during which you always reminded me of my identity in Christ and the power of the tongue. Your presence in my life made my birthdays unforgettable, and no housemate will ever compete! To my lifelong friends including Amanda Tukwasiibwe, Jovia Tugume, Rachael Mubeezi, Phoebe Mutebi, the Kaitare's, my 'Grambling' family, my Squad (Mrs. Georgette Mashugri, Mrs. Melody Haguma, Mrs. Elizabeth Irakiza, Mrs. Eva Muranga, Mrs. Daphne Amenya, and Mrs. Amanda Akatukunda), and everyone who believed in me and saw the greatness within me, I am immensely grateful.

My family, the Mitooma crew, the Nyarubona clan, my Bulindicians, played a vital role in ensuring that I never doubted my ability to make it to the finish line. To my godparents, uncle Gibson Nsimiire, auntie Anne Rugege, and auntie Rosemary Nyarubona, I am forever grateful for your love, support, and guidance through all the seasons of life. To every auntie, uncle, and cousin who supported me by sending a word of encouragement, timely bible verse, or much needed joke, I appreciate you.

To my amazing big brothers, Edwin Ruyonga, Ivan Kasagama, and Henry Nyabongo, thank you for being such compassionate and supportive brothers. Despite the usual sibling squabbles, I know that you always have my back. I love you all so much, and I am grateful for your constant encouragement and belief in me.

My sisters, Eva Kababiito, Stella Komukyeya, Pamela Katooro, Kate Kabahuma, Betsy (Choco) Komuntale, and Stephanie Mutegeki, you each played a pivotal role in my growth and success. For every phone call, zoom meeting, prayer, laughter, night out, investment, and declaration, this testimony is ours. With each of you by my side, whether physically or in spirit, I had my own army propelling me forward with a strength I cannot fathom. I am grateful for the combined faith of sisters like you.

Having been raised by a single mother whose sacrifices cannot be expressed in words or fully comprehended by me, I honor my mama. Hon. Canon. Jovah Kamateeka, my mother, thank you for being my confidant, prayer partner, voice of reason, and biggest supporter. I could spend all my days expressing my gratitude and still fall short. This accomplishment is as much yours as it is mine, for without your example of experiential faith and trust in the Almighty, I cannot imagine where I would be today. I love you so much.

Finally, I acknowledge that I am a product of God's grace and love. All I do is to give Him glory. Therefore, I thank God for all the people who walked this journey with me, for all the ways this work came together, all the times I was reminded of His promises of a hope and future, and for all the hills and valleys which shaped me.

Tukutendereza (1 John 2:20)

ABSTRACT

East African countries have set universal energy access goals for electricity and clean cooking access while also committing to reducing carbon emissions. However, there are three critical gaps in our understanding of how to meet these goals while upholding climate commitments. First, there is a need for high-resolution open-source models that provide a clear representation of who lacks access considering tier access levels in rural and urban East Africa. Second, we require a better understanding of the level of access countries aim to achieve and the resulting implications. Finally, as countries work towards achieving universal energy access targets in a carbon-constrained world, there is an opportunity to incorporate the demand for transportation services into energy planning efforts at an early stage. Addressing these gaps is crucial to creating effective policies that promote equitable access to clean energy sources and mitigate the adverse impacts of climate change in East Africa.

This dissertation is motivated by the fact that inadequate data hinders the ability of stakeholders to make sound decisions at the national level regarding issues such as energy poverty. Chapter 2 of this dissertation introduces the first version of the Temoa East Africa model, offering a detailed parameterization of supply, electricity, building, and transport sector technologies, and includes carbon removal and hydrogen production pathways. The open-source model seeks to encourage further collaboration and sustainable energy planning in East Africa, with future work aiming to incorporate more detailed transportation service demand data, electric vehicle charging station characterizations, and extended technology characterizations to the industrial and agricultural sectors.

Concurrently, climate change is a reality and there is an immediate and urgent call to reduce human-made emissions. Therefore, energy access goals must be met in a carbon-constrained world. Using the Temoa East Africa model, Chapter 3 examines the impacts of different energy access definitions on final energy consumption, least-cost technology mix, emissions, and costs by access tier group. The findings emphasize the importance of identifying obstacles and opportunities for each access tier to progress towards equitable energy access and low-carbon pathways, including policies that promote access to modern, efficient, and cleaner energy sources, particularly in rural households. The shift towards natural gas and electricity in higher access tier groups can facilitate progress towards a low-carbon energy system, dependent on renewable energy and carbon removal

technologies. Decentralized technologies such as off-grid solar PV will play a pivotal role in increasing electricity access to households in sparsely populated rural areas.

Countries in Africa are not responsible for the vast majority of global carbon dioxide (CO₂) emissions, accounting for only 4% of cumulative emissions. Nevertheless, in the long term, continued reliance on fossil fuels and traditional biomass traps these nations, particularly the poorest in rural areas, in a continued cycle of energy poverty. Chapter 4 develops a long-term land-based passenger and freight transport demand model to support open-modeling community efforts in developing countries, including East Africa. The findings suggest that middle-income countries in Asia such as India and China show the highest expected transport demand across all scenarios. Low-income countries in the sub-Saharan African region are likely to experience the largest growth in demand for passenger and freight transport services. Sustainably meeting this growing demand will require the adoption of data-driven transport planning tools and leveraging cross-linkages across other energy sectors such as electricity.

Overall, this dissertation provides valuable insights into sustainable energy planning in East Africa, highlighting the need for data-driven tools and policies that promote equitable access to clean energy sources while meeting clearly defined universal energy access targets. Achieving universal energy access goals that enable development while adhering to climate mitigation targets necessitates the use of open-source data-driven tools to aid country-level decision-making in East Africa.

CONTENTS

1 INTRODUCTION	1
1.1 PROGRESS IN ENERGY ACCESS TARGETS IN SUB-SAHARAN AFRICA	2
1.2 ELECTRICITY AND CLEAN COOKING ACCESS TARGETS IN EAST AFRICA	4
1.3 MEASURING ELECTRICITY AND CLEAN COOKING ACCESS.....	7
1.4 USING ENERGY MODELING TOOLS TO EXPLORE ENERGY TRANSITION PATHWAYS IN AFRICA.	8
1.5 REVIEWING THE LITERATURE ON ENERGY ACCESS STUDIES IN DEVELOPING COUNTRIES.....	10
1.6 EXPANDING MODELING EFFORTS TO HARD-TO-DECARBONIZE SECTORS IN DEVELOPING COUNTRIES.	11
1.7 RESEARCH OBJECTIVE AND MODELING FRAMEWORK	13
2 TEMOA EAST AFRICA: BUILDING THE DATABASE	16
2.1 ENERGY LANDSCAPE IN EAST AFRICA	16
2.2 ENERGY SYSTEM MODELING IN SUB-SAHARAN AFRICA	17
2.3 RESEARCH OBJECTIVE AND MODELING FRAMEWORK	23
2.4 PRIMARY ENERGY RESOURCE CHARACTERIZATION FOR EAST AFRICA	25
2.5 ENERGY CONVERSION TECHNOLOGY CHARACTERIZATION FOR EAST AFRICA	27
2.6 END-USE TECHNOLOGY CHARACTERIZATION	30
2.7 CARBON DIOXIDE REMOVAL	32
2.8 CONCLUSION.....	32
3 USING TEMOA EAST AFRICA MODEL: UNIVERSAL ENERGY ACCESS IMPACTS UNDER DIFFERENT SCENARIOS IN EAST AFRICA.....	34
3.1 RESEARCH GOALS AND MODELING FRAMEWORK.....	34
3.2 DEMAND SCENARIOS.....	36
3.3 SERVICE DEMAND FUNCTIONS.....	46
3.4 DEMAND PROJECTIONS.....	48
3.5 MODELING RESULTS AND DISCUSSION	50
3.6 CONCLUSION.....	66
4 LAND-BASED PASSENGER AND FREIGHT TRANSPORTATION SERVICES DEMAND MODEL	68
4.1 LITERATURE REVIEW: TRANSPORT DEMAND MODELS	70
4.2 RESEARCH GOALS AND MODELING FRAMEWORK	70
4.3 REGRESSION MODEL	71

4.4 DEMAND SCENARIOS.....	77
4.5 RESULTS AND DISCUSSION.....	79
4.6 CONCLUSION.....	86
5 CONCLUSION AND FUTURE WORK.....	90
5.1 REVISITING THE DISSERTATION RESEARCH GOALS	91
5.2 LIMITATIONS.....	93
5.3 FUTURE WORK	94
6 REFERENCES	96
7 APPENDICES.....	115

LIST OF TABLES

TABLE 1.1 SOCIOECONOMIC INDICATORS IN 2020. KENYA HAD THE LARGEST ECONOMY AMONG THE SIX COUNTRIES, AND THE ONLY COUNTRY IN 2020 WITH ELECTRICITY ACCESS RATES HIGHER THAN SUB-SAHARAN AFRICAN REGIONAL AVERAGE. NONE OF THE COUNTRIES IN HAVE ACHIEVED UNIVERSAL ENERGY ACCESS (DATA SOURCE: (THE WORLD BANK GROUP 2022)).	4
TABLE 1.2 INTERMEDIATE LEVELS OF ENERGY ACCESS BETWEEN NO ACCESS AND FULL ACCESS IN THE MULTI-TIER FRAMEWORK (BHATIA AND ANGELOU 2015; THE WORLD BANK ET AL. 2020).	7
TABLE 2.1 MAIN FEATURES OF BOTTOM-UP AND TOP-DOWN ENERGY MODELS. ADAPTED FROM DIOHA (2017)	18
TABLE 2.2 EXAMPLES OF ENERGY SYSTEM APPLICATIONS IN SUB-SAHARAN AFRICA USING DIFFERENT MODELS. WHILE NOT EXHAUSTIVE, REVIEW OF MODELING APPLICATIONS IN THE REGION SHOWS FOCUS ON ELECTRICITY EXPANSION.	20
TABLE 2.3 PRIMARY SOURCES OF DATA FOR MODEL CHARACTERIZATION	24
TABLE 2.4 CO ₂ EMISSION FACTORS	26
TABLE 2.5 END-USE TRANSPORT SECTOR TECHNOLOGY OPTIONS	32
TABLE 3.1 SUMMARY OF SSP STORYLINES AND ASSUMPTIONS FOR QUANTIFICATION OF THE SOCIOECONOMIC PROJECTIONS WITH RATIOS PROVIDED BY IIASA (CUARESMA 2017; JIANG AND O'NEILL 2017; KC AND LUTZ 2017; RIAHI ET AL. 2017). THIS STUDY USED THREE SSPs IN THE TEMOA EAST AFRICA HOUSEHOLD MODEL.	39
TABLE 3.2 SUMMARY TABLE OF ACCESS SCENARIO ASSUMPTIONS	40
TABLE 3.3 SCENARIO ASSUMPTIONS OF POPULATION SHARE BY ACCESS TIER	42
TABLE 3.4 HOUSEHOLD ELECTRICITY CONSUMPTION IN MULTI-TIER FRAMEWORK. THE STUDY EXCLUDED LIGHTING AND FOOD PROCESSING FROM THE MULTI-TIER MATRIX ASSUMPTIONS TO ESTIMATE ELECTRICITY DEMAND FOR APPLIANCES. DATA SOURCE: (BHATIA AND ANGELOU 2015)	48
TABLE 3.5 HOUSEHOLD SERVICE DEMAND IN 2050 UNDER THE REGRESSION DEMAND SCENARIO ASSUMPTIONS ACROSS EACH SCENARIO	49

TABLE 3.6 COST OF ABATEMENT BETWEEN SDG2050- SCENARIOS UNDER 100% TIER 3+ ACCESS ASSUMPTIONS IN SSP3	65
TABLE 4.1 MODEL FIT RESULTS IN COMPARISON WITH SELECTED MODEL HIGHLIGHTED YELLOW. THE ASTERISKS RELATE TO THE STATISTICAL SIGNIFICANCE (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$), WHILE THE BRACKETED RANGES REPRESENT 95% CONFIDENCE INTERVALS CLUSTERED AT THE COUNTRY LEVEL.....	74
TABLE 4.2 SUMMARY OF SSP STORYLINES AND ASSUMPTIONS FOR QUANTIFICATION OF THE SOCIOECONOMIC PROJECTIONS WITH RATIOS PROVIDED BY IIASA (CUARESMA 2017; JIANG AND O’NEILL 2017; KC AND LUTZ 2017; RIAHI ET AL. 2017).	77
TABLE 7.1 FUEL RESERVE ASSUMPTIONS ON SUPPLY-SIDE (DATA SOURCE: (ALLINGTON ET AL. 2022))......	117
TABLE 7.2 POWERPLANT RENEWABLE ENERGY POTENTIAL BY COUNTRY (DATA SOURCE: (ALLINGTON ET AL. 2022; IRENA 2021)).....	118
TABLE 7.3 POWERPLANT TECHNOLOGY PARAMETER INPUTS AND ASSUMPTIONS.....	118
TABLE 7.4 POWERPLANT FUEL PRICE PROJECTION ASSUMPTIONS IN MILLION USD\$2020 PER PJ (DATA SOURCE: (ALLINGTON ET AL. 2022; IRENA 2021))	119
TABLE 7.5 DIRECT FUEL USE TO HOUSEHOLD PRICE ASSUMPTIONS IN MILLION USD\$2020 PER PJ (DATA SOURCES: (DIOHA AND KUMAR 2020A; DAGNACHEW ET AL. 2020)).....	120
TABLE 7.6 PRICE PROJECTIONS FOR FUEL USE IN TRANSPORT SECTOR IN MILLION USD\$2020 PER PJ (KHALILI ET AL. 2019).....	120
TABLE 7.7 SUMMARY OF FINAL ENERGY CONSUMPTION IN PJ BY ACCESS LEVEL IN 2050 AND SSP3	126
TABLE 7.8 SUMMARY OF DATA SOURCES FOR LOG-LINEAR MODEL INPUTS.....	129
TABLE 7.9 VARIANCE COVARIANCE MATRIX USING SELECTED LOG-LOG MODEL.	132
TABLE 7.10 CORRELATION MATRIX FOR SELECTED EXPLANATORY VARIABLES. EVIDENCE OF HIGH CORRELATION BETWEEN POPULATION AND URBAN SHARE.	132
TABLE 7.11 FUNCTION FORM OF DOUBLE LOG MODELS IN EXPLORATORY ANALYSIS.....	133
TABLE 7.12 LOG-LOG PASSENGER AND FREIGHT MODELS. FIT RESULTS IN COMPARISON WITH SELECTED MODEL HIGHLIGHTED YELLOW. THE BRACKETED RANGES REPRESENT	

95% CONFIDENCE INTERVALS POOLED AT THE COUNTRY LEVEL ACCOUNTING FOR AUTOCORRELATION.	133
TABLE 7.13 THE AVERAGE OF THE R-MSE FOR MODEL 1 IS LOWER THAN MODEL 2. ALTHOUGH RESULTS ARE SIMILAR, MODEL 1 HAS LOWER AVERAGE ERRORS AND IS A BETTER PREDICTOR OF OUT-OF-SAMPLE DATA COMPARED TO MODEL 2.	136
TABLE 7.14 SUMMARY OF GLOBAL LAND-BASED TRANSPORT DEMAND RESULTS BY 2050. WE HIGHLIGHT THE HIGHEST SCENARIO-BASED PROJECTIONS.	136

LIST OF FIGURES

FIGURE 1.1 DISSERTATION OVERVIEW BY CHAPTER SHOWN WITHIN TEMOA EAST AFRICA VERSION 1 STRUCTURE.	14
FIGURE 2.1 EXAMPLE NETWORK DIAGRAM OF AN ENERGY SYSTEM IN TEMOA	23
FIGURE 2.2 OVERVIEW OF TEMOA EAST AFRICA VERSION 1.	25
FIGURE 2.3 POWER PLANT TECHNOLOGIES IN THE TEMOA EAST AFRICA DATABASE.....	28
FIGURE 2.4 PATHWAYS FOR HYDROGEN PRODUCTION IN TEMOA EAST AFRICA MODEL ..	30
FIGURE 2.5 END-USE TECHNOLOGIES BY HOUSEHOLD ENERGY SERVICES	31
FIGURE 3.1 TEMOA EAST AFRICA HOUSEHOLD MODEL STRUCTURE REPRESENTATION	36
FIGURE 3.2 SUMMARY OF 36 SCENARIO MATRIX IN THIS ANALYSIS. WE RUN 3 SSPs UNDER 2 EMISSIONS SCENARIOS FOR EACH ENERGY ACCESS DEFINITION. THE INCREMENTAL CHANGE IS HIGHLIGHTED IN BOLD COLORS.	37
FIGURE 3.3 TECHNOLOGY CHOICE BY ACCESS TIER. UNDERLINED TEXT REPRESENTS SELECT TECHNOLOGIES AVAILABLE BY ACCESS LEVEL.	38
FIGURE 3.4 REGIONAL SERVICE DEMANDS IN SSP3 REVEAL A SHIFT IN TIER ACCESS LEVELS UNDER THE ACCESS SCENARIO ASSUMPTIONS BY 2030 AND 2050.....	49
FIGURE 3.5 REGIONAL FINAL ENERGY CONSUMPTION IN (A) RURAL AND (B) HOUSEHOLDS BY FUEL IN THE SSP3 AND REGRESSION SCENARIOS.....	51
FIGURE 3.6 REGIONAL FINAL ENERGY CONSUMPTION IN THE UNCONSTRAINED AND NET ZERO CO ₂ EMISSIONS BY 2050 EMISSIONS SCENARIOS UNDER ALL THE ACCESS SCENARIOS IN SSP 3	52
FIGURE 3.7 PERCENTAGE OF FINAL ENERGY CONSUMPTION BY DEMAND SERVICE IN HOUSEHOLDS WITH TIER 1+ ACCESS IN ALL SCENARIOS UNDER SSP 3 ASSUMPTIONS	55
FIGURE 3.8 LEAST COST TECHNOLOGY MIX IN SDG2050- SCENARIOS UNDER SSP 3.....	56
FIGURE 3.9 REGIONAL INSTALLED CAPACITY AND POWER GENERATION IN 2020, 2030, AND 2050 UNDER SSP 3 ASSUMPTIONS	59
FIGURE 3.10 SHARE OF CENTRALIZED VERSUS DECENTRALIZED TECHNOLOGY BY POWER GENERATION IN RURAL AND URBAN HOUSEHOLDS IN 2050 UNDER SSP3 ASSUMPTIONS.	60

FIGURE 3.11 CUMULATIVE CO ₂ EMISSIONS BETWEEN 2020 AND 2050 INCLUDING BIOGENIC EMISSIONS FROM BIOMASS UNDER LINEAR REDUCTION TO NET ZERO BY 2050 CONSTRAINTS.....	62
FIGURE 3.12 TOTAL COST TO MEET HOUSEHOLD ENERGY DEMAND IN SSP3 BETWEEN 2020 AND 2050.....	64
FIGURE 4.1 SUMMARY OF COUNTRY-LEVEL LOG-LOG MODEL FOR ESTIMATING TRANSPORTATION DEMAND. THE GREY SIDE BOXES SHOW TITLE LABELS, AND THE BLUE BOXES REPRESENT INPUTS TO THE MODEL WHILE THE GREEN IS THE OUTPUT. THE YELLOW BOXES ARE REPRESENTATIVE OF THE MODEL.....	71
FIGURE 4.2 LOG-LINEAR MODEL PERFORMANCE BY THE PREDICTOR. THE RESIDUAL SCATTER PLOTS ABOVE SHOW THE DEPENDENT VARIABLES AGAINST THE REGRESSORS IN EACH FACET (INCOME, POPULATION, AND URBAN POPULATION). THE PREDICTED VALUES ARE SHADED BLACK, WHILE THE CONTINUOUS INTENSITY SCALE FROM BLUE TO RED SHOWS ACTUAL DATA (COLORED) VALUES. WE USE A BLUE, WHITE, AND ORANGE COLOR RANGE TO REPRESENT UNDER-, EXACT, AND OVER-ESTIMATES.	76
FIGURE 4.3 GLOBAL HISTORICAL TRENDS (GREY: WORLD BANK DATA) AND SSP PROJECTIONS (COLORED: IIASA DATASET) OF TRANSPORT DEMAND DRIVERS. THE HISTORICAL INCOME TREND SHOWS DEGROWTH IN THE YEARS CORRESPONDING TO THE GREAT RECESSION (~2008) AND THE COVID-19 PANDEMIC (~2020).....	78
FIGURE 4.4 PASSENGER AND FREIGHT DEMAND PROJECTIONS BY SSP SCENARIOS. THESE RESULTS FOR THE ANNUAL LAND-BASED PASSENGER DEMAND IN BILLION PKM AND LAND-BASED FREIGHT TRANSPORT IN BILLION TKM ON A GLOBAL SCALE AGGREGATED ACROSS EACH OF THE SSP SCENARIOS.	79
FIGURE 4.5 SCENARIO-BASED PROJECTIONS OF LAND-BASED TRANSPORTATION DEMAND BY (A) REGION AND (B) INCOME GROUP. THE PANELS IN EACH PLOT REPRESENT THE LAND-BASED PASSENGER SCENARIO-BASED PROJECTIONS ON THE RIGHT AND THE LAND-BASED FREIGHT DEMAND ON THE LEFT, WITH THE RESULTS GROUPED AND FACETED BY SCENARIO. PANEL (A) SHOWS THAT SOUTH ASIA AND EAST ASIA & PACIFIC REGIONS WILL HAVE THE HIGHEST ABSOLUTE DEMAND FOR LAND-BASED TRANSPORT BY 2050. IN PANEL (B), WE OBSERVE THE MOST CONSIDERABLE DEMAND IN MIDDLE-INCOME COUNTRIES.....	82

FIGURE 4.6 LAND-BASED PASSENGER (A) AND FREIGHT (B) TRANSPORT DEMAND BY COUNTRY IN 2050 ACROSS EACH SCENARIO. THE COLORING REPRESENTS ABSOLUTE DEMAND IN 2050 FOR EACH COUNTRY. COUNTRIES WITH MISSING DATA ARE SHADED GRAY.	83
FIGURE 4.7 SHARE OF GLOBAL LAND-BASED TRANSPORTATION DEMAND BY REGION AND INCOME GROUP. THE PANELS IN EACH PLOT REPRESENT THE SHARE OF THE SCENARIO-BASED PROJECTIONS ON GLOBAL LAND-BASED PASSENGER AND FREIGHT TRANSPORT BY REGION ON THE RIGHT AND BY INCOME-GROUP ON THE LEFT, WITH THE RESULTS GROUPED AND FACETED BY SCENARIO. PANEL (A) SHOWS THAT THE SHARE OF TRANSPORT DEMAND FROM EAST ASIA & PACIFIC, NORTH AMERICA, AND EUROPE & CENTRAL ASIA START DECREASING BY 2030, WHILE THE SHARE FROM SOUTH ASIA AND SUB-SAHARAN AFRICA START INCREASING. IN PANEL (B), THE LOWER AND UPPER-MIDDLE-INCOME COUNTRIES ACCOUNT FOR OVER 60% OF GLOBAL TRANSPORT DEMAND.	84
FIGURE 4.8 ANNUAL GROWTH RATES IN LAND-BASED PASSENGER (A) AND FREIGHT (B) TRANSPORT DEMAND BY COUNTRY. THE COLOR SHARING OF THE COUNTRIES REPRESENTS THE ANNUAL GROWTH RATE OVER THE 30 YEARS BETWEEN 2020 AND 2050.	85
FIGURE 7.1 TOTAL INSTALLED ELECTRICITY CAPACITY (GW) IN SIX COUNTRIES BETWEEN 2000 AND 2020.....	118
FIGURE 7.2 SOCIOECONOMIC DEMAND DRIVERS FOR ALL SIX COUNTRIES UNDER 3 SSPs. THE GREY LINE IS HISTORICAL DEMAND, THE BLUE LINE IS SSP2 AND THE LOWER AND UPPER BOUNDS ARE LABELLED, SSP1 AND SSP3.....	122
FIGURE 7.3 GDP PROJECTIONS BY COUNTRIES UNDER THE 3 SSPs. THE GREY LINE IS HISTORICAL DEMAND, THE BLUE LINE IS SSP2, AND THE LOWER AND UPPER BOUNDS ARE SSP3 AND SSP1.	123
FIGURE 7.4 POPULATION AND URBANIZATION PROJECTIONS BY COUNTRY UNDER THE SSPs. THE GREY LINE IS HISTORICAL DEMAND, THE BLUE LINE IS SSP2, AND THE LOWER AND UPPER BOUNDS ARE SSP1 AND SSP3.....	123
FIGURE 7.5 THE SHARE OF POPULATION WITH ACCESS TO ELECTRICITY BY TIER IN RURAL AND URBAN HOUSEHOLDS IN 2020 (DATA SOURCE: (FALCHETTA ET AL. 2020)). ...	124

FIGURE 7.6 HISTORICAL ELECTRICITY ACCESS LEVELS BY COUNTRY BETWEEN 2000 AND 2020 (DATA SOURCE: (THE WORLD BANK GROUP 2022)).	125
FIGURE 7.7 HISTORICAL ELECTRICITY ACCESS LEVELS BY RURAL AND URBAN SHARE BETWEEN 2000 AND 2020 (DATA SOURCE: (THE WORLD BANK GROUP 2022)).	125
FIGURE 7.8 HISTORICAL TOTAL ELECTRICITY GENERATION BY FUEL IN EACH COUNTRY (DATA SOURCE: (IRENA 2023; 2021)).	126
FIGURE 7.9 TECHNOLOGY CHOICE BY ACCESS TIER. HIGHLIGHTED PATHWAYS ARE FOR SELECT TECHNOLOGIES AVAILABLE BY ACCESS LEVEL.	126
FIGURE 7.10 POWER PLANT CAPACITY IN 2050 BY COUNTRY IN SSP3	127
FIGURE 7.11 POWER GENERATION IN SSP3 BY COUNTRY	128
FIGURE 7.12 LAND-BASED TRANSPORT DEMAND AND LOG TRANSPORT DEMAND FOR ALL COUNTRIES IN HISTORICAL ITF DATASET.	129
FIGURE 7.13 HISTORICAL LAND-BASED PASSENGER TRANSPORT DEMAND TRENDS BETWEEN 1990 AND 2018 FOR THE FIRST 36 COUNTRIES.	130
FIGURE 7.14 HISTORICAL LAND-BASED FREIGHT DEMAND BETWEEN 1990 AND 2018 FOR 36 OUT OF THE 43 AVAILABLE COUNTRIES IN ITF OECD-DATASET.	131
FIGURE 7.15 CORRELATION MATRIX FOR EXPLANATORY VARIABLES INCLUDING GDP PER CAPITA, LAND AREA, AND POPULATION DENSITY.	131
FIGURE 7.16 TEST FOR AUTOCORRELATION AMONG RESIDUALS USING ACF() FUNCTION IN THE (A) PASSENGER AND (B) FREIGHT DOUBLE LOG REGRESSION MODELS.	134
FIGURE 7.17 COMPARISON OF GLOBAL AGGREGATE MODEL RESULTS WITH ITF GLOBAL LAND-BASED PASSENGER DEMAND. THE ITF LAND-BASED PASSENGER DEMAND BY 2050 LIES WITHIN THE SCENARIO-BASED PROJECTIONS.....	137
FIGURE 7.18 COMPARISON OF GLOBAL AGGREGATE MODEL RESULTS WITH ITF GLOBAL LAND-BASED PASSENGER AND FREIGHT DEMAND. THE ITF HAS HIGHER 2015 REFERENCE/BASELINE AND MUCH HIGHER DEMAND FOR GLOBAL LAND-BASED FREIGHT BY 2050.	137

LIST OF APPENDICES

APPENDIX 1 TEMOA EAST AFRICA	116
APPENDIX 2 IMPACT ASSESSMENT UNDER DIFFERENT SCENARIOS	121
APPENDIX 3 LAND-BASED TRANSPORTATION DEMAND MODEL.....	129

I dedicate this dissertation to my heavenly Father – all of this is for your glory.

I also dedicate this feat to my earthly mother, Hon. Jovah Kamateeka;, because of your unshakeable faith in God and your sacrifices as a single mother, I have the opportunity to live a life beyond anything I could ever have imagined – I am eternally grateful.

“The [one] who moves a mountain begins by carrying away small stones”.

- Confucius [modified]

1 INTRODUCTION

With rapid urbanization, exponential population growth, and rising economic activity in 2020, many countries in the East Africa are falling behind in their energy access targets. This problem is especially severe in rural areas and among households in lower levels of energy access. Access to energy is essential for socioeconomic development because it enables improved education, health care, and other services (Li, Bae, and Rishi 2023; Tiba and Belaid 2021). At the household level, end-use energy services include lighting, cooking, water heating, space heating and cooling, and transport mobility that are made possible through energy access.

Moreover, attaining universal energy access is a crucial component of the 2030 Agenda for Sustainable Development's broader sustainable development agenda. 193 United Nations member states signed the 2030 Agenda for Sustainable Development, which identifies 17 Sustainable Development Goals (SDGs) with 169 targets and 247 indicators to ensure sustainable development for all (IEA et al. 2022). This was done to resolve the world's toughest challenges. This dissertation focuses on universal energy access (SDG 7) and urgent climate action (SDG 13), both with linkages to progress in other SDGs (McCollum et al. 2018). SDG 7, which calls for universal access to affordable, reliable, sustainable, and modern energy for all, and SDG 13, which calls for immediate climate action to reduce the effects of human-made emissions on the earth's climate (IEA et al. 2022).

1.1 Progress in energy access targets in sub-Saharan Africa

According to the International Energy Agency (IEA) (2020a) a definitive description for energy access is households with reliable and affordable access to electricity and clean cooking, initially with access levels sufficient to provide a basic bundle of energy services, and then evolving to higher levels of access over time. This definition is the benchmark to measure progress in the first SDG 7 target, 7.1, which seeks to ensure *"universal access to affordable, reliable, and modern services"* (IEA et al. 2022). The two indicators to measure progress in SDG target 7.1 are share of population with access to electricity (7.1.1) and share of population primarily relying on clean cooking fuels and technologies (7.1.2).

Since the start of the global COVID-19 pandemic, 90 million people in developing countries have rejoined the electricity access deficit with disparities in access gains leaving 77% of the global electricity access deficit or 568 million people in 2020, residing in sub-Saharan Africa (IEA et al. 2022; Li, Bae, and Rishi 2023). Due to the sparse population and low consumption patterns, achieving universal energy access goals in rural areas is more difficult, therefore, decentralized technologies will play a crucial role in closing the electricity access gap in rural areas specifically.

Similar access disparities exist between clean cooking access in rural and urban households. In 2020, 2.4 billion people lacked access to clean cooking fuels and technologies, with 86% of the urban population had access to cooking solutions, while only 44% of rural households had access to clean cooking fuels and technologies globally (IEA et al. 2022). The sub-Saharan African region was the only region with an increasing clean cooking deficit (IEA et al. 2022; Pachauri et al. 2021). The absence of access to cooking solutions has devastating health consequences, particularly for rural populations. In addition, the continued use of unsustainable cooking fuels contributes to greenhouse gas emissions (Sterman et al. 2022; Cherubini et al. 2011; Haberl 2013), exacerbating the impacts of climate change and widening inequalities (Soergel et al. 2021).

Therefore, the COVID-19 pandemic and looming energy crisis reversing energy access growth rates and progress, sub-Saharan African countries will require an annual investment of 12 USD\$ billion to 41 USD\$ billion is necessary to achieve SDG 7 targets (IEA et al. 2022; Mbanda et al. 2021). Domestic capital alone will be insufficient to meet the SDG 7.1 target in line with climate action; policies must be redesigned to attract

private investments and strengthen international support for countries to achieve their ambitious goals (Szabó et al. 2016). For instance, current international financing for electricity access totals 8 USD\$ billion in East Africa, which is insufficient to close the access gap (Hafner et al. 2019; Li, Bae, and Rishi 2023; Mbanda et al. 2021).

Simultaneously addressing SDG 13, to achieve global stabilization goals to limit temperature rise to 1.5-degrees Celsius (°C) would require the world to reach net-zero CO₂ emissions by 2050 (IPCC 2018). Meeting demand for energy services in low-income and developing countries will not result in a significant increase in emissions in the short term (Poblete-Cazenave et al. 2021; IPCC 2022a). Long-term, however, fossil-fuel lock-in and continued use of unsustainable cooking fuels will amplify sustainability issues making it more difficult to meet climate mitigation targets. To achieve ambitious objectives consistent with climate action, policymakers must also revise policies to attract private investments and bolster international support, while avoiding overly ambitious mitigation policies that can push more people into poverty (Soergel et al. 2021).

Historically, African countries have not been responsible for the majority of global carbon dioxide emissions, however, their continued reliance on fossil fuels and traditional biomass places these countries, especially the poorest in rural areas, in a continued cycle of poverty facing the disproportionate impacts of climate change (Soergel et al. 2021). An additional consideration is the treatment of biomass combustion as carbon neutral, which may be a significant flaw in current emission accounting frameworks. Due to its high carbon content, lower efficiency, and uncertain regeneration, fuelwood generates more CO₂ emissions compared to fossil fuel combustion, such as a natural gas, (Sterman et al. 2022; Cherubini et al. 2011; Haberl 2013). Policymakers must investigate all fuel and technology mixes to leverage the untapped regional potential.

Africa has the largest gap between resource availability and energy access levels (Mbanda et al. 2021), and the continent's resource abundance presents an opportunity for policymakers to implement effective strategies that take into account different access levels and a medium to long-term outlook across rural and urban areas of East Africa. Access to clean energy, defined in this thesis as the combined efforts to achieve SDG 7 and SDG 13, is essential for addressing inequality and eradicating poverty on a long-term basis. To accomplish these objectives, this thesis advocates for targeted policies that address barriers to accelerated energy access, in rural and urban areas by considering access levels through time.

1.2 Electricity and clean cooking access targets in East Africa

Universal energy access is a crucial goal for all nations, and the East African region is no exception. However, despite the region's stated goals of achieving minimum access levels by 2030, there is still a significant energy access deficit in East Africa, with five of the world's top 20 countries with the largest energy access deficits, located in this region (IEA et al. 2022). This dissertation focuses on six East African countries – Burundi, the Democratic Republic of Congo, Kenya, Rwanda, Tanzania, and Uganda.

As shown in Table 1.1, the majority of the population in the Eastern Africa countries resided in rural areas in 2020. By 2050, Eastern Africa is projected to house 5% of Africa's population, and the urban population share in these countries will surpass the rural population share (United Nations Department of Economic and Social Affairs 2022a). However, there are significant challenges to achieving universal energy access in these countries.

Table 1.1 Socioeconomic indicators in 2020. Kenya had the largest economy among the six countries, and the only country in 2020 with electricity access rates higher than sub-Saharan African regional average. None of the countries in have achieved universal energy access (Data source: (The World Bank Group 2022)).

Average	GDP (Million USD2015)	GDP per capita (USD2015)	Population (Million people)	Urban population share (%)	Rural population share (%)	Electricity access in 2020 (%) [urban/rural]	Clean cooking access in 2020 (%) [urban/rural]
World	82,037,256	10,489	7,820	56%	44%	90.4% [97%/83%]	69.6% [86%/48%]
SSA	1,811,680	1,574	1,151	41%	59%	48.2% [29%/78%]	17.6% [35%/6%]
Burundi	3,218	263	12.2	14%	86%	11.7% [64%/3%]	0.2% [0.2%/0.2%]
Congo, Dem. Rep.	45,260	487	92.9	46%	54%	19% [41%/1%]	3.7% [8.3%/0.5%]
Kenya	84,054	1,616	52	28%	72%	71% [94%/63%]	19.5% [45%/5%]
Rwanda	10,807	822	13.1	17%	83%	46.6% [86%/38%]	2.4% [9.9%/0.3%]
Tanzania	61,523	1,027	61.7	35%	65%	40% [73%/22%]	4.5% [11%/1%]
Uganda	40,768	918	44.4	25%	75%	42% [70%/33%]	0.5% [1%/0.2%]

The following sub-sections provide additional context and justification for the analysis of energy access and climate mitigation in the six countries.

Burundi

Burundi, for instance, has the lowest rates of electricity and clean cooking access among the six countries, and relies largely on traditional biomass to satisfy its energy needs (Ministry of Planning and Communal Development - Burundi 2011). In response, the country has established goals to achieve 30% electricity access by 2030 and to replace all traditional biomass cookstoves by 2030 (United Nations Climate Change (UNCC) 2023). According to a recent review of sustainable energy planning in East Africa, Burundi lags behind in its use of analytical tools to develop comprehensive energy access policies and strategies (Manirambona, Talai, and Kimutai 2022).

Democratic Republic of Congo

Likewise, the Democratic Republic of Congo has one of the world's largest energy access deficits, with 75 million people lacking access to electricity and 89 million lacking access to clean cooking fuels (Table 1.1) (World Bank 2020a). While the country has considerable renewable energy potential, political unrest and limited data on energy access trends at a granular level presents challenges for planning (World Bank 2020a). The country's natural assets, such as rainforests, currently absorb two thirds of Africa's carbon annual emissions but meeting needs of rapidly growing population threaten the country's carbon sink estimated as around 42 giga-tonnes carbon dioxide (120% of global emissions in 2020) (International Monetary Fund African Department 2022).

Kenya

Kenya, on the other hand, has made notable progress in achieving electricity and clean cooking access, with a national electrification plan and a mid-term and long-term power development plans in place (Lecaros et al. 2021). Kenya's 2018 National Electrification Plan positions the country as a regional pioneer in advancement of electricity access. In addition to the electrification plan, the Kenyan government developed a least cost power development plan using modeling tools to inform power capacity expansion efforts (Republic of Kenya 2018). Kenya leads the region in electricity and clean cooking energy access (Table 1.1), and the country aims to achieve 100% minimum electricity access by 2025.

Rwanda

Rwanda has made significant strides in the development of policy and regulatory frameworks for sustainable energy, achieving the highest Regulatory Indicators for Sustainable Energy (RISE) score in East Africa and dominating the African region (World Bank 2020b). The Ministry of Infrastructure's Energy Sector Strategic Plan outlines implementation measures to achieve 100% minimum electricity access by 2024. The Rwanda Energy Group (REG) has developed the Least Cost Power Development Plan (LCPDP) for the electricity sector out to 2040, which provides a systematic strategy to increase tariff affordability by prioritizing investment in cost-effective generation options (Rwanda Energy Group 2019; Rwanda Energy Group (REG) 2022).

Tanzania

Vision 2025 of the United Republic of Tanzania is to achieve a high quality of life for all citizens with a strong and competitive economy (Ministry of Finance - The United Republic of Tanzania 2011). In 2020, less than 50% of the country's population had electricity access, and less than 5% has access to clean cooking fuels and technologies (Table 1.1). Tanzania's Sustainable energy for all (SE4All) action plan seeks to increase minimum energy access rates to over 75% by 2030 (Ministry for Energy and Minerals - The United Republic of Tanzania 2015).

Uganda

Uganda's nation vision 2040 aims to transform the low-income country into a “competitive upper-middle-income country” by 2040 with universal access to affordable energy facilitate accelerated industrialization (Ministry of Finance 2007). The country's SE4All action plan targets over 98% electricity access and over 99% non-solid cooking fuel by 2030 (Ministry of Energy and Mineral Development - Republic of Uganda 2015). Uganda's utility, UMEME, is the only utility in the region with positive net revenue gains (Falchetta, Hafner, and Tagliapietra 2020). The nation implemented an electric cooking tariff in 2021 to stimulate electricity demand and encourage a transition from traditional biomass use (ERA 2021).

There are significant challenges to achieving universal electricity and clean cooking access in the six countries amplified by the disparities between rural and urban areas. The

following sub-sections provide the gap analysis to anchor this dissertation's contributions to the existing literature on energy access and climate mitigation in East Africa.

1.3 Measuring electricity and clean cooking access

Access to energy has traditionally been measured based solely on the percentage of population with access to a particular energy form, without considering usage patterns, service quality, or reliability. In 2015, the World Bank introduced the multi-tier framework (MTF), which defines access to electricity along a continuum of energy services across five tiers, each with assigned thresholds for various metrics, including affordability, availability, reliability, legality, safety, quality, efficiency, convenience, health impact, and capacity (Bhatia and Angelou 2015). Additionally, the multi-tier framework has been extended to measure to clean cooking fuels and technologies with access levels aligned with the International Workshop Agreement (IWA) on cookstove performance metrics. The access deficit and five tier access levels are defined Table 1.2.

Table 1.2 Intermediate levels of energy access between no access and full access in the multi-tier framework (Bhatia and Angelou 2015; The World Bank et al. 2020).

Tier	7.1.1. Electricity access ("connected" vs. "not connected")	7.1.2. Clean cooking access ("solid" vs. "nonsolid" fuels)
Access deficit (No access)	In the no electricity access level, a connection to electricity is not available	This tier represents lowest performance of cookstoves including traditional biomass technologies such as three stone methods.
Tier 1 (Minimum access)	A minimum availability of 4 hours of electricity per day per household with capacity for only radio, phone charging, and task lighting	The lowest intermediate level of access meets performance cookstove efficiency of >10%
Tier 2 (Transition)	Similar to Tier 1 with minimum of 4 hours per day of electricity availability. However higher capacity allowing for low-load appliances like television	In this transition access level for cooking, access improvements with cookstove efficiency >20%.
Tier 3 (Intermediate)	At least 8 hours of electricity available per day and medium-load appliances like refrigerators and water pumps	This intermediate level with access to higher efficiency technologies compared to Tier 2 using efficiency technologies >30%
Tier 4 (Modern energy)	At least 16 hours of electricity availability per day and high load appliances like washing machines and microwaves	Clean cooking technologies in this tier meet >40% cookstove efficiency
Tier 5 (Full access)	In highest access level, electricity is available at least for 23 hours per data with very high-load like air conditioners and space heaters.	The full access energy level represents highest level of access across the six metrics with cookstove efficiency >50%

1.4 Using energy modeling tools to explore energy transition pathways in Africa.

Models of energy systems serve as valuable quantitative tools to aid policymakers in evaluating various energy planning options. Following the oil crisis of the 1970s, the scope of modeling applications broadened to provide insights for both short- and long-term energy planning (Bhattacharyya and Timilsina 2010a; Lopion et al. 2018). Over time, the emphasis of energy system modeling shifted from cost analysis to broader applications, such as examining climate change and environmental implications research (Pfenninger, Hawkes, and Keirstead 2014).

An energy system is defined by Pfenninger *et al* (2014) as “*the process chain from extraction of primary energy to the use of final energy to supply services and goods*”. Pfenninger *et al* (2014) assessed four categories of energy models (energy system optimization, energy system simulation, power system, and electricity market models) in their highly cited review. The review emphasized the importance of comprehending the assumptions underlying modeling results to enhance decision-making. This highlighted the need for greater transparency through open modeling communities to elucidate the theoretical and hypothetical assumptions underlying results and analyses. However, these models have high data requirements and frequently struggle to account for disparities among sub-groups in developing nations.

Differences between energy modeling in developed and developing countries include the absence of legacy systems, the opportunity to leverage technological advancements, disparities between rural and urban areas, heavy reliance on traditional biomass, varying levels of energy access, and uneven income distribution (Ouedraogo 2017a; Bhattacharyya and Timilsina 2010b; van Ruijven et al. 2008). Numerous researchers have examined energy modeling in context of addressing energy challenges, recognizing the limitations posed by data scarcity in developing countries (Pfenninger, Hawkes, and Keirstead 2014; Lopion et al. 2018; Prina et al. 2020; Bhattacharyya and Timilsina 2010a). Typically, these models fall into one of three approaches: top-down, bottom-up, or hybrid modeling.

In Africa, an increasing number of research studies are simulating energy transition pathways under various scenarios using bottom-up energy system models (Dioha and Kumar 2020a; Ouedraogo 2017a; Mentis et al. 2017; Falchetta, Hafner, and Tagliapietra

2020; Taliotis et al. 2014; Sridharan et al. 2019; Republic of Kenya - Ministry of Energy 2021; Rwanda Energy Group 2019). Energy transition pathways refer to the different strategies and technologies to transition from fossil-fuel dependence to more sustainable energy systems reliant on renewable energy sources that p global temperatures from rising above pre-industrial levels of 2 degrees Celsius (IEA and IRENA 2017). Existing literature focuses predominantly on the electricity sector and uses geographic information system approaches to support electrification strategies in Africa (Mentis et al. 2017; Taliotis et al. 2016; Falchetta, Hafner, and Tagliapietra 2020; Sridharan et al. 2019).

For instance, Mentis *et al* (2017) created an open-source spatial electrification tool (OnSSET) to investigate least-cost electrification strategies for 44 sub-Saharan Africa countries in order to demonstrate the relevance of system modeling approaches electricity planning using local context information. Their study of various 100% Tier 1+ electrification scenarios by 2030 revealed that decentralized technologies played a critical role in meeting low electricity demand, whereas higher consumption led to a transition towards mini-grid and centralized technologies (Mentis et al. 2017).

Similarly, Falchetta *et al* (2020) investigated least-cost universal access trajectories in East Africa (including Burundi, Kenya, Rwanda, Tanzania, and Uganda) for various consumption electricity tier access levels and generation mix scenarios up to 2030 using the same OnSSET model. The authors estimated an annual regional investment need of between \$57 billion and \$110 billion for new connections, in addition to \$2.7 billion annually to meet the growing demand of the already electrified population. Falchetta *et al* (2020) concluded that sufficient alignment between national and international financing efforts, and policy intervention could make East Africa's goal achieving 100 percent Tier 1+ electricity access by 2030 feasible.

Sridharan *et al* (2019) used long-term electricity supply and water management system models to assess the vulnerability of the East Africa Power Pool to climate change effects, taking into account the potential critical role of hydropower in meeting future electrification demands in the East African region. Using the Open-Source Energy Modeling System (OSeMOSYS) model for the electricity expansion analysis, the study accounted for wide uncertainty in climate, long-term temperature, and precipitation, and concluded that the region requires climate-proofing to avoid adverse outcomes in arid climates.

However, these previous studies are predominately concerned with electricity capacity expansion and do not consider technology choice in end-use demand sectors such as building and transportation.

1.5 Reviewing the literature on energy access studies in developing countries.

Numerous studies have investigated energy access scenarios for achieve Sustainable Development Goal (SDG) 7 targets, but most have focused on electricity access as an SDG 7.1 indicator with minimal representation of the tier access levels in developing countries (Li, Bae, and Rishi 2023; Dioha and Kumar 2020a; Morrissey 2019; Dagnachew et al. 2019). Li *et al* (2023) explored the relationship between energy aid and electricity access in sub-Saharan Africa and discovered that aid to low-income countries facilitates low-carbon pathways while indirectly attaining universal energy access. However, Li *et al* (2023) did not examine tier-specific access levels and only focused on electricity access. Other studies that included representation of tier access levels in sub-Saharan Africa primarily focused on indicator 7.1.1 using least-cost electrification modeling approaches without considering technology choice for access deficits or representing tier levels in household services like cooking (Szabó et al. 2016; Mentis et al. 2017; Dagnachew et al. 2017).

While there is a growing body of work assessing global impacts of universal energy services with an emphasis on tier access levels, the six countries in the regional scope of this dissertation and their tier access levels are inadequately represented in the existing literature (Poblete-Cazenave et al. 2021; Pachauri et al. 2021; Dagnachew et al. 2020). Poblete-Cazenave *et al* (2021) presented a bottom-up residential household and energy demand model using discrete choice methods to estimate appliance demand by fuel and explored pathways to achieve ambitious and optimistic energy access targets under different scenarios. While Poblete-Cazenave *et al* (2021) modeled end-use services by income group in rural and urban areas under the three Shared Socioeconomic Pathways (SSPs) and policy scenarios, the authors did not consider individual technology choice focusing only on aggregate service-based demand and only used representative countries in Africa, none of which included the six East African countries in this regional scope, thus failing to fully capture the heterogeneity across the region.

Pachauri *et al* (2021) used MESSAGEix-GLOBIOM micro-data technology choice and demand models under the first three shared socioeconomic pathways (SSPs), which we describe in detail in later chapters, along with climate mitigation scenario policy considerations taking into account rural and urban disparities and income distribution to examine global clean cooking access out to 2050. The regression-based analysis by Pachauri *et al* (2021) only included estimates from seven African countries including Kenya and Uganda. Their study did not evaluate tier access or technology selection in relation to the end-use energy service demand of households.

Most recently a study by Rammelt *et al* (2022) on the impacts of universal energy access found that meeting the energy needs of the world's poorest in developing and low-income countries, particularly in rural Africa, had minimal short-term environmental impacts compared to the wealthiest 1%-4%. The study assessed impacts on two access levels: dignity (equivalent to Tier 1 access with an annual 365 kilowatt hour (kWh) per household) and capability (equivalent to Tier 3 access with an average annual energy use of 1,250 kWh per household), assuming universal energy access was achieved in 2018. However, the study only considered electricity in the household energy consumption assumptions and did not capture heterogeneity across rural and urban areas, nor did it investigate the impacts on technology choice, emissions, and cost of closing the energy gap after 2018.

As population growth continues to outpace electricity and clean cooking access rates in the East African region, more aggressive and targeted policies are necessary to address barriers to accelerated energy access at different access level tiers, in accordance with global climate action.

1.6 Expanding modeling efforts to hard-to-decarbonize sectors in developing countries.

Simultaneously, SDG 13 aims to reduce the adverse effects of human-induced emissions in the earth's climate (IEA *et al*. 2018), which can be disastrous for vulnerable economies in developing countries (IPCC 2023). In the short term, meeting demand for energy services in low-income and developing countries will significantly increase emissions (Poblete-Cazenave *et al*. 2021; IPCC 2022a). However, long-term fossil fuel lock-in and continued use of sustainable cooking fuels will strain the earth's systems, making global targets to limit temperature rise to 1.5°C pre-industrial levels even less achievable (IPCC 2022a). The Paris Agreement's principle of common but differentiated responsibility

assigns developed and heavily industrialized countries a greater share of responsibility for pursuing a sustainable future (United Nations 2015). However, developing countries, particularly those in Africa, bear a heavier burden of impacts and share the common goal of achieving the SDGs.

In the Paris agreement, in support of SDG 13, countries committed to limiting temperature change relative to pre-industrial temperature to “*well below 2°C*” (United Nations 2015). Since the ratification of the agreement, there has been a growing consensus on the need to limit temperature changes to 1.5°C to limit catastrophic climate change. Meeting these temperature targets would necessitate achieving global net-zero CO₂ emissions by 2050 (IPCC 2018) (IPCC 2022a). Consequently, achieving net-zero CO₂ emissions would likely require avoiding any significant growth in new emissions sources.

Transportation is the fourth largest source of greenhouse gas (GHG) emissions worldwide (Lamb et al. 2021). In 2018, the transportation sector accounted for 30% of the total final energy consumed and 14% of global direct carbon dioxide (CO₂) emissions (IEA 2019c; 2020b; Lamb et al. 2021). The transportation sector is also experiencing a 2% annual growth rate in emissions, which poses a challenge for meeting the climate stabilization goals set forth in the Paris Agreement (Lamb et al. 2021).

Looking ahead, simultaneously achieving universal energy access in a carbon constrained world will require addressing the hard-to-decarbonize transport sector. On a global scale, the ITF (2019; 2021a) and the World Economic Council (2011) rely on historical data and scenario frameworks to project transportation services demand on a global and regional scale to 2050. The IEA's mobility model (MoMo) for global transport was published in the Energy Technology Perspectives (ETP) report, covering 22 regions and countries (IEA 2017; 2020b). Unfortunately, country-level estimates from these modeling efforts are not freely available.

Therefore, existing country-specific transport demand models focus on countries with robust historical datasets, making it challenging to estimate the service demand for developing countries. The electricity generation mix in East Africa is dominated by hydropower, whereas the transport sector is heavily dependent on fossil fuels. Lack of demand data is a key empirical barrier in understanding the barriers and opportunities for low-carbon transportation systems in developing countries.

1.7 Research objective and modeling framework

Summarizing the literature presented in this introduction studies on access to energy in sub-Saharan Africa have focused on least cost electrification modeling approaches, with only limited consideration for the technology choices available to household with different tier access level beyond 2030. In addition, few of these have examined access to clean cooking using energy models. Consensus exists among researchers that data paucity, reliability, and quality issues have hampered country-level analysis Africa, resulting to incomplete and inaccurate evaluations (Mentis et al. 2017; Ouedraogo 2017a; Dioha and Kumar 2020a).

To bridge these gaps in the literature, this dissertation focuses on universal energy access in rural and urban households in East Africa across different tiers of the multi-tier framework (Bhatia and Angelou 2015), with a particular emphasis on (i) parity considerations in access levels, (ii) heterogeneity within countries in the rural and urban areas, and (iii) time-bound indicators to track population progress in achieving minimum access and the evolution of access within the multi-tier framework.

East African countries have set universal energy access targets and climate commitments, which this dissertation addresses through three primary research goals. The research goals for this thesis are (i) to improve open-source modeling tools to explore energy transition or transformation pathways at the country level in rural and urban East Africa, (ii) to examine impacts of hypothetical pathways achieving universal energy access goals across tier access levels in rural and urban East African households, and (iii) to provide long-term land-based passenger and freight transport demand estimates on a country-level to support open modeling community efforts in developing countries including those in East Africa.

To achieve these research goals, this dissertation contributes to the literature in three ways as depicted in the Figure 1.1 document overview. First, this dissertation introduces the first version of the Temoa East Africa model to improve relevance of open modeling community efforts (Chapter 2). Second, it captures within country heterogeneity across access levels and assesses impacts in different energy access targets in a least-cost optimization household model (Chapter 3). Lastly, it expands efforts to a hard-to-decarbonize sector by extending knowledge of potential demand for transportation services to identify opportunities for universal access to modern energy services and mitigation potential in East Africa out to 2050 (Chapter 4).

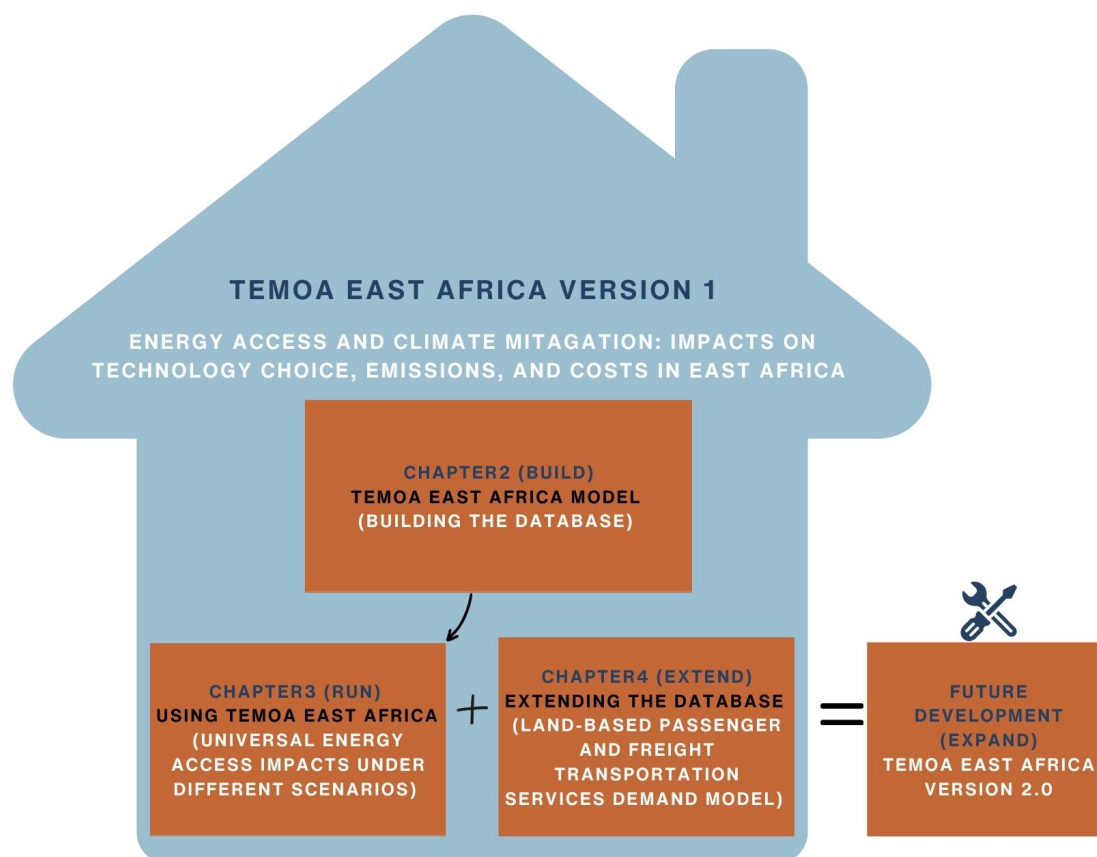


Figure 1.1 Dissertation overview by chapter shown within Temoa East Africa version 1 structure.

A key contribution of this dissertation is a country-level model that address gaps in tiered access considerations in the end-use technology characterization using an established modeling framework. Chapter 2 of this dissertation introduces the Temoa East Africa model, building on previous work by using an established modeling framework with resource and technology parameters for rural and urban households in six East African countries. In this chapter, we present version 1 of a Temoa input database for East Africa that characterizes the primary energy resources, conversion technologies, and end-use technologies for buildings and transportation. This chapter details the parameterization of the inaugural Temoa East Africa model that uses an established modeling framework. In addition to using the multi-tier framework, the model parameterization represents end-use services by access tier in rural and urban areas whereby select technology options are available by the level of access.

In Chapter 3, the study uses the Temoa East Africa model to assess the impacts of different access definitions on least-cost technology choice, emissions, and costs in the

rural and urban households under different scenarios. This chapter builds on the work of Rammelt et al (2022) by exploring the impact of different energy access definitions and estimating energy service demand using a scenario-based approach. Our analysis considers three simplified access levels to track progress towards SDG target 7.1 across indicators, 7.1.1 and 7.1.2 Tier 0 represents households lacking access to electricity and clean cooking fuel and technologies, while Tier 1+ but not Tier 3 access level represents the minimum access level as a transition towards full access. Lastly, households with Tier 3 and higher access levels have achieved modern energy access, with the capability to overcome poverty. This analysis also provides additional clarity on who (tier access level), where (rural and urban areas), and when (by 2030 and out to 2050) universal energy access can be achieved under different CO₂ emissions pathways.

Lastly, Chapter 4 presents an inclusive demand model for land-based passenger and freight transport on a country-level, using the countries in East Africa as a case study. This chapter considers future scenarios to model the long-term demand for land-based passenger and freight services and how they will affect energy access needs. This dissertation contributes to the expansion of modeling efforts in East Africa by extending open modeling community efforts and providing insights into the potential nature of long-term demand in the hard-to-decarbonize transport sector, with a focus on developing countries. The key contribution of this study is the development of a demand model to estimate a demand model to estimate long-term land-based passenger and freight transportation service demand at the country-level.

In summary, this dissertation focuses on addressing three core areas in progressing the East African region's movement towards achieving universal energy access while adhering to climate goals, we build a model using established modeling framework, we use this open-source modeling tool with different access tiers to account for different access scenarios and the impacts of varying degrees of household energy access definitions, and lastly, our key contribution in the transportation sector is to provide a demand model for land-based passenger and freight transport on a country level including the countries in East Africa.

2 TEMOA EAST AFRICA: BUILDING THE DATABASE

2.1 Energy landscape in East Africa

East Africa boasts of a wealth of renewable energy sources; yet the region's adoption of low-carbon technologies remains limited due to financial management and resource allocation challenges. According to the International Energy Agency (IEA 2019a), East Africa possesses the highest theoretical solar energy potential on the continent, in addition to proven fossil fuel reserves. Tanzania has discovered large natural gas reserves (Amanam 2017), while Rwanda and the Democratic Republic of Congo rely on methane gas from Lake Kivu, and Uganda's Albertine Graben area also has natural gas (Munyaneza, Wakeel, and Chen 2016; Safari 2010; EIA 2016b). Despite oil discoveries in Uganda and Kenya, development progress of these discoveries has been slow due to geopolitical and economic challenges, resulting in the majority of oil products being imported (EIA 2016a; 2016b).

Hydropower is the dominant power source in East Africa, yet it is location-specific and highly sensitive to precipitation variability (Sridharan et al. 2019). In addition, hydropower plants have lengthy construction lead times, making it challenge to construct large to medium power plants (greater than 10 megawatts) (IRENA 2012). Solar penetration has increased in the region, with Kenya leading the solar PV market in East Africa (Adwek et al. 2020; Muchunku et al. 2018). Kenya's geothermal power plant is also the largest on the African continent in 2019, and the East African region possesses abundant geothermal resources (IEA 2019a).

Despite hydropower dominating the power sector, East Africa remains heavily dependent on biomass for its total final energy needs (Bär, Heinemann, and Ehrensperger 2017; Clemens et al. 2018). However, the unsustainable use of biomass resources is associated with indoor air pollution and adverse health effects, mostly affecting women and children. Traditional biomass, including charcoal and fuel wood, when combusted for cooking, is inefficient and produces high levels of pollution. In contrast, modern cooking solutions,

such as biogas and improved cookstoves, offers more efficient energy alternatives that reduce pollution from combustion of traditional biomass (Clemens et al. 2018). According to the World Health Organization (WHO), electricity, liquified petroleum gas (LPG), solar, biogas, and natural gas are considered clean cooking fuels (WHO and SEI 2021).

Modeling energy systems can aid in evaluating and optimizing different pathways towards sustainable energy, as well as quantifying the potential impacts of energy transition strategies (Pfenninger, Hawkes, and Keirstead 2014). Therefore, this chapter introduces the inaugural version of an energy system model for East Africa in an effort to increase the relevance of modeling applications in investigating energy transition pathways for Africa.

2.2 Energy system modeling in sub-Saharan Africa

Since the 1970s, energy system models have been used to analyze energy futures of countries and regions. These models can be classified as either as bottom-up or top-down models. Bottom-up, engineering-based models provide a detailed technical representation of the energy system, but typically do not account for economic feedbacks. Top-down macroeconomic models, on the other hand, include aggregate representations of the energy system and have a more detailed characterization of macroeconomic factors that affect energy balances. Integrated Assessment Models (IAMs), which are extensively used for climate modeling, rely on the top-down approach. Nonetheless, they lack a detailed depiction of sectoral end-uses and energy technologies (Gambhir et al. 2019). Table 2.1 summarizes the main features of bottom-up and top-down energy models from a systematic review of energy system models (Dioha MO 2017). As shown in Table 2.1, bottom-up energy models are useful for representing “best performance” pathways and “what if” analyses, thereby providing valuable insights for system planning, environmental and energy policy design, and R&D needs. However, applications of bottom-up energy models in least-developed countries in sub-Saharan Africa have been more limited than in other regions. Table 2.2 provides a summary of recent research focused on energy modeling in sub-Saharan Africa, which includes approximately 30 studies that have used energy system models in the context of the region.

Table 2.1 Main features of bottom-up and top-down energy models. Adapted from Dioha (2017)

Bottom-up Energy Models	Top-down Energy Models
Examples: SWITCH (Carvallo et al. 2017), TIAM-ECN model (van der Zwaan et al. 2018), OSeMOSYS (Howells et al. 2011), TIMES-MARKAL, Temoa (Decarolis, Hunter, and Sreepathi 2020; Hunter, Sreepathi, and DeCarolis 2013)	Examples: GCAM (Joint Global Change Research Institute (JGCRI) 2017; Calvin et al. 2019), GEM-E3 (P Capros et al. 2013), PRIMES (Pantelis Capros et al. 2018; E3Mlab/NTUA 2016)
Use an “engineering approach.”	Use an “economic approach.”
Allow for a detailed description of technologies	Cannot explicitly represent technologies in detail
Use disaggregated data for exploring purposes	Use aggregated data for predicting purposes
Directly assess costs of technological options	Endogenize behavioral relationship
Represent supply technologies using disaggregated data, but energy demand is exogenous to the model	Determine energy demand through aggregate economic indicators (i.e., GDP) but vary in their representation of supply.

Several studies have examined the power sector in sub-Saharan Africa, such as the low carbon expansion of the power sector in Kenya (Carvallo et al. 2017), decarbonization pathways for the power sector in Ghana (Diawuo et al. 2020), electrification pathways in Uganda (Trotter, Cooper, and Wilson 2019), and electrification pathways in Kenya (Nandi Moksnes et al. 2017). Van der Zwaan (2018) evaluated renewable energy deployments in 17 African regions using a bottom-up optimization approach, the Times Integrated Assessment Model of the Energy research Center of the Netherlands (TIAM-ECN) model. Ouedraogo (2017a) used the Low Emissions Analysis Platform (LEAP) model, a simulation-based tool, to develop energy demand scenarios and associated emissions for a business-as-usual scenario, as well as alternative scenarios focusing on expanding energy access and renewable energy policies. In addition, Allington et al (2022) and Pappis et al (2022) used the open-source OSeMOSYS model to explore electrification and decarbonization pathways in developing regions, including African countries.

However, the majority of previous research in sub-Saharan Africa has been focused on the electricity sector and neglecting other sectors like transportation, industry, and commerce which can be more challenging to characterize. Furthermore, research has focused primarily on deploying conventional renewable energy sources, overlooking the potential for other low-carbon energy sources. A systematic review of energy system models for sub-Saharan Africa suggests that existing modeling approaches struggle to capture specific features of energy systems in sub-Saharan Africa (Bhattacharyya and

Timilsina 2010a; Dioha MO 2017; Ouedraogo 2017b). Nevertheless, future work should strive to expand the modeling capabilities for the region (Dioha MO 2017).

Model selection depends on the study's objective and purpose (Bhattacharyya and Timilsina 2010a; Dioha MO 2017; Yoro et al. 2021; Lopion et al. 2018).

Selecting an appropriate model is based on the research objective, data availability, and computational resources. Different models have varying degrees of classifications that determine the trade-offs between geographic scope, sectoral coverage, time horizon, analytical approach, to mention but a few (Prina et al. 2020). For instance, while the OnSSET model has gained popularity in short-term electrification planning due to its high spatial resolution, the Tools for Energy Model Optimization and Analysis (Temoa) is more suited for long-term optimization across multiple sectors but has a lower space geospatial resolution (Table 2.2). However, all models require trade-offs that may affect accuracy, such as spatial and temporal trade-offs that can impact computational time (Lopion et al. 2018; Prina et al. 2020). When selecting models to assess the impact of achieving energy access goals in developing countries, there is a challenge as many existing energy models originated from developed countries and may not capture unique features of energy systems in African countries such as unreliable grids, low energy access rates, uneven income distribution, and heavy reliance on traditional biomass. Therefore, research objectives drive model selection by informing the necessary trade-offs since no single model can accurately depict reality while balancing complexity, transparency, and treatment of uncertainty (Prina et al. 2020; Pfenninger, Hawkes, and Keirstead 2014).

In this study, bottom-up models that allow for rich technology characterization in end-use sectors were better suited to the modeling objective. Applying bottom-up models to investigate energy transitions enables the selection of the least-cost technology mix, representing energy flow from supply to demand, with enhanced technology detail (Prina et al. 2020). Groissböck (2019) identified Temoa, the OSeMOSYS, and the solar and wind energy integrated with transmission and convention sources (SWITCH) tools as the three preferred open-source tools for bottom-up energy system analysis. While the SWITCH model focuses on the electricity sector, Temoa and OSeMOSYS share similar structures, capturing the entire energy system. Recent advancements have given Temoa a computational edge over OSeMOSYS, rendering it more suitable for larger energy system analyses using open-source solvers (Nicoli et al. 2022).

Table 2.2 Examples of energy system applications in sub-Saharan Africa using different models. While not exhaustive, review of modeling applications in the region shows focus on electricity expansion.

reference	Model (Approach)	Regional scope	Sectoral scope	Time horizon	Demand	Scenarios	Indicator 7.1.1. electricity access	Indicator 7.1.2. clean cooking access	Tier access levels	Key findings
(IEA 2022b)	IEA World Energy Model	Africa	Residential, commercial, transport, industry, agriculture	2020 - 2050	Final energy consumption with rural urban split	Sustainable Africa scenarios including universal access scenario by 2030	●	●		Exploring sustainable Africa scenario (SAS), the IEA World Energy Outlook found an investment need of \$25 billion annually, 1% of current global energy investments, to achieve universal energy access in Africa by 2030
(Parzen et al. 2022) [preprint]	PyPSA-Africa (optimization)	Africa with Nigeria- specific data	Electricity	2020 - 2060	Aggregate national electricity consumption	2 historical system and net zero by 2060 scenarios in Nigeria	●			Arguing for need to develop open-source modeling tools to explore net zero carbon pathways in Africa, model demonstrates regional electricity expansion model for long term energy planning with high geospatial resolution
(Dioha and Kumar 2020b)	TIMES-Nigeria (Optimization)	Nigeria	Residential, commercial, transport,	2010 - 2050	30 demand services with rural- urban split	4 NDC scenarios	●	●		Exploring low carbon transitions, study found residential sector holds highest potential to dampen growth in demand however, electricity sector should

			industry, agriculture						be focus of CO ₂ emission mitigation
(Remy and Chattopadhyay 2020)	World Bank Electricity Planning Model (optimization)+	11 countries in Eastern Africa Power Pool (EAPP)	Electricity	2020 - 2030	Aggregate national electricity consumption	3 business-as- usual (BAU), shallow and tight levels of integration scenarios under varying low carbon targets	●		Investigating impacts of regional power integration, research found region experienced highest benefits with lower cost of mitigation under integration and 30% CO ₂ emission reduction targets
(Dagnachew et al. 2017)	IMAGE-TIMER (Least-cost rural electrification)	Sub-Saharan Africa	Electricity	2010 - 2030	Electricity consumption by access tier in MTF with rural electrification focus	2 baseline SSP2 and universal access scenarios	●	●	Exploring Universal energy access under different demand scenarios for rural electrification, researchers found decentralized electricity technologies will play key role in achieving universal energy access targets in sub-Saharan Africa with off-grid supporting areas with low consumption
(Ouedraogo 2017a)	LEAP (simulation)	52 African countries	Residential, commercial, transport, industry, agriculture	2010 - 2040	Final energy demand by technology with rural-urban split	4 business-as- usual (BAU), moderate to high access levels, and energy efficiency policy scenarios	●	●	Modeling demand for energy services under different climate constraints to identify synergies and trade-offs between universal energy access and climate mitigation targets.

Energy Access and Climate Mitigation: Impacts on Technology Choice, Emissions, and Costs in East Africa

(Carvallo et al. 2017)	SWITCH-Kenya (optimization)	Kenya	Electricity	2020 - 2035	Electricity consumption with rural-urban split	14 scenarios including Business-as-usual (BAU), varying load profiles, geothermal costs and potential, storage, low coal, and emission reductions	●	Exploring low carbon pathways for Kenya under different climate mitigation scenarios, study found Kenya depends on large geothermal capacity expansion that is sensitive to operational degradation. A generation with zero carbon technologies increased levelized cost of electricity by \$3/MWh to \$7/MWh.
(Taliotis et al. 2016)	OSeMOSYS (optimization)	47 countries in Africa	Electricity	2010 - 2040	Electricity consumption with rural-urban split	2 reference and enhanced trade scenarios	●	Examining the relationship between electricity investment and regional power trade at higher geospatial resolution to identify opportunities for lower electrification costs, research found trade significantly altered generation mix within countries

2.3 Research Objective and Modeling Framework

Temoa is an open-source bottom-up linear optimization energy system modeling tool. The model represents the flow of energy from the supply sector through the transformation or conversion sectors to the demand sectors, as shown in Figure 2.1. In the model, commodities represent energy forms and process conversions are technologies (DeCarolis et al. 2016). Temoa optimizes the installation and use of energy technologies to minimize the discounted cost of energy supply while adhering to user-defined constraints. The main advantage open-source modeling interface is its ability to facilitate third-party replication of analyses. Additionally, the interface supports for scenario-based analysis which is an added benefit.

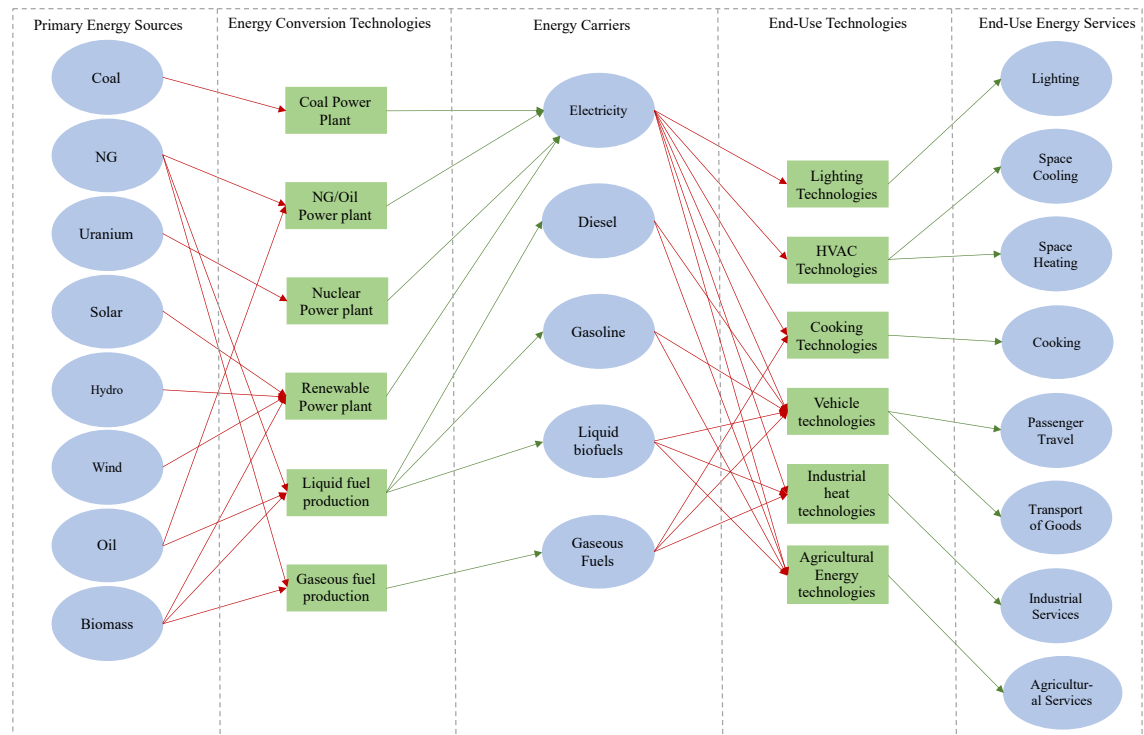


Figure 2.1 Example network diagram of an energy system in Temoa

While the mathematical formulation and code of Temoa exist, there was no structured database characterizing the East African energy system. This chapter presents version 1 of a Temoa input database for East Africa that characterizes the primary energy resources, conversion technologies, and end-use technologies for buildings and transportation for six countries in East Africa. The input database's model time horizon is in 5-year increments, and it is structured in a way that can be used to analyze energy systems in East Africa. The primary data sources used for parameterization of the Temoa East Africa

database are summarized in Table 2.3, with comprehensive tables in the Appendix, and the model architecture is depicted in Figure 2.2.

Table 2.3 Primary sources of data for model characterization

Category	Resource/Technology	Parameters	Data sources
Historical data in base year	Rural and urban electricity access rates		(The World Bank Group 2022)
	Share of population in each tier		(Bhatia and Angelou 2015; Falchetta et al. 2020)
	Existing power plant capacity		(IRENA 2023; 2021)
Primary energy supply	Solid biomass		
	Biogas		
	Domestic and imported natural gas	Fuel price projections and supply limits	(Allington et al. 2022; Dioha and Kumar 2020b; Dagnachew et al. 2020)
	Imported liquid fuel (kerosene, diesel, LPG)		
	Domestic and imported oil		
	Imported coal		
	Nuclear fuel		
Electricity conversion technologies	Renewables (solar, wind, hydro, geothermal)		
	Imported petroleum fuels (diesel, gasoline)		
	Off-grid technologies (diesel genset, stand-alone solar home systems)		
	Biomass power plants	Capital costs, fixed O&M costs, efficiency, technology lifetime, emissions factors, capacity factors	(Allington et al. 2022; IRENA 2021; Rubin, Davison, and Herzog 2015)
	Coal (steam turbine, IGCC*, CCS**)		
	Concentrated Solar PV (CSP) with storage		
	Diesel centralized power plants		
	Hydropower plants (Large (>100MW) and medium (10 – 100MW))		
	Natural gas (combustion, combined cycle, CCS) power plants	(IPCC 2022b; 2019) (IRENA 2021)	
	Heavy fuel oil power plant		
	Centralized solar PV power plant		
	Nuclear power plants		
	Geothermal power plants		
Hydrogen production	Wind (onshore and offshore) power plants		
	Mini-grid technologies (small hydro (<10MW))		
Hydrogen production	Electrolysis	Cost, efficiency, technology lifetime	(Khalili et al. 2019; Venkatesh, Sinha, and Jordan 2022; IEA 2019b)
	Natural gas steam methane reforming (SMR)		
Removal technologies	Biomass with CCS	Cost, carbon removal efficiency, technology lifetime	(Venkatesh, Sinha, and Jordan 2022; Keith et al. 2018)
	Direct Air Capture (DAC)		
Building end-use technology	Cooking technologies	Cost, efficiency, technology lifetime	(Dagnachew et al. 2020; Dioha and Kumar 2020b; Energy Information Administration (EIA) 2023)
	Lighting technologies		
Transport end-use technology	Water heating technologies		
	Electricity appliances (excluding lighting, water heating, and cooking)		
	Passenger vehicles (2-Wheeler vehicles, light-duty vehicles, public buses)	Cost, efficiency, technology lifetime	(Khalili et al. 2019; Dioha and Kumar 2020b)
	Freight vehicles (Medium-duty vehicles, heavy-duty vehicles)		

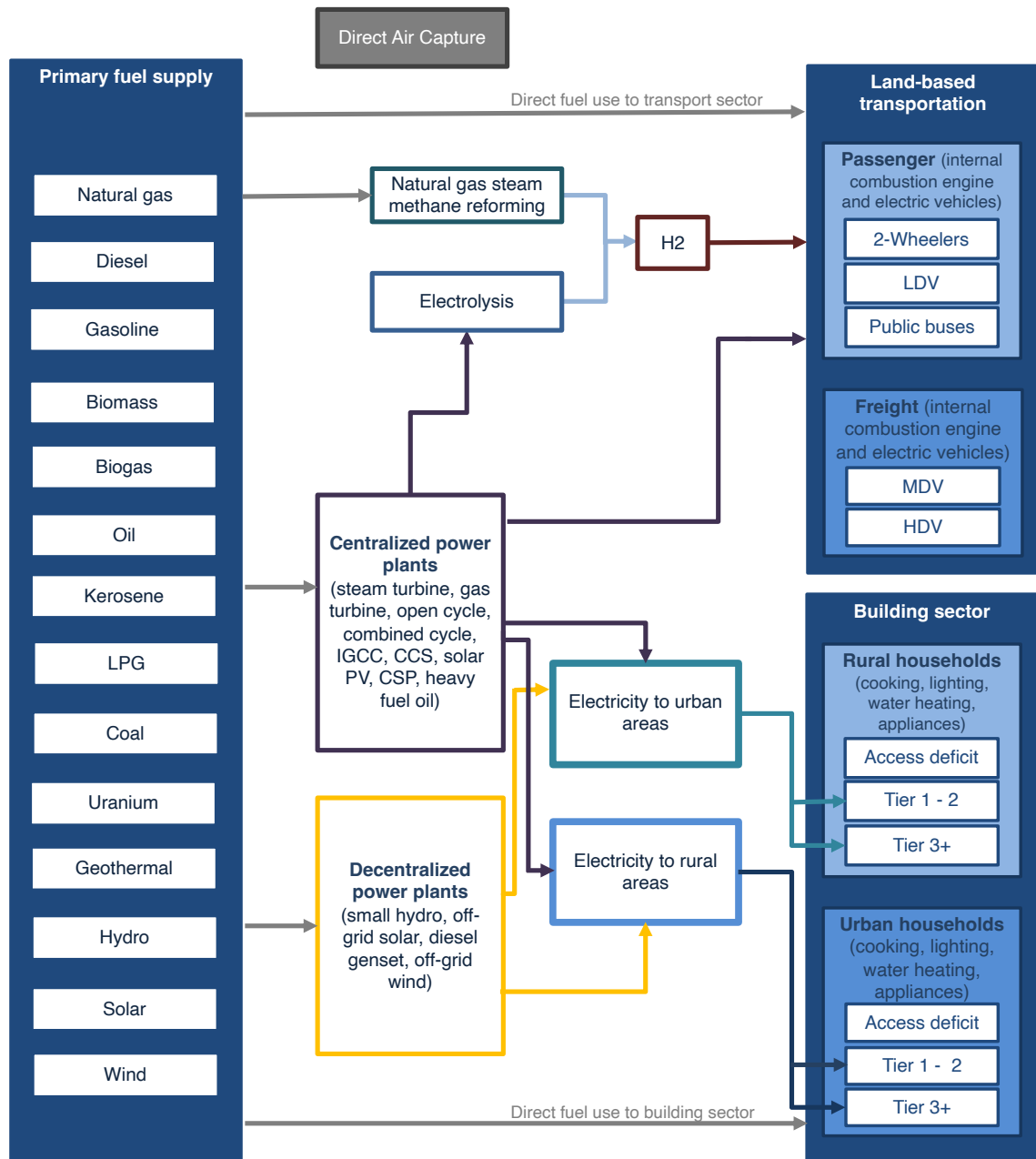


Figure 2.2 Overview of Temoa East Africa version 1.

2.4 Primary Energy Resource Characterization for East Africa

In this study, we characterized primary energy sources by focusing on imported and domestic fuel commodities and technologies. The primary data points for the supply-side include fuel price projections, carbon dioxide (CO₂) emissions activity, and non-renewable proven reserves. We collected data on these aspects for both renewable and non-renewable energy sources. Non-renewable sources in the model consist of fossil fuels (coal, natural gas, liquified petroleum gas, refined oil products) and nuclear energy, while renewable sources encompass hydropower, solar energy, geothermal, biomass, and wind. We assumed a constant domestic biomass resource price due to data scarcity issues.

We derived fossil fuel proven resource data from the Allington publication and the International Renewable Energy Agency (IRENA) online data query tool, IRENASTAT (Allington et al. 2022; IRENA 2023; 2021). The refined oil products in the model include heavy fuel oil, kerosene, diesel, and gasoline. We characterized petroleum-based fuels in East Africa as primary energy sources using Africa-specific data from peer-reviewed literature (Allington et al. 2022; Khalili et al. 2019). Although the parameterization defined fuel prices, the model does not incorporate fuel cost curve information due to limited reliable data from the region.

Our study used 5-year region-specific fuel price projections for imported and domestically extracted primary fuels until 2050. We assumed no fuel costs for renewable sources such as solar, wind, geothermal, and water. The study did not account for heterogeneity in fuel prices across all modeled countries, an assumption based on publicly available information. Furthermore, our modeling approach constrained the annual use of non-renewable energy resources based on 2020 proven reserves (Allington et al. 2022; IRENA 2023; 2021). We used supplemental fuel price data from a Nigerian residential energy model for supplying primary fuels to the end-use building sector, such as kerosene for lighting or LPG for water heating (Dioha and Kumar 2020b; 2020a).

We relied on emission activity data from the Intergovernmental Panel on Climate Change (IPCC)'s emission factors database (IPCC 2019; 2022b) and tracked biogenic CO₂ emissions from biomass (Table 2.4). The Temoa parameterization modeled CO₂ emissions activity in kilo-tonnes (kt) CO₂ per petajoule (PJ). The IPCC emission factor database provided emission factors by fuel, which we assumed would remain constant until 2050 (IPCC 2022b; 2019).

Table 2.4 CO₂ emission factors

Emission commodity	Fuel	Emission activity (kt CO ₂ /PJ)	Source
CO ₂	Coal	94.6	(IPCC 2022b; 2019)
	Diesel	74.1	
	Oil	73.3	
	Kerosene	71.9	
	LPG	63.1	
	Natural gas	56	
Biogenic CO ₂	Biomass	100	

We model biogenic emissions as a separate emissions commodity, allowing modeling approaches to treat biofuel combustion as carbon neutral. The annual CO₂ emission results are dependent on the fuel mix.

2.5 Energy Conversion Technology Characterization for East Africa

Power plants

Power plants in this study convert primary fuels into electricity, with the model optimizing the least-cost technology mix to meet electricity demand. It is important to note that the Temoa model endogenously calculates electricity demand based on the demand for final energy services. The model measures electricity generation technology capacity in gigawatts (GW) and generation activity in petajoules (PJ).

Households' access to electricity refers to connections either from the centralized grid network of decentralized mini-grid and off-grid systems for households with Tier 1+ access (Figure 2.3). Centralized grid technologies use high voltage transmission lines for long-distance electricity transmissions and distribution lines for delivering electricity at lower voltage and shorter distances. Using IRENA costs, we estimate higher costs of transmission for rural area expansion, primarily because these areas are farther from the existing grid (IRENA 2021). Off-grid technologies, such as off-grid solar photovoltaic (PV), off-grid wind, and diesel generators, supply electricity directly to end-users without incurring transmission or distribution costs. Mini-grid small hydropower plants, less than 10MW, only incur distribution costs (Mentis et al. 2017). The model represents transmission and distribution losses as technology efficiency in aggregated rural and urban transmission and distribution networks separately. The modeling approach only fixes the share of grid to off-grid connections in the base year, 2020 (IRENA 2023). We aggregated existing capacity by technology in each country.

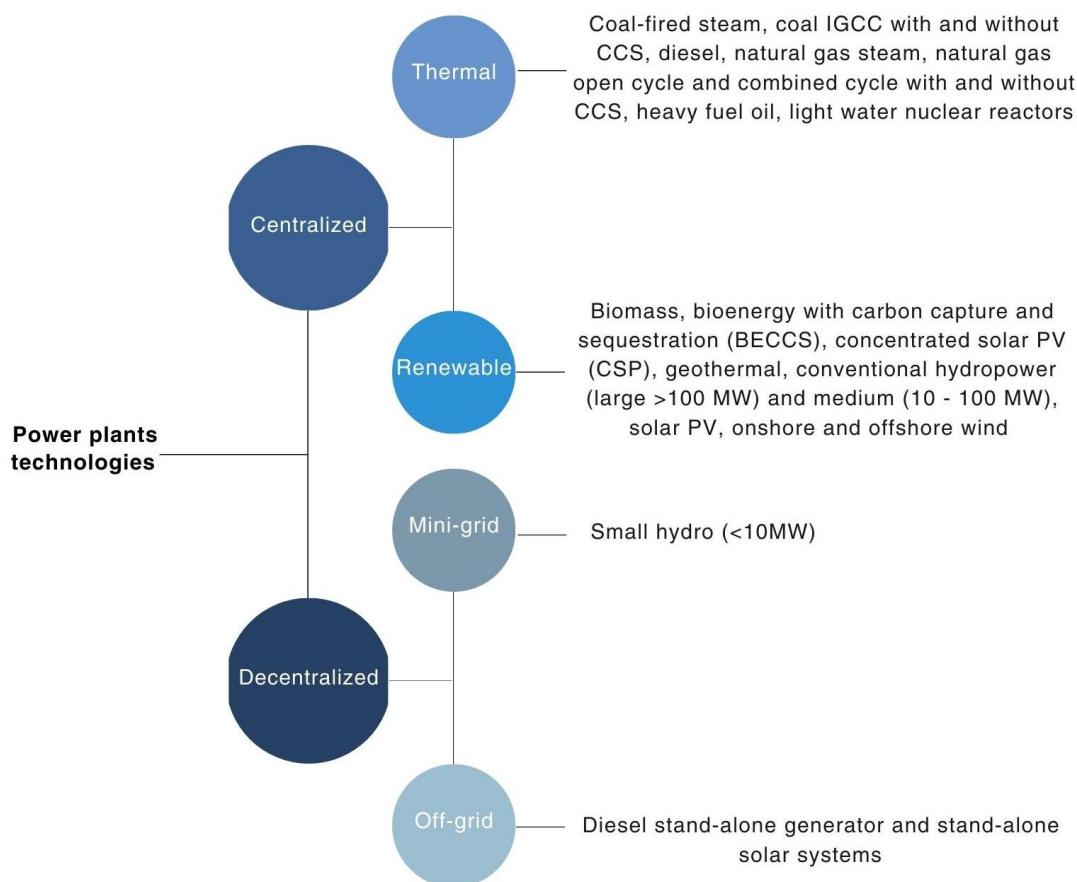


Figure 2.3 Power plant technologies in the Temoa East Africa database

Techno-economic parameters for power generation technologies include capital cost, fixed cost, existing capacity, efficiency, technology lifetime, and capacity factor. Power generation technologies depicted in Figure 2.3 encompass steam-fired, gas-fired combustion turbines, combined cycle, integrated gasification carbon capture (IGCC), and carbon capture and sequestration (CCS). Data on existing power plants by country and vintage year is available in the Allington et al. starter kit and IRENASTA online tool (Allington et al. 2022; IRENA 2023). We used country-level capacity factor data from IRENA for renewable energy technologies (IRENA 2021) and lower national average transmission costs from Allington et al. (2022) for urban areas. We assume higher transmission costs from IRENA’s Eastern Africa Power Pool publication and peer-reviewed literature for rural connections (IRENA 2021; Mentis et al. 2017).

The capital cost, an initial investment for building power generation technology, is annualized using a global discount rate of 5% in the model. The fixed costs, annualized operation and maintenance (O&M) costs are incurred regardless of electricity generated,

covering labor and equipment maintenance costs. Variable costs, O&M costs per unit of electricity generated, are the only costs in million USD (\$M) per PJ. While non-renewable technology capital costs remain constant throughout the modeling horizon, renewable energy technologies experience cost reductions over time.

The modeling approach also defines technology and region-specific efficiency, capacity factor (including availability factor), technology lifetime, emission reduction from CCS, existing installed capacity, and annual constraints on maximum and minimum activity for each power generation technology. Efficiency measures electricity generated divided by the maximum output, accounting for conversion process losses. The capacity factor represents the duration a power plant generates electricity over a given time, such as 33% for a power plant operating 8 hours in a day with 24 hours. Location-specific renewable power plants, like wind turbines, solar PV, and hydropower, usually have lower capacity factors, depending on weather variability. The technology lifetime represents the operational duration, in years, before a technology requires replacement. We approximated renewable capacity using a higher capacity factor activity based on the maximum available capacity for each country from the IRENA Renewable Energy Roadmap (Remap) country-level data (IRENA 2021).

Carbon capture and sequestration (CCS) technologies

The inclusion of carbon capture and sequestration (CCS) technologies in the least-cost optimization model presents a challenge for regions such as East Africa where empirical data is scarce. East Africa does not currently have any commercialized carbon capture and sequestration (CCS) power plants. Despite this, the model includes CCS for coal, natural gas, and biomass power plants as a technology option. However, due to the absence of empirical data on technology costs in the region, techno-economic data for emission reduction and cost implications are sourced from existing literature in developed countries (Rubin, Davison, and Herzog 2015). Limited information on the storage potential for CCS technologies in the region mean we assume no limits on sequestration capacity. The CCS technologies are modeled with a 90% carbon efficiency.

Hydrogen production

To produce hydrogen for the transport sector, we considered two processes: electrolysis and natural gas steam methane reforming, as shown in Figure 2.4. In the electrolysis process, we modeled hydrogen production from electricity, using data from the Open

Energy Outlook (OEO) database (Venkatesh, Sinha, and Jordan 2022). For the natural gas steam methane reforming representation, we simplified the process by using imported natural gas SMR, and we based our costs and efficiencies on OEO database that references the IEA Future of Hydrogen report (IEA 2019b). We assumed that only electricity from centralized technologies would be available to meet the demand for electricity in the transport sector.

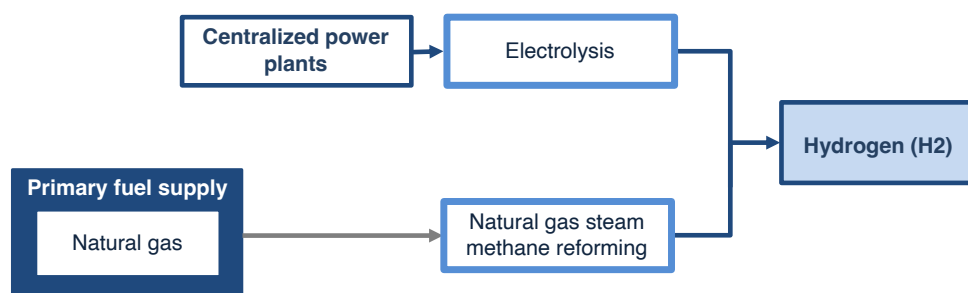


Figure 2.4 Pathways for hydrogen production in Temoa East Africa model

2.6 End-Use Technology Characterization

Energy system models aim to represent demand for energy services through the entire economy. However, the current version of the model does not include all end-use sectors.

Buildings

To meet energy demands for the building sector, the database covers technologies for different levels of energy access as defined by the World Bank's Multi-Tiered Energy Access Framework. We gathered data on investment cost, efficiency, and lifetime for each building technology for this study, and Figure 2.5 illustrates the technology options for each access tier. However, households without access to electricity cannot use electric technologies.

The model includes traditional stoves, improved modern stoves, and electric stoves as cooking technology options. We assume that households in higher consumption tiers (Tier 3+) have access to clean cooking and do not primarily rely on traditional biomass. For lighting, the database includes incandescent bulbs, light emitting diodes (LEDs), Compact Fluorescent Light (CFLs) bulbs, and solar lamps. Gas-fired (LPG and natural gas) and solar-powered water heaters are the technology options available to all households. Electric heaters are only available to households with access to electricity.

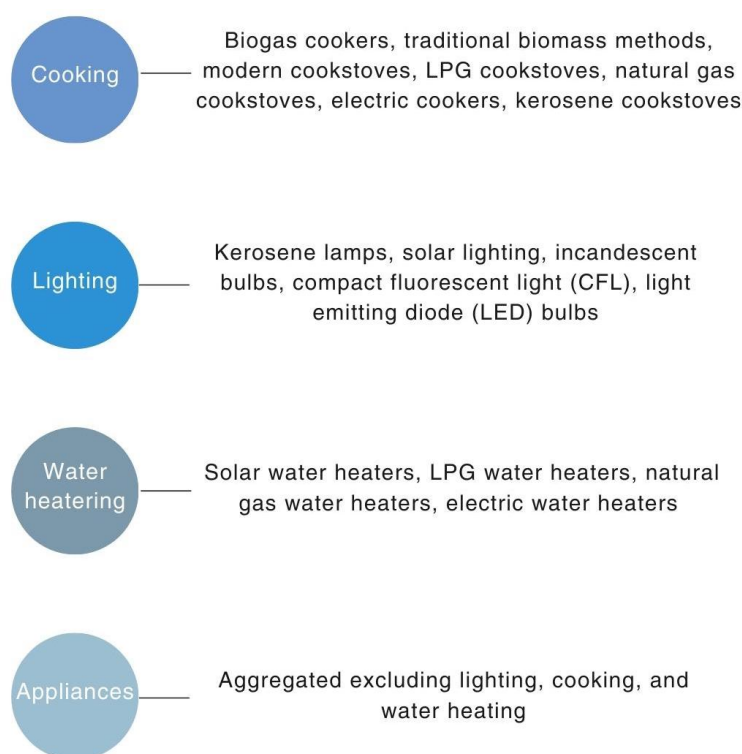


Figure 2.5 End-use technologies by household energy services

Due to limited literature on household appliances' granularity, we model this category as an aggregate electricity demand, rather than service demand with a unique set of technology options. We relied on techno-economic data from peer-reviewed literature as reported by Dagnachew et al. (2020) and Dioha and Kumar (2020b).

Transportation

To model the transport sector, we parameterized passenger and freight technologies options, as shown in Table 2.5, and include five vehicle categories: 2-wheeler vehicles (2WV), light-duty vehicles (LDV), public buses (BUS), medium-duty vehicles (MDV), and heavy-duty vehicles (HDV). For passenger transport, the model includes 2-wheeler motorcycles, light-duty vehicles, and buses. The 2-wheeler motorcycle vehicle technologies consist of diesel, gasoline, and electric vehicles (EVs). The technology options for full light-duty vehicles and buses include diesel and gasoline internal combustion engines, battery-powered EVs, and hydrogen fuel vehicles. For land-based road freight, the technologies considered are diesel, gasoline, and EVs for medium-duty class 6 trucks and medium-duty vehicles.

Table 2.5 End-use transport sector technology options

	Passenger			Freight	
	2-Wheeler	Light-duty vehicle	Public buses	Medium-duty vehicle	Heavy-duty vehicle
Diesel	×	×	×	×	×
Diesel Hybrid		×	×	×	×
Gasoline	×	×	×		
EV: 100-mile range	×	×	×	×	×
Hydrogen Fuel Cell		×		×	×

2.7 Carbon Dioxide Removal

According to the latest IPCC climate assessment report (2023), achieving the 1.5-degree target will likely require carbon dioxide removal (CDR) measures. CDR can be achieved through nature-based solutions or carbon capture processes like direct air capture. Although these technologies have not been implemented on a large scale, we have included them in the Temoa input database for East Africa. Our modeling approach incorporates bioenergy with carbon capture and sequestration (BECCS) and direct air capture (DAC) technology as negative emissions technologies. The cost and efficiency assumptions underlying these carbon removal technologies are based on peer-reviewed publication and assumptions from the OEO database (Venkatesh, Sinha, and Jordan 2022; Keith et al. 2018).

2.8 Conclusion

This chapter presented the first iteration of the Temoa East Africa database, offering a comprehensive depiction of supply, electricity, building, and transport sector technologies. Using the Temoa modeling framework, the model serves as a high-resolution, open-source energy model for six East African countries: Burundi, the Democratic Republic of Congo, Kenya, Rwanda, Tanzania, and Uganda. With the Temoa East Africa version 1, we accomplished our research objective of creating a widely accessible, high-resolution energy database and model, specifically designed for energy system analysis.

Future iterations will strive to improve the database by disaggregating transportation service demand based on access level across rural-urban divisions and incorporating electric vehicle charging station characterizations. In addition, the forthcoming Temoa East Africa version will encompass the industrial and agricultural sectors, boasting

extensive technology characterizations. The long-term objective is to continuously expand the modeling scope to encompass end-use technologies in all major energy system sectors, ultimately publishing the complete model database in a publicly accessible GitHub repository. By providing an open-source model, we hope to encourage further research and collaboration on sustainable energy planning in East Africa.

3 USING TEMOA EAST AFRICA MODEL: UNIVERSAL ENERGY ACCESS IMPACTS UNDER DIFFERENT SCENARIOS IN EAST AFRICA

3.1 Research Goals and Modeling Framework

Energy system models provide insights into how the energy system could evolve. However, they often fail to adequately capture disparities within sub-groups in developing countries. The objective of this chapter is to assess the impacts of meeting SDG 7.1 under different socioeconomic and access tier scenarios that simulate various universal energy access definitions in rural and urban East Africa. This regional analysis includes Burundi, the Democratic Republic of Congo, Kenya, Rwanda, Tanzania, and Uganda.

To leverage energy system modeling tools, this study first estimated the household demand for energy services in rural and urban areas in East Africa under different scenarios. Second, using the Temoa East Africa model parameterization described in Chapter 2, the analysis evaluated the least cost fuel and technology mix to meet the household demand for energy services in different access tiers under hypothetical access scenarios by the year 2050 in rural and urban communities. The study relies on the multi-tier framework definition of energy access, with five energy access tiers representing an access continuum based on energy consumption levels (as shown in Table 1.2).

This study used the detailed representation of the primary fuel supply, electricity, and household end-use services in Temoa East Africa, which we parameterized in Chapter 2 of this thesis. This study examined the impact of varying energy access definitions on technology choice, costs, and emissions of for countries in East Africa. The primary fuels

on the supply side either meet the demand, for end-use services in the households directly or in a converted energy form, such as electricity. However, the transport characterization and hydrogen production pathways from Chapter 2 were not included in this study because the current version of the database does not adequately represent electricity access levels to meet the transportation sector's electricity demand.

Figure 3.1 depicts a modified network diagram of the modeling framework used in this chapter, focusing solely on meeting household energy demand. The modified modeling framework does not include the transport characterization and hydrogen production pathways. Energy supply from primary fuels through the electricity conversion processes is endogenous to the model, going to the end-use energy demand in each tier access level by energy service in rural and urban areas. The modeling objective of this work was to minimize the total cost of meeting household energy demand under different socioeconomic and policy scenarios out to 2050, with a time horizon of 30 years between 2020 and 2050, using 5-year increments. Temoa solves assuming perfect foresight across the time horizon based on demand exogenously specified in each period. It is important to consider the key assumptions underlying the model when interpreting the results. we assume a global discount rate of 5% with all monetary units are denominated in USD\$2020, and each annual period is represented by two time slices (day and night in one season).

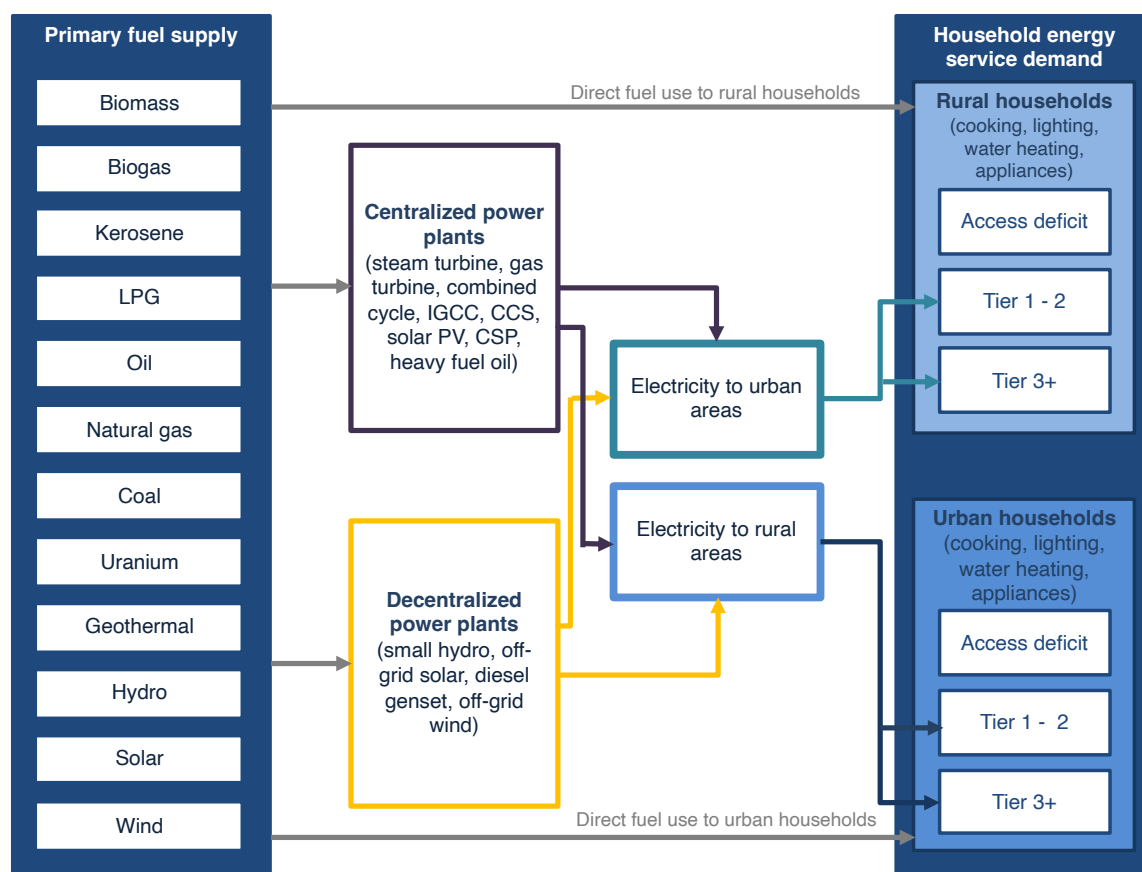


Figure 3.1 Temoa East Africa household model structure representation

3.2 Demand scenarios

The objective of this study was to identify the most cost-effective fuel and technology options to meet household demand for energy services over a 30-year period (2020 – 2050), with scenarios analyzed in 5-year increments. To do this, we used a total of 36 scenarios, which included three shared socioeconomic pathways (SSPs), six access scenarios, and two CO₂ emissions scenarios (Figure 3.2). The *Uncon* scenario had no emissions constraints, while the *NZ2050* scenario simulated linear CO₂ reductions by 2050 to explore trade-offs between universal energy access and climate mitigation.

In the *NZ2050* scenario, we tracked biogenic CO₂ emissions from biomass burning throughout the energy system. While the approach treated combustion of biomass and biofuels as carbon neutral, we tracked the biogenic CO₂ emissions throughout the energy system to examine the evolution under various scenarios. These emissions were not restricted by carbon constraints, which only limit fossil fuel CO₂ emissions.

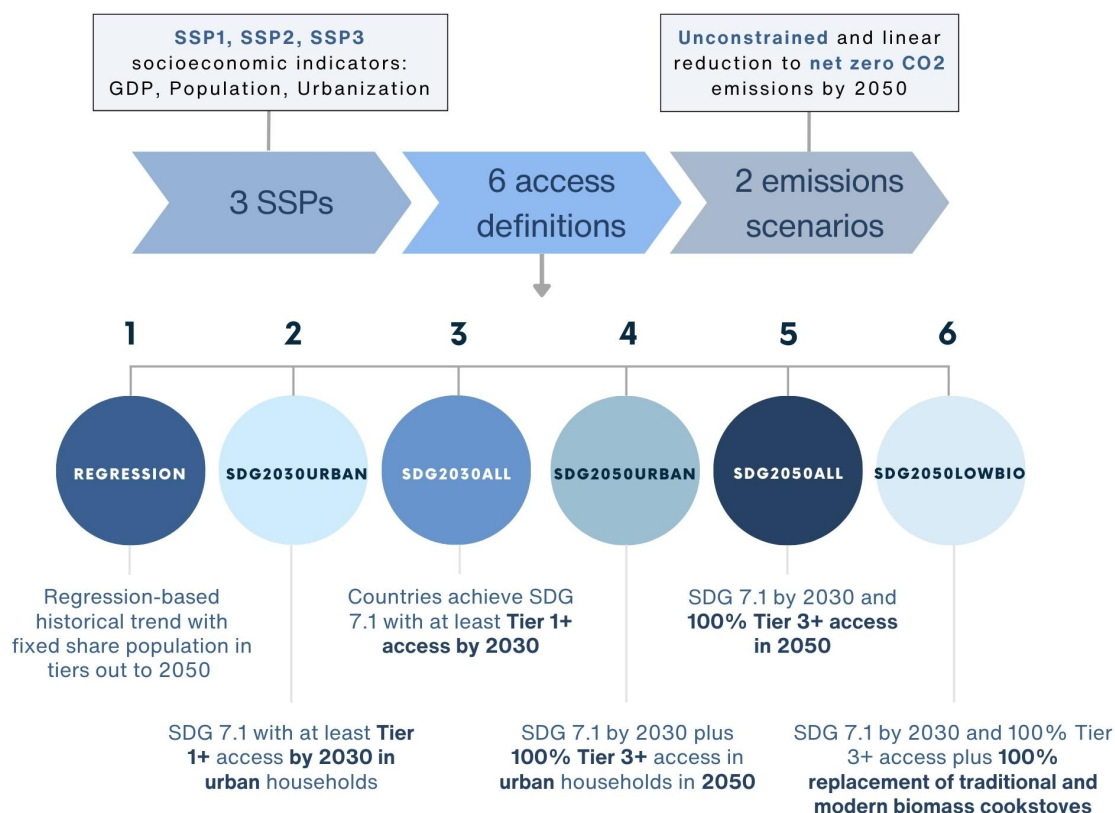


Figure 3.2 Summary of 36 scenario matrix in this analysis. We run 3 SSPs under 2 emissions scenarios for each energy access definition. The incremental change is highlighted in bold colors.

We used the Temoa East Africa household model to analyze energy access, dividing the end-use services by access to electricity and cooking solutions in rural and urban areas, and disaggregating the end-use services by tier using the multi-tier framework, which defines energy access levels as depicted in Table 1.2 (Bhatia and Angelou 2015). We defined the demand for cooking, lighting, water heating, and other appliances in rural and urban households in 5-year increments until 2050. We relied on socioeconomic indicators such as population, economic activity or income levels, urbanization, and other factors that determine energy consumption levels to determine future service demands. Our demand functions for service demand in developing countries were based on peer-reviewed literature (Levesque et al. 2018; Daioglou, van Ruijven, and van Vuuren 2012; Dioha and Kumar 2020b). Figure 3.3 depicts the disaggregated demand for cooking, lighting, appliances, and water heating by access tier in rural and urban areas.

	COOKING	LIGHTING	WATER HEATING
Access deficit	Biogas cookers, <u>traditional biomass methods</u> , <u>modern biomass cookstoves</u> , LPG cookstoves, natural gas cookstoves, kerosene cookstoves	<u>Kerosene lamps</u> , solar lighting	Solar water heaters, LPG water heaters, natural gas water heaters
Tier 1 - 2	Biogas cookers, <u>modern biomass cookstoves</u> , LPG cookstoves, natural gas cookstoves, <u>electric cookers</u> , kerosene cookstoves	<u>Kerosene lamps</u> , solar lighting, <u>incandescent bulbs</u> , <u>compact fluorescent light (CFL)</u> , <u>light emitting diode (LED) bulbs</u>	Solar water heaters, LPG water heaters, natural gas water heaters, <u>electric water heaters</u>
Tier 3 - 5	Biogas cookers, LPG cookstoves, natural gas cookstoves, <u>electric cookers</u> , kerosene cookstoves	Solar lighting, <u>incandescent bulbs</u> , <u>compact fluorescent light (CFL)</u> , <u>light emitting diode (LED) bulbs</u>	Solar water heaters, LPG water heaters, natural gas water heaters, <u>electric water heaters</u>

Figure 3.3 Technology choice by access tier. Underlined text represents select technologies available by access level.

Shared socioeconomic pathways (SSPs)

To explore a broad range of plausible scenarios for long-term demand for energy services, this study used the shared socioeconomic pathways (SSPs) as a framework for scenario analysis. The SSPs are not predictions of the future, but rather exploratory narratives of plausible “what-ifs,” based on alternative socioeconomic futures in the absence of climate policy intervention (IPCC 2018). Therefore, we used the SSPs socioeconomic data to develop demand scenarios for the building sector end-use services in rural and urban areas (Schandl et al. 2020; Riahi et al. 2017). We estimated household demand for energy services in rural and urban areas on a tiered access level out to 2050 under three of the five global representative SSPs in Table 3.1 (IIASA 2018; Riahi et al. 2017). These SSPs covered a wide range of socioeconomic pathways from low to high economic growth, low to high population, and low to high urbanization. Specifically, SSP 2 followed historical trends, while SSP 3 assumed rapid population growth in developing countries and served as the benchmark demand scenario for this analysis. In addition, we included SSP 1, which presumed high economic activity growth. More detailed historical and SSP projections for each country by indicator are available in Appendix Figure 7.2, Figure 7.3 and Figure 7.4.

Table 3.1 Summary of SSP storylines and assumptions for quantification of the socioeconomic projections with ratios provided by IIASA (Cuaresma 2017; Jiang and O'Neill 2017; KC and Lutz 2017; Riahi et al. 2017). This study used three SSPs in the Temoa East Africa household model.

Scenario	Name (Description)	GDP [Ratio 2050:2010]	Population [Ratio 2050:2010]	Urban share [Ratio 2050:2010]
SSP1	Sustainable – Taking the Green Road (Inclusive and sustainable pathway)	High economic growth [4.3]	Low population growth [1.2]	High urbanization [1.9]
	Middle-of-the-road (Path follows historical patterns)	Uneven income growth [3.4]	Moderate population growth [1.3]	Moderate urbanization [1.8]
SSP3	Regional rivalry – A Rocky Road (Low international priority and regional cooperation)	Slow income growth [2.7]	High population growth in developing countries. [1.5]	Low urbanization [1.6]
	Inequality – A Road Divided (Increasing regional inequalities within and across countries)	Unequal GDP growth [3.3]	Moderate population growth [1.3]	High urbanization [2.0]
SSP5	Fossil-fueled development (Rapid global technological progress and heavy fossil-fuel deployment)	High economic growth [5.4]	Global population peaks [1.3]	High urbanization [1.9]

Access scenarios

To address data scarcity and quality challenges, this study developed target-based scenarios for population share in each access tier out to 2050. Previous studies in the region relied on econometric approaches using historical data (Ouedraogo 2017a; Dioha and Emodi 2019). This study explored six hypothetical national energy access scenarios informed by who, where, and when aspects to clarify the definition of energy access (see Table 3.2 for summary). The share of the population in each tier was assumed to remain constant across time, except for the two scenarios that explore 100% Tier 3+ access in 2050.

In the reference scenario, *Regression*, this study used logistic regression analysis adapted from Dioha and Kumar (2020b)'s TIMES Nigeria model. We rely on historical data from the World Bank on electricity access rates between 2000 and 2020 in rural and urban areas for each country as shown Appendix Figure 7.7 (IEA et al. 2018; The World Bank

Group 2022). This study estimated the share of the population with access to electricity in rural and urban areas, equation (1) is the resulting logistic function, where k and x_0 are the regression coefficients, and i is the rural or urban population share. Our analysis confirmed a strong positive relationship between Gross Domestic Product (GDP) and electricity access rates, as seen in previous studies (Dioha and Kumar 2020b).

$$Elc_i = \frac{1}{1 + e^{-(k+GDPx_0)}} \quad (1)$$

Due to limited historical data availability, we assumed a 5-year 2% annual growth rate for Burundi and a 1% annual growth rate in the Democratic Republic of Congo for the population share with access to electricity in rural areas until 2050 assuming the two countries follow same growth trend in available historical data. For clean cooking access, there is also scarce country-level data, so we assumed a 5% growth rate in rural areas and a 10% growth rate in urban areas of the share of population with access to cooking solutions.

Table 3.2 Summary table of access scenario assumptions

Scenario	Electricity and clean cooking access	Incremental change from the preceding scenario
Regression	Regression-based historical trends with a fixed share of the population in access tiers	
SDG2030urban	SDG 7.1 by 2030 in urban areas	100% Tier 1+ electricity and clean cooking access in urban areas by 2030
SDG2030All	Countries achieve SDG 7.1 by 2030	100% Tier 1+ electricity and clean cooking access in rural areas by 2030
SDG2050urban	SDG 7.1 by 2030 plus 100% Tier 3+ access in urban areas by 2050	100% Tier 3+ electricity and clean cooking access in urban areas
SDG2050All	SDG 7.1 by 2030 plus 100% Tier 3+ by 2050	100% Tier 3+ electricity and clean cooking access in rural areas by 2050
SDG2050LowBio	SDG 7.1 by 2030 and 100% Tier 3+ electricity and clean cooking access in rural and urban areas by 2050	100% replacement of traditional biomass and modern biomass cookstoves

Our modeling assumptions keep the population distribution across tiers for electricity and clean cooking was fixed according to 2020 data (Falchetta et al. 2020; Bhatia and Angelou 2015). In the *SDG2030urban* access scenario, we assumed that by 2030, 100% of the

urban population in East Africa would achieve at least minimum access to electricity and clean cooking access (Tier 1+), while in rural areas, access and population share across access tiers follows the *Regression* scenario.

In the *SDG2030All* access scenario, we assumed that all the East African countries would achieve 100% minimum access to electricity and clean cooking access (Tier 1+) in both rural and urban areas by 2030.

In the *SDG2050urban* access scenario, we maintained the *SDG2030All* assumptions and additionally assumed that by 2050, 100% of the urban population in East Africa would achieve higher access tiers for electricity and clean cooking (Tier 3+). Before 2050, the scenario assumptions follow *SDG2030All* scenario.

The *SDG2050All* access scenario, we explored the impact of achieving higher access tiers (Tier 3+) for 100% of the rural and urban population in East Africa by 2050. Before 2050, the assumptions track the *SDG2030All* scenario.

We also considered the *SDG2050LowBio* access scenario, which explores the impact of replacing traditional and modern biomass cookstoves with alternative options. This scenario assumes the *SDG2050All* access scenario in addition to a complete replacement of traditional biomass cookstoves. To address the unsustainable use of traditional biomass, we investigated an alternative pathway where traditional biomass is not an option, while still maintaining the *SDG2050All* assumptions. Excluding modern biomass as an end-use fuel option in this scenario aims to pay closer attention to reducing reliance on the household use of biomass.

Table 3.3 Scenario assumptions of population share by access tier.

subregion	SDG 7.1 indicator	Access tier	Country	Regression		SDG2030urban		SDG2030All		SDG2050urban		SDG2050All	
				Upper limit (SSP3) 2030	Upper limit (SSP3) 2050	Upper limit (SSP3) 2030	Upper limit (SSP3) 2050	Upper limit (SSP3) 2030	Upper limit (SSP3) 2050	Upper limit (SSP3) 2030	Upper limit (SSP3) 2050	Upper limit (SSP3) 2030	Upper limit (SSP3) 2050
Rural households	7.1.1 access to electricity	Access deficit	Burundi	95%	90%	95%	90%	0%	0%	0%	0%	0%	0%
			Congo, Dem. Rep.	98%	96%	98%	96%	0%	0%	0%	0%	0%	0%
			Kenya	6%	0%	6%	0%	0%	0%	0%	0%	0%	0%
			Rwanda	67%	20%	67%	20%	0%	0%	0%	0%	0%	0%
			Tanzania	74%	33%	74%	33%	0%	0%	0%	0%	0%	0%
			Uganda	49%	1%	49%	1%	0%	0%	0%	0%	0%	0%
		Tier 1 – 2	Burundi	5%	10%	5%	10%	100%	100%	100%	100%	100%	0%
			Congo, Dem. Rep.	1.4%	3%	1.4%	3%	73%	73%	73%	73%	73%	0%
			Kenya	40%	43%	40%	43%	43%	43%	43%	43%	43%	0%
			Rwanda	10%	25%	10%	25%	31%	31%	31%	31%	31%	0%
			Tanzania	16%	40%	16%	40%	60%	60%	60%	60%	60%	0%
			Uganda	40%	79%	40%	79%	80%	80%	80%	80%	80%	0%
		Tier 3+	Burundi	0%	0%	0%	0%	0%	0%	0%	0%	0%	100%
			Congo, Dem. Rep.	0.6%	1%	0.6%	1%	27%	27%	27%	27%	27%	100%
			Kenya	54%	57%	54%	57%	57%	57%	57%	57%	57%	100%
			Rwanda	23%	55%	23%	55%	69%	69%	69%	69%	69%	100%

Chapter 3: Using Temoa East Africa Model: Universal energy access impacts under different scenarios in East Africa

	7.1.2 access to clean cooking	Access deficit	Tanzania	10%	27%	10%	27%	40%	40%	40%	40%	40%	100%
			Uganda	11%	20%	11%	20%	20%	20%	20%	20%	20%	100%
			Burundi	95%	80%	95%	80%	0%	0%	0%	0%	0%	0%
			Congo, Dem. Rep.	95%	80%	95%	80%	0%	0%	0%	0%	0%	0%
			Kenya	85%	60%	85%	60%	0%	0%	0%	0%	0%	0%
			Rwanda	95%	80%	95%	80%	0%	0%	0%	0%	0%	0%
			Tanzania	95%	80%	95%	80%	0%	0%	0%	0%	0%	0%
			Uganda	95%	80%	95%	80%	0%	0%	0%	0%	0%	0%
		Tier 1 – 2	Burundi	5%	20%	5%	20%	100%	100%	100%	100%	100%	0%
			Congo, Dem. Rep.	4%	15%	4%	15%	73%	73%	73%	73%	73%	0%
			Kenya	6%	17%	6%	17%	43%	43%	43%	43%	43%	0%
			Rwanda	4%	19%	4%	19%	97%	97%	97%	97%	97%	0%
			Tanzania	3%	12%	3%	12%	60%	60%	60%	60%	60%	0%
		Tier 3+	Uganda	4%	16%	4%	16%	80%	80%	80%	80%	80%	0%
			Burundi	0%	0%	0%	0%	0%	0%	0%	0%	0%	100%
			Congo, Dem. Rep.	1%	5%	1%	5%	27%	27%	27%	27%	27%	100%
			Kenya	8%	23%	8%	23%	57%	57%	57%	57%	57%	100%
			Rwanda	1%	1%	1%	1%	3%	3%	3%	3%	3%	100%
			Tanzania	2%	8%	2%	8%	40%	40%	40%	40%	40%	100%
			Uganda	1%	4%	1%	4%	20%	20%	20%	20%	20%	100%

Energy Access and Climate Mitigation: Impacts on Technology Choice, Emissions, and Costs in East Africa

Urban households	7.1.1 access to electricity	Access deficit	Burundi	36%	27%	0%	0%	0%	0%	0%	0%	0%	0%
			Congo, Dem. Rep.	59%	1%	0%	0%	0%	0%	0%	0%	0%	0%
	Tier 1 – 2		Kenya	6%	0%	0%	0%	0%	0%	0%	0%	0%	0%
			Rwanda	14%	5%	0%	0%	0%	0%	0%	0%	0%	0%
			Tanzania	27%	2%	0%	0%	0%	0%	0%	0%	0%	0%
			Uganda	30%	0%	0%	0%	0%	0%	0%	0%	0%	0%
			Burundi	32%	37%	50%	50%	50%	50%	50%	0%	50%	0%
			Congo, Dem. Rep.	25%	59%	60%	60%	60%	60%	60%	0%	60%	0%
			Kenya	38%	40%	40%	40%	40%	40%	40%	0%	40%	0%
			Rwanda	7%	8%	8%	8%	8%	8%	8%	0%	8%	0%
			Tanzania	37%	49%	50%	50%	50%	50%	50%	0%	50%	0%
			Uganda	25%	36%	36%	36%	36%	36%	36%	0%	36%	0%
	Tier 3+		Burundi	32%	36%	50%	50%	50%	50%	50%	100%	50%	100%
			Congo, Dem. Rep.	16%	40%	40%	40%	40%	40%	40%	100%	40%	100%
			Kenya	56%	60%	60%	60%	60%	60%	60%	100%	60%	100%
			Rwanda	79%	87%	92%	92%	92%	92%	92%	100%	92%	100%
			Tanzania	36%	49%	50%	50%	50%	50%	50%	100%	50%	100%
			Uganda	45%	64%	64%	64%	64%	64%	64%	100%	64%	100%
	7.1.2 access to clean cooking	Access deficit	Burundi	95%	60%	0%	0%	0%	0%	0%	0%	0%	0%

Chapter 3: Using Temoa East Africa Model: Universal energy access impacts under different scenarios in East Africa

	Tier 1 – 2	Congo, Dem. Rep.	85%	65%	0%	0%	0%	0%	0%	0%	0%	0%
		Kenya	40%		0%	0%	0%	0%	0%	0%	0%	0%
		Rwanda	70%	30%	0%	0%	0%	0%	0%	0%	0%	0%
		Tanzania	65%	20%	0%	0%	0%	0%	0%	0%	0%	0%
		Uganda	90%	95%	0%	0%	0%	0%	0%	0%	0%	0%
		Burundi	3%	20%	50%	50%	50%	50%	50%	0%	50%	0%
		Congo, Dem. Rep.	9%	21%	60%	60%	60%	60%	60%	0%	60%	0%
		Kenya	24%	40%	40%	40%	40%	40%	40%	0%	40%	0%
		Rwanda	22%	53%	76%	76%	76%	76%	76%	0%	76%	0%
		Tanzania	17%	40%	50%	50%	50%	50%	50%	0%	50%	0%
	Tier 3+	Uganda	4%	18%	36%	36%	36%	36%	36%	0%	36%	0%
		Burundi	2%	20%	50%	50%	50%	50%	50%	100%	50%	100%
		Congo, Dem. Rep.	6%	14%	40%	40%	40%	40%	40%	100%	40%	100%
		Kenya	36%	60%	60%	60%	60%	60%	60%	100%	60%	100%
		Rwanda	8%	17%	24%	24%	24%	24%	24%	100%	24%	100%
		Tanzania	18%	40%	50%	50%	50%	50%	50%	100%	50%	100%
		Uganda	6%	32%	64%	64%	64%	64%	64%	100%	64%	100%

3.3 Service demand functions

Using socioeconomic data from the shared socioeconomic pathways (SSPs) (Cuaresma 2017; Jiang and O'Neill 2017; KC and Lutz 2017; Riahi et al. 2017), this section describes the demand function for each end-use service. Using the scenario-based gross domestic product (GDP), population, and urban share projections, we use the demand functions to estimate the long-term demand for household energy services by tier in rural and urban areas out to the year 2050. The equations we used in this analysis are based on a previous study (Levesque et al. 2018).

Cooking. The demand for cooking is strongly linked to population growth and has been found to be independent of income levels in previous studies (Dioha and Kumar 2020a; Levesque et al. 2018; Daioglou, van Ruijven, and van Vuuren 2012). While, energy intensity ranges from 0.5 to 5 GJ/capita/year in the literature (Dioha and Kumar 2020b; Levesque et al. 2018; Daioglou, van Ruijven, and van Vuuren 2012), we assumed a constant energy intensity of 1.8 gigajoules (GJ) per capita per year. Equation (2) represents the cooking demand function where the i value is the population in rural or urban households, and j represents population distribution in each tier. We disaggregate demand by tier to separate the technologies available to meet demand as depicted in Figure 3.3. We used the multi-tier framework which employs the International Workshop Agreement technical guidelines (Bhatia and Angelou 2015) to provide different technology options for each access level (Table 1.2).

$$Cooking_{i,j} (GJ) = 1.8 \times Population_{i,j} \quad (2)$$

Water heating. The East African region has tropical weather with an equatorial climate, which makes hot water a luxury rather than a necessity in the area. Additionally, it has been noted that the demand for hot water is positively associated with income levels (Dioha and Kumar 2020a; Levesque et al. 2018). This study estimated the demand for water heating in East Africa based on the assumption that 10% of the rural population and 20% of the urban population use hot water in 2020. To estimate demand for hot water, we used equation (3), where I represents the income per capita and ϕ_n represents the coefficient (Levesque et al. 2018). The ϕ_n factors vary across each SSP to represent different climate futures (Levesque et al. 2018).

$$Water\ heating_{i,j} (GJ) = \frac{\phi_1}{1 + \exp\left(\frac{\phi_2 - I}{\phi_3}\right)} \times Population_{i,j} \quad (3)$$

Lighting. We used equation (4) to project floor space, and equation (5) estimate the long-term demand for lighting services under the SSPs when calculating lighting demand out to the year 2050. I represents per capita GDP in equation (4), while D represents the population density. To estimate the demand for lighting, this study assumed an average lighting need of 80 lumens per square meter (m^2) based on a recent Ghanaian study (Diawuo et al. 2019). To estimate service demand, we assumed 80 lumens/ m^2 for urban households from the Ghanaian city study, but used lower lighting need of 25 lumens/ m^2 for rural households (Lighting Global 2013). The multi-tier framework provides data on average lighting hours per tier (Bhatia and Angelou 2015), which we use in equation (5) to estimate the service lighting demand in gigaumen-years.

$$Floor\ space, FS (m^2\ per\ capita) = \alpha I^\beta D^\gamma \quad (4)$$

$$\begin{aligned} &Lighting\ (Gigalumen - year) \\ &= FS (m^2\ per\ capita) \times Population_{i,j} \times \frac{lumens}{m^2} \\ &\times (lighting\ hours_j \times 365)hrs \times \frac{year}{8760hrs} \times \frac{gigalumens}{10^9lumen} \end{aligned} \quad (5)$$

Electric Appliances. We used equation (6), to estimate the aggregate household demand for electricity for other appliances accounting for the proportion of rural and urban populations with access to electricity in each of the five tiers. In equation (6), i represents the rural or urban share, and j is the population distribution in each tier. The thresholds for household consumption levels by tier are from the multi-tier framework (as shown in Table 3.4). Household characteristics data are from the national Demographic and Health Survey (DHS) program (DHS Program - USAID 2022). We hold the average number of people per household in 2020 constant through to 2050.

Other electric appliances_{i,j} (PJ)

= (Population_{i,j} / Average people per household)

× Electricity access share(%) × Distribution across tiers (%)

× Electricity consumption by tier (kWh/household)

(6)

Table 3.4 Household electricity consumption in Multi-Tier Framework. The study excluded lighting and food processing from the multi-tier matrix assumptions to estimate electricity demand for appliances. Data source: (Bhatia and Angelou 2015)

	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5
Annual consumption levels (kWh per household)	>=4.5	>=73	>=365	>=1,250	>=3,000
Radio	1.5	5.8	5.8	5.8	5.8
Air circulation		29.2	87.6	175.2	262.8
Television		14.6	29.2	29.2	29.2
Washing machine			182.5	182.5	182.5
Refrigerator				657	657
Iron				120.5	120.5
Air conditioner					1,642.5
Total (excluding lighting and cooking)	1.5	50	305	1,1170	2,900

3.4 Demand projections

In the *Regression* scenario, the demand for energy services increases across all the scenarios, with cooking accounting for the highest share of demand. Table 3.5 shows that demand for household energy services is increasing in both rural and urban areas as projected income and population increase in each SSP. Under SSP 1, rural demand for cooking decreases by 9%. The share of the population in urban areas increases in all three SSPs with higher urbanization in SSP1. Rapid urbanization partially explains the diminishing growth rate in rural areas as more people migrate to urban areas.

Table 3.5 Household service demand in 2050 under the Regression demand scenario assumptions across each scenario

Service	rural			urban		
	SSP1	SSP2	SSP3	SSP1	SSP2	SSP3
Cooking (PJ)	192	435	805	298	373	375
Lighting (Gigalumen-years)	1.02	6.41	11.1	10.8	38.2	35.4
Water heating (PJ)	15.4	25.7	37.9	38.3	35.3	27.1
Other appliances (PJ)	62.5	71.3	75.6	168	194	227

Figure 3.4 depicts the energy demand for household services by access tiers under all the access scenarios in SSP 3. The demand estimates do not change between *SDG2050All* and *SDG2050LowBio* scenarios, so we only present *SDG2050All* (Figure 3.4). The scenarios assume meeting SDG target 7.1 Tier 1+ by 2030 and access level Tier 3+ by 2050, as described in Table 3.2, which explains the observations in the years 2030 and 2050.

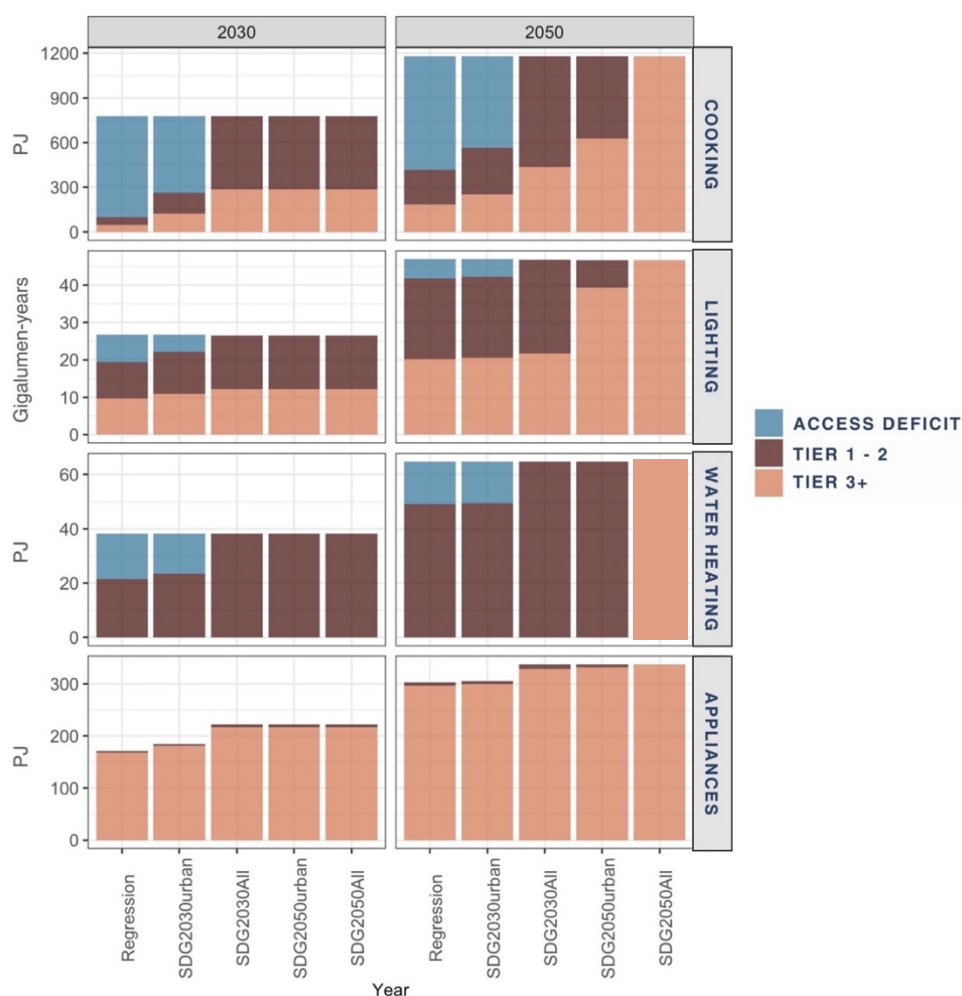


Figure 3.4 Regional service demands in SSP3 reveal a shift in tier access levels under the access scenario assumptions by 2030 and 2050.

3.5 Modeling Results and Discussion

This section presents the final energy consumption, least-cost technology mix, emission activity, and total costs for each scenario by tier access group to show the impact of changing universal access definitions. The access deficit represents the proportion of the population that lacks access to electricity and clean cooking, the Tier 1 – 2 access level represents the share of the population with minimal energy access, and the Tier 3+ access level is the higher consumption tier in the proportion of population that has access to electricity and clean cooking. The section presents the tier results as regional totals for rural and urban households.

First, this section discusses the results of the six access scenarios (*Regression*, *SDG2030urban*, *SDG2030All*, *SDG2050urban*, *SDG2050All*, *SDG2050LowBio*), three SSPs (*SSP1*, *SSP2*, *SSP3*), and two emission reduction scenarios (*Uncon* and *NZ2050*). It begins by presenting the final energy consumed by fuel and technology choice to meet service demand in rural and urban households. The section concludes with a discussion of the cost and emission implications under the different energy access scenarios, along with corresponding results. The net zero CO₂ emission constraints only apply to CO₂ emissions from fossil fuels.

Final household energy consumption by fuel

For all the six countries combined, the final energy consumption in rural households is 2.4 times higher than urban households in 2050 under the reference *Regression* scenario. This result may be explained by the larger share of access deficit in rural households rely on inefficient technologies. In the *Regression* scenario, traditional biomass is currently the primary source of energy, meeting over 99% of final energy consumption in households in the access deficit. In contrast, modern biomass is used for 91% of final energy consumption in Tier 1 – 2 households, while LPG and electricity over meet 90% of final energy consumption in Tier 3+ households under the SSP 3 unconstrained emissions scenario in 2050. Figure 3.5 provides a visual representation of the regional distribution of final household energy consumption in both rural and urban areas by access tier and optimal fuel choice using SSP 3 projections. The x-axis of the figure represents years from 2020 to 2050, while the color scale indicates which fuel meets household energy demand. The vertical panels illustrate the unconstrained (*Uncon*) and linear reduction to net zero by 2050 (*NZ2050*) scenarios across the years. The study found

that electricity is not a feasible option for minimum access levels, Tier 1 – 2, because modern biomass is a cheaper alternative. In contrast to lower tiers (Tier 0 and Tiers 1 - 2), we assume higher tiers (Tiers 3 - 5) do not primarily rely on biomass, making traditional and biomass cookstoves unavailable options for this population in the model, therefore electric technologies become increasingly viable to meet demand from Tier 3+ households. The figure also shows that in 2050, the total final household energy consumption in rural households (3,599 PJ) is approximately two times higher than in urban households (1,454 PJ) driven by the continued reliance on efficient traditional biomass methods in rural areas.

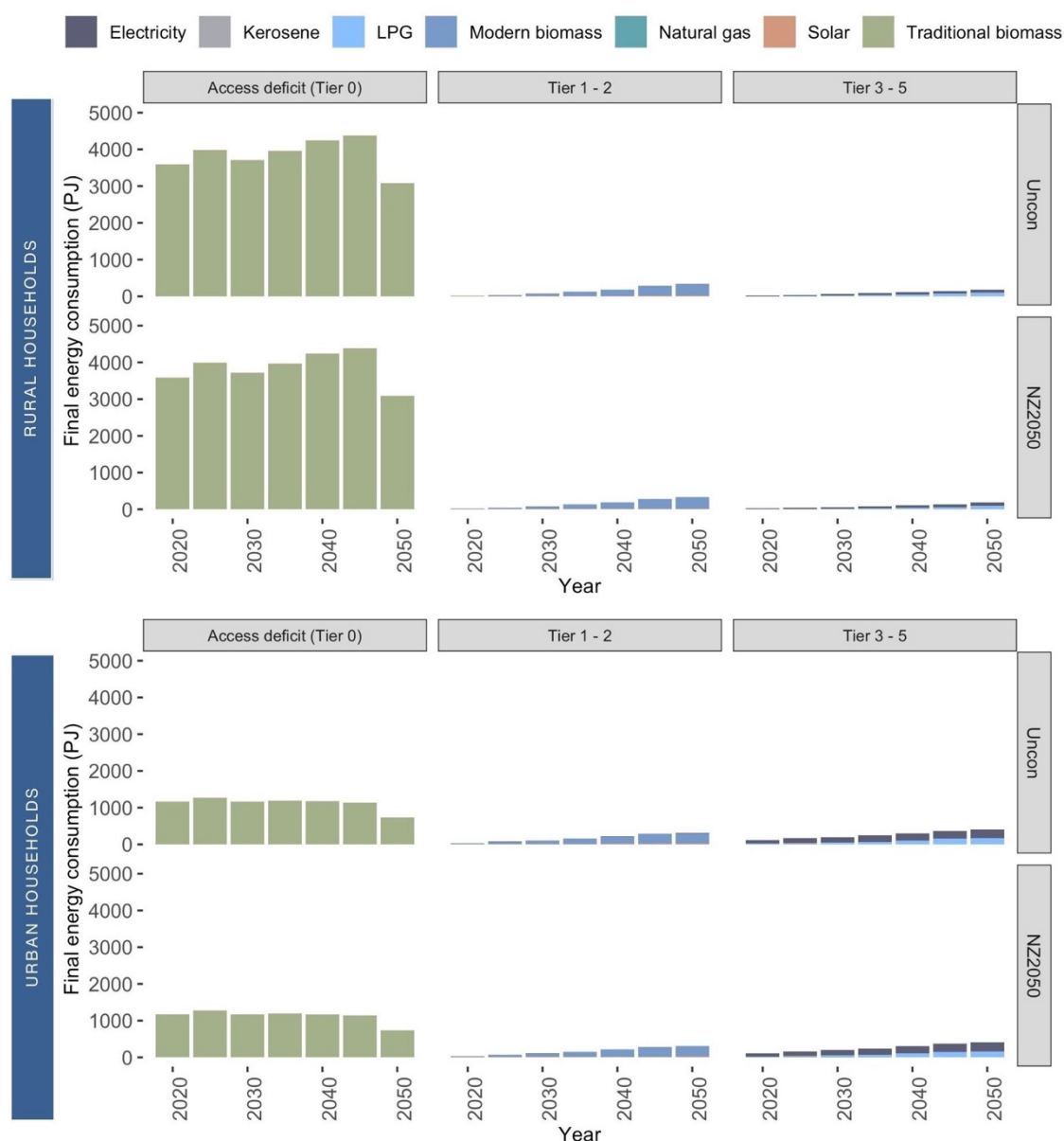


Figure 3.5 Regional final energy consumption in (a) rural and (b) households by fuel in the SSP3 and Regression scenarios

Figure 3.6 compares the regional final energy consumption in the years 2030 and 2050 across the two emissions scenarios, namely, unconstrained (*Uncon*) and net zero CO₂ emissions (*NZ2050*) emissions scenarios, for each of the access scenarios in SSP 3. The bars depicting final energy consumption for each tier are color-coded based on the fuel source. The x-axis represents the two emissions scenarios, while the y-axis shows the annual final energy consumption in petajoules per year.

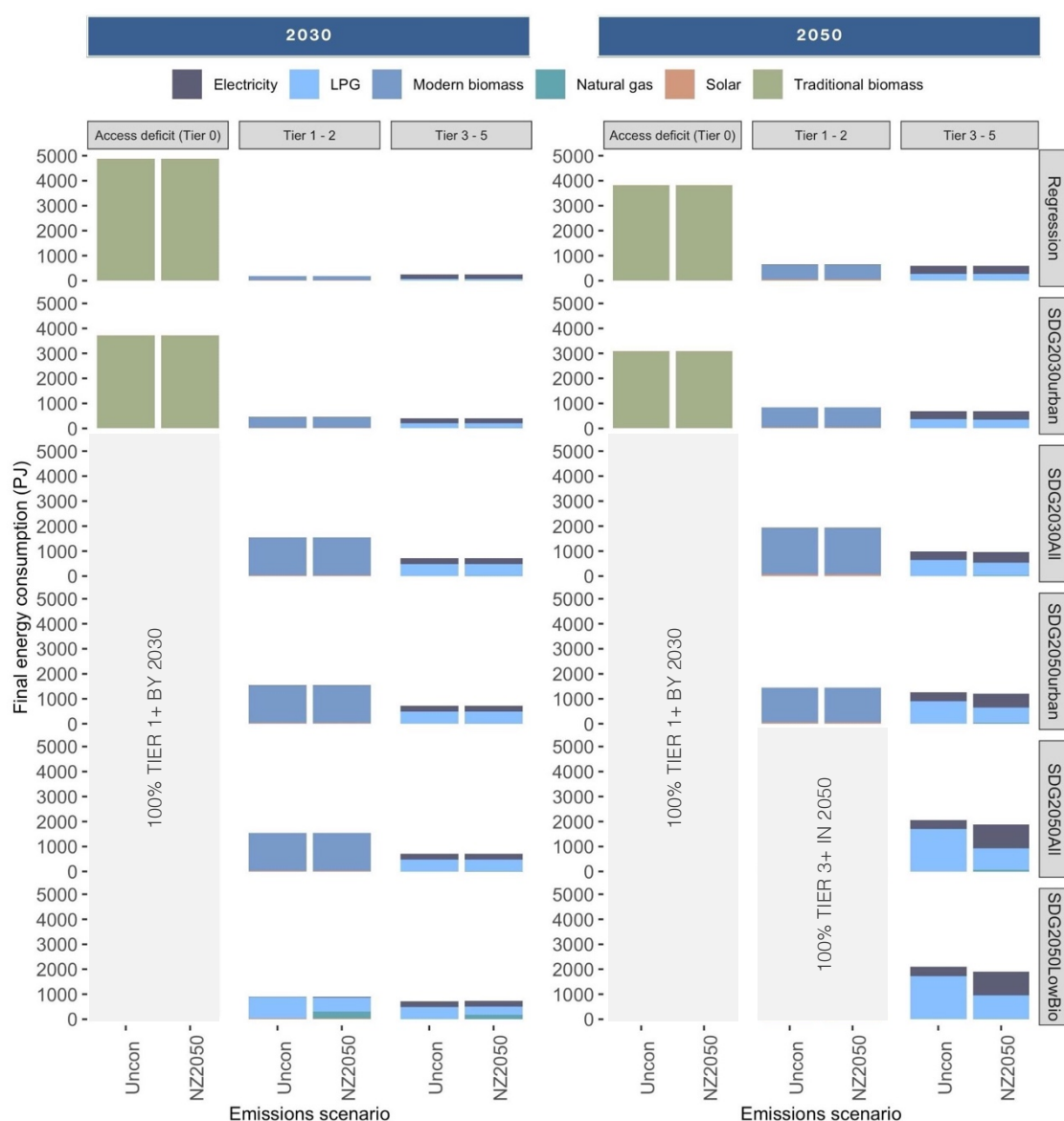


Figure 3.6 Regional final energy consumption in the unconstrained and net zero CO₂ emissions by 2050 emissions scenarios under all the access scenarios in SSP 3

Access deficit (Tier 0). In the *Regression* scenario, rural households have higher share of final household energy consumption compared to urban households (Figure 3.5). This

result is due to the greater share of access deficit in rural areas and their reliance on inefficient technologies. Under the *Uncon* and *NZ2050* emissions scenarios, the access deficit population continues to rely primarily on traditional biomass in all the scenarios (Figure 3.6). The modeling results imply that without penalizing biogenic CO₂ emissions, the population with access deficits will continue to rely on inefficient household. This observation highlights the need for policies that promote access to modern, efficient, and cleaner energy sources, particularly for rural populations. Without such interventions, it is unlikely that energy poverty and its associated negative impacts on health and environment can be addressed effectively.

Minimum access (Tier 1 – 2). Figure 3.6 demonstrates that electric technologies are not the least cost option for households in the minimum access tiers (Tier 1 – 2). Instead, modern biomass serves as the primary source of energy, accounting for 91% of the total final energy consumed for this tier access level under *Regression* and SSP 3 scenarios. In *SDG2030All* scenario, where 100% of households achieve Tier 1+ access by 2030, the share of modern biomass increases to 96% by 2050 in SSP 3, as shown in Figure 3.6. This shift towards modern biomass as a cleaner energy source indicates the need to internalize the cost of externalities arising from biogenic CO₂ emissions, despite the promotion of transitioning from traditional biomass. As we work towards achieving sustainable energy transitions, it is essential that the externalities of the end-use energy sectors, such as biogenic CO₂ emissions from households, are internalized to encourage the adoption of electric technologies.

Higher access level (Tier 3 – 5). Figure 3.6 indicates that for households with higher energy consumption, Tier 3+, LPG is the dominant fuel choice in the *Uncon* emission scenario under the *Regression* scenario. However, under the *SDG2050All* scenario, where 100% of households achieve Tier 3+ access in 2050, LPG is partially replaced by an increasing share of natural gas. The net zero CO₂ emissions, *NZ2050* scenario, assumes the implementation of policies to reduce CO₂ emissions. Under the *NZ2050* scenario in the *SDG2050All* scenario, the increasing share of natural gas is primarily because natural gas has a lower emission factor of 56 kilo-tonne(kt) CO₂/PJ compared to LPG (63 ktCO₂/PJ) (Table 2.4). Moreover, the share of electricity as an alternative fuel increase by 2050, meeting 62% of household energy service demand in *SDG2050All* under the *NZ2050* emissions scenario. In contrast, in the *SDG2050LowBio* scenario, the share of electricity decreases to 50% as biomass with carbon capture and sequestration (BECCS)

displaces CO₂ emissions from LPG. This observation is crucial as it emphasizes the need to promote the adoption of cleaner fuels such as electricity, for households with higher energy consumption to reduce CO₂ emissions. The shift towards natural gas and electricity can facilitate progress towards a low-carbon energy system, which is essential in mitigating the adverse impacts of climate change. This result is dependent on the use of renewable energy and carbon removal technologies.

It is crucial to note that this analysis assumes 100% electricity and clean cooking access for all scenarios, except for *Regression* and *SDG2030urban* scenario, where electricity as an alternative fuel is only available to the population with access. In the following sub-section, we examine the changes in the least cost technology mix for each end-use demand service.

Final household energy consumption by technology choice

This analysis highlights the importance of clarity in defining the level of access countries are trying to achieve to better inform energy policies for target groups. This sub-section examines the population with access to electricity and clean cooking by 2030 and beyond. As the reliance of the access deficit population on traditional cookstoves, solar lighting, and solar water heaters persists across all scenarios, we will not delve further into this tier group. Instead, we concentrate on the population that has access to electricity and clean cooking. We assume 100% Tier 1+ access in *SDG2030All* scenario and 100% Tier 3+ access in the *SDG2050All* and *SDG2050LowBio* scenarios, as described in Table 3.2.

As household demand for end-use energy services increases across all the SSPs, cooking accounts for most of the final energy consumption. Figure 3.7 shows the percentage of final energy consumption by demand service for households with Tier 1+ access in all scenarios under SSP 3 assumptions. The figure illustrates the proportion of energy consumed by different household activities, including cooking, lighting, water heating, and other appliances. The figure highlights the differences in energy consumption patterns across different services and scenarios for each tier group. For instance, cooking is the primary energy-consuming activity in households with Tier 1+ access accounting for over 90% of the total energy consumption in 2050, except in the *SDG2050-* scenarios.

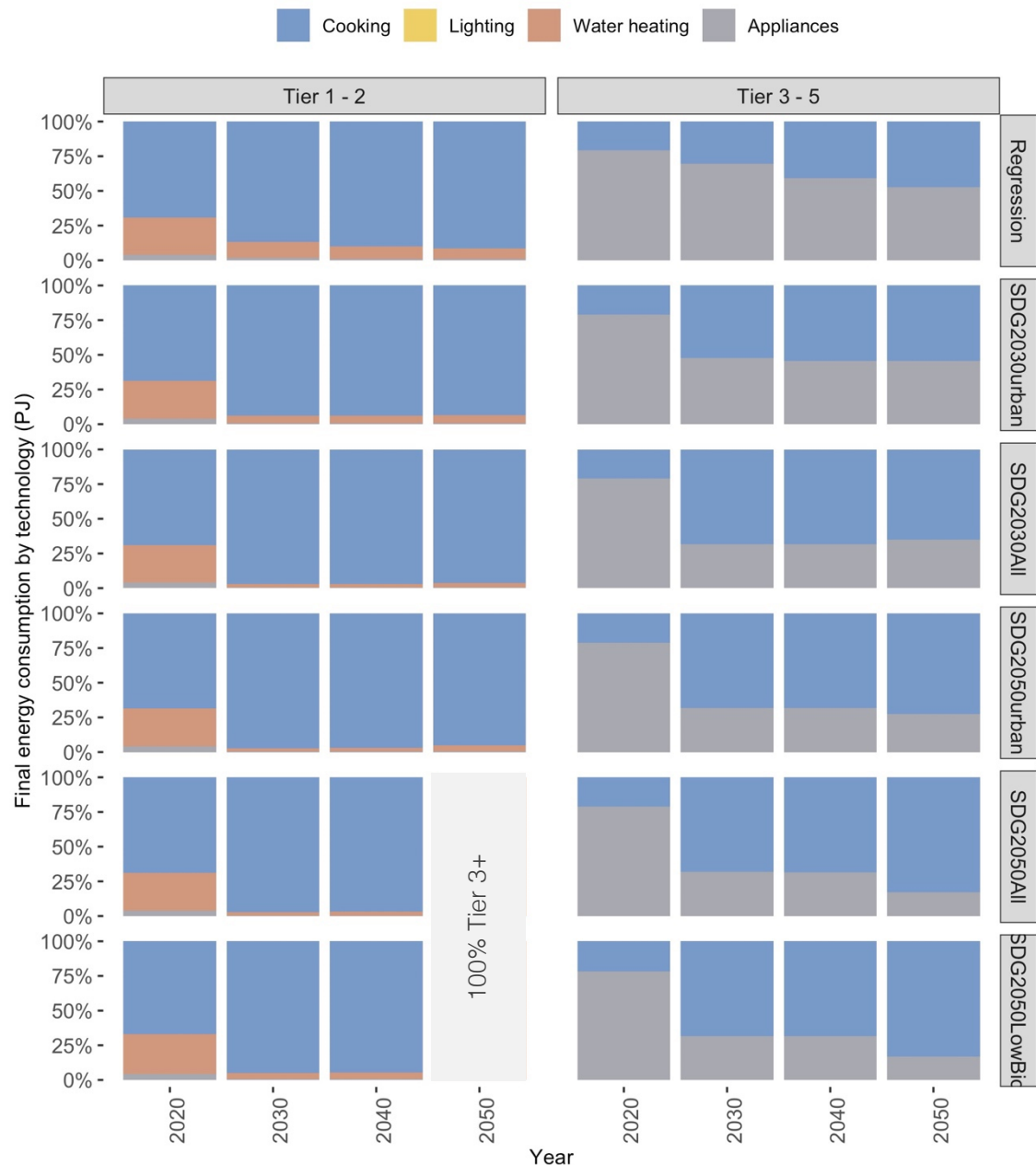


Figure 3.7 Percentage of final energy consumption by demand service in households with Tier 1+ access in all scenarios under SSP 3 assumptions

Figure 3.8 displays the least cost technology mix meeting the final household energy consumption in *SDG2050All* and *SDG2050LowBio* scenarios, where we assume 100% Tier 3+ access in 2050. The figure illustrates the distinctions in energy consumption patterns for households with Tier 1+ energy access under SSP 3 assumptions. By identifying obstacles and opportunities for each access level, policymakers can progress towards low-carbon pathways that ensure equitable access to clean energy while mitigating adverse impacts of climate change.

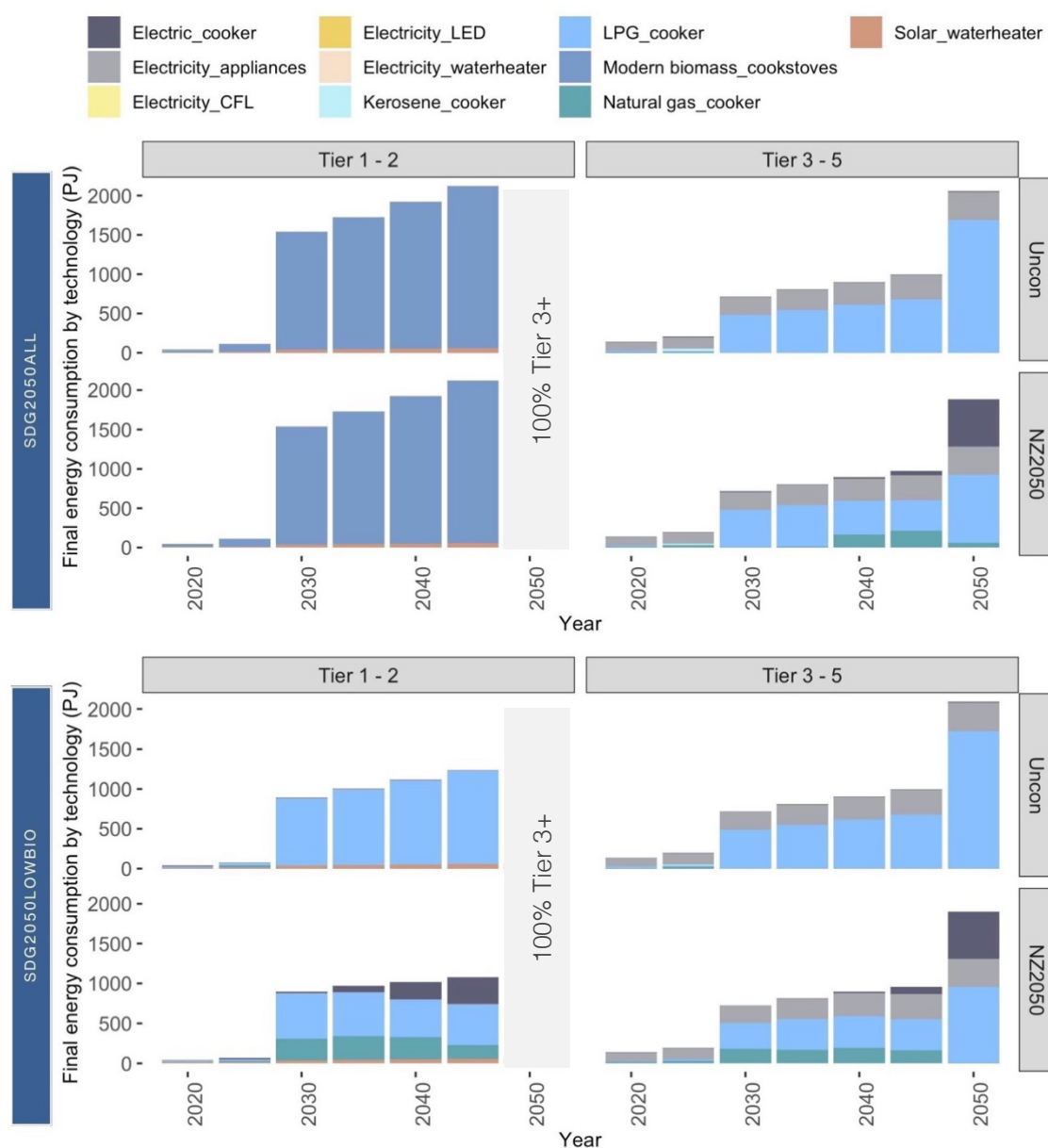


Figure 3.8 Least cost technology mix in SDG2050- scenarios under SSP 3

Cooking

Final energy consumption for cooking in rural and urban households is the largest share of final energy consumption, irrespective of the access tiers. In *SDG2050All*, LPG cookstoves meet 99% of 1,703 PJ of final energy consumption for cooking under *Uncon* emissions scenarios by 2050. In comparison, LPG cookers meet 57% of 1,529 PJ of final energy cooking consumption, while electric cookers meet 39% of final energy demand for cooking in 2050 under SSP 3 assumptions and *NZ2050* scenario. In the most extreme simulation of universal energy access without biomass as a cooking fuel option, *SDG2050LowBio* scenario, final energy consumption increases to 1,739 PJ under *Uncon*

scenario and 1,547 PJ under *NZ2050* scenario compared to *SDG2050All* results. LPG cookstoves meet over 61% (949 PJ) of the total cooking final energy consumption (1,547 PJ) under the *NZ2050* scenario in 2050. Without biomass cookstoves, emission reduction targets under *NZ2050* scenario results in a partial shift from LPG to electricity cookers in 2050.

Although the World Health Organization consider modern biomass and LPG cookstoves as clean cooking solutions, East African countries should consider strategies such as subsidized tariffs for electric cookers to avoid fossil-fuel lock-in in the long-term and align national universal energy access targets with global mitigation targets, even at the household level. The availability of affordability electricity is essential for this purpose, particularly since renewable energy sources will likely generate electricity.

Lighting

In households with access to electricity, all the scenarios exhibit a shift from compact fluorescent light (CFL) bulbs to more efficient light-emitting diode (LED) bulbs (Figure 3.8). This shift is attributed to technology efficiency improvements, longer lifespan, and declining costs, projected in the Temoa East Africa model database. LED bulbs have a longer lifespan and higher efficiency compared to CFL bulbs, making them a more cost-effective lighting option by 2050. Energy-efficient lighting can help reduce energy consumption. However, LED bulbs have higher upfront costs compared to CFL lighting, this suggests the need for policies to subsidize the more efficient lighting technology.

Water heating

In all the scenarios, solar water heaters are the most cost-effective technology to meet the demand for water heating. Since there are no tier level assumptions for water heating, we assign the share of the population with electricity access to the minimum access level (Tier 1 – 2), representing Tier 1+ access. As a result, solar water heaters meet 100% of the final energy consumed for hot water in all households. The technology is suitable for the East African region that boasts of ample solar potential throughout the year (Hafner et al. 2019). Policymakers can encourage the adoption of solar water heaters by offering innovative financing models to reduce up-front costs for households, especially without access to electricity.

Capacity and power generation for household electricity demand

The modeling results suggest that household demand for lighting and appliances in 2050 is met entirely by electricity in households with Tier 1+ access. There is also a partial shift to electric cookers leading to an increase in electricity demand under the *NZ2050* scenario. To further examine the impact of different energy access definition on the least cost fuel mix in the power sector, we analyzed installed capacity and generation mix by fuel for meeting household electricity demand. The results shown in Appendix Figure 7.10 and Figure 7.11. reveal that the capacity and generation mix in the power sector vary by country depending on the level of energy access assumed. The resource potential, capacity factors, and existing capacity inputs vary by country, explaining the national results. The key findings in this section highlight 100% renewable energy generation mix to meet households demand and show decentralized off-grid technologies play a critical role in meeting electricity demand in rural households.

Figure 3.9 displays the regional installed capacity and power generation in 2020, 2030, and 2050 under SSP 3 assumptions. The x-axis represents two emissions scenarios, and the y-axis displays the installed capacity in gigawatts (GW) and power generation in terawatt-hours (TWh), with fuel sources color-coded under each access scenario. The results indicate that in the unconstrained emissions scenarios, centralized solar PV satisfies most of the electricity demand between 2030 and 2050, with capacity additions ranging from 7.9 GW (*NZ2050*) to 9.1 GW (*Uncon*) in the *Regression* scenarios. In the most ambitious *SDG2050-* scenarios, where 100% Tier 3+ access is assumed in 2050, solar PV capacity increases by 6 GW (*NZ2050*) and 9.7 GW (*Uncon*) in the *SDG2050All* scenario between 2030 and 2050 (Figure 3.9). Under the *NZ2050* scenario without biomass as a cooking fuel option, the solar capacity increases by 8.6 GW from centralized solar PV and 4.7 GW in off-grid solar PV between 2030 and 2050. Furthermore, the biomass with carbon capture and sequestration (BECCS) generation share increases in the *NZ2050* scenario from 18 TWh in the *Regression* scenario to 57 TWh in *SDG2050All* and 41 TWh in *SDG2050LowBio* in 2050 under SSP 3 high population growth scenario assumptions (as depicted in Figure 3.9). These findings have important implications for policymakers and renewable energy investors in East Africa, who must prioritize investment in renewable energy sources and phase out fossil fuel sources in order to meet achieve universal energy access while adhering to national and global climate targets.

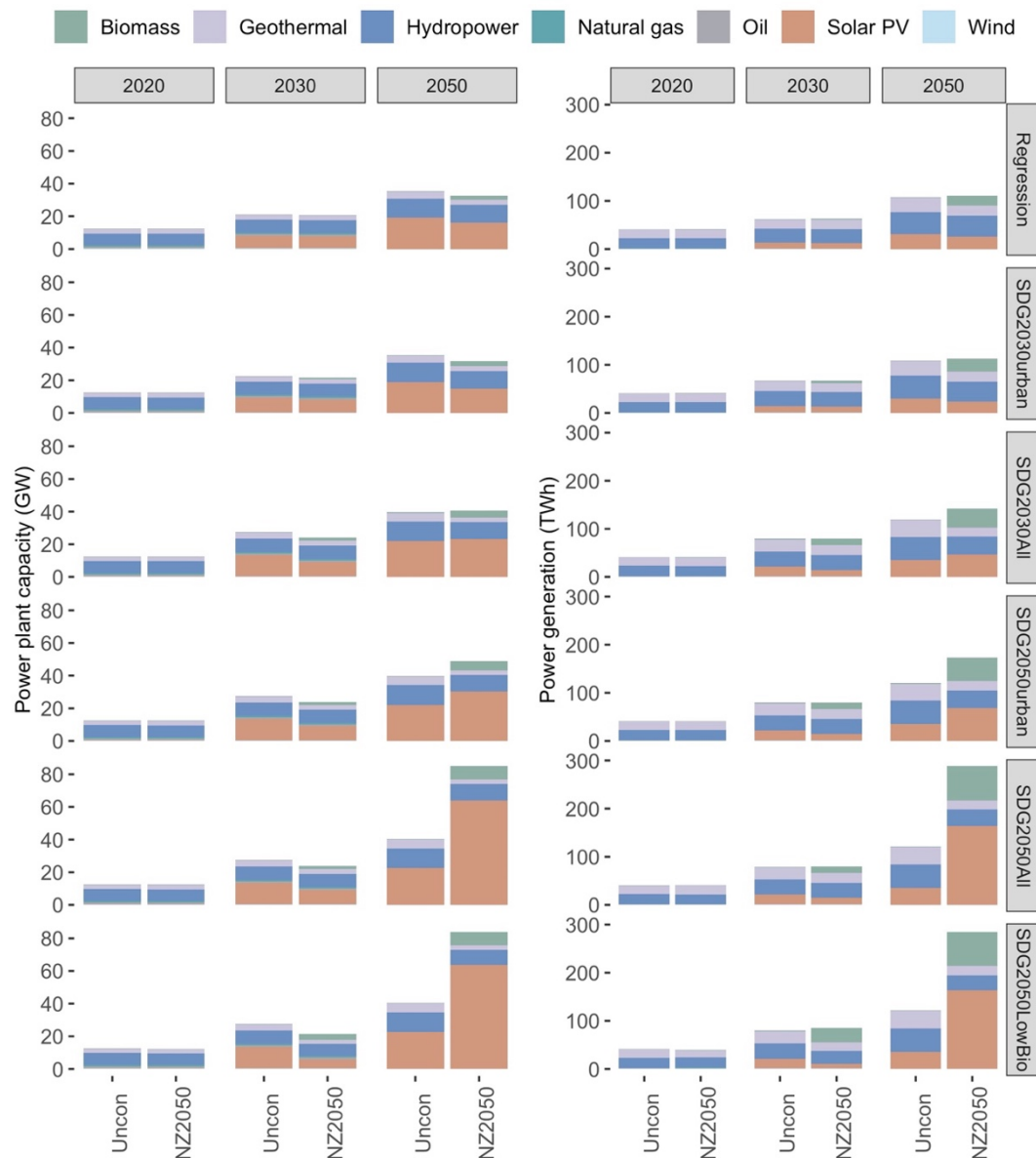


Figure 3.9 Regional installed capacity and power generation in 2020, 2030, and 2050 under SSP 3 assumptions

The results demonstrate that decentralized generation technologies are essential in meeting electricity demand in rural areas, and their share increases with the rising demand for electricity under 100% Tier 3+ access scenarios (*SDG2050-*). Figure 3.10 depicts the distribution of electricity generation technologies in rural and urban households in 2050 under SSP 3 high population growth assumptions, indicating the share of centralized versus decentralized generation technologies. We observe decentralized technologies being used when the electricity demand in rural areas exceeds approximately 25 TWh (Figure 3.10). Decentralized technologies, such as off-grid options and mini-grids, can help overcome the challenges of extending centralized power grids to remote and rural

areas. These technologies can also provide energy access to households in areas where centralized grids may not be feasible or cost-effective. Furthermore, decentralized systems can increase energy security and resilience by reducing reliance on single centralized power sources. This finding is in line with previous studies that have highlighted the importance of decentralized generation technologies, particularly in rural areas (Mentis et al. 2017; Dagnachew et al. 2017). Therefore, the findings reemphasize the need to prioritize decentralized technologies in rural electrification plans and strategies to ensure universal access to electricity.

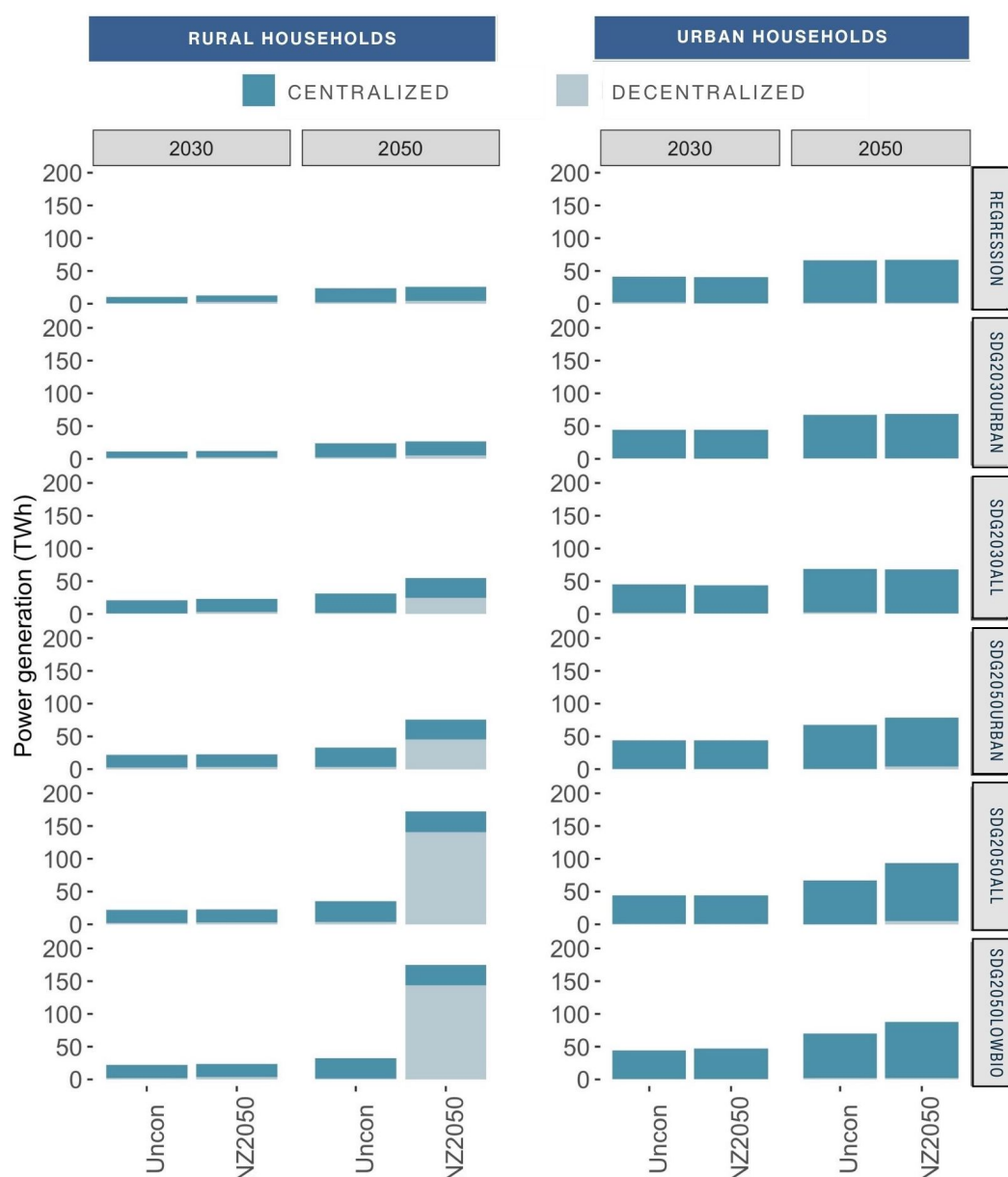


Figure 3.10 Share of centralized versus decentralized technology by power generation in rural and urban households in 2050 under SSP3 assumptions.

Emissions in 2050

This study emphasizes the need for well-defined policies to achieve universal energy access beyond 2030 for both rural and urban households, considering potential shifts in technology choices under different scenario assumptions. This section examines the CO₂ emissions and associated costs under the various universal energy access scenarios.

Under the unconstrained scenario (*Uncon*), LPG is the primary fossil fuel source of CO₂ emissions. In our simulations, the range of CO₂ emissions from fossil fuels in 2050 under the *SDG2050All* scenario is 43 million-tonne (Mt) CO₂ (SSP 1) to 105 Mt CO₂ (SSP 3). In 2022, with global CO₂ emissions totalling 36.8 Gt CO₂; the building sector accounted for roughly 8% of these emission (IEA 2022a). Notably, the six East African countries in our analysis contributed only 44 Mt CO₂ in 2021, representing less than 1% of global CO₂ emissions (Maigné et al. 2021). Despite the region's negligible contribution to global CO₂ emissions, it remains vital to prevent rising emissions in the region.

In the *Regression* scenario, the total cumulative regional CO₂ emissions from fossil fuel and biogenic sources range between 8.5 gigatonnes (Gt) CO₂ and 18.3 CO₂ from both rural and urban households between 2020 and 2050. However, cumulative CO₂ emissions decrease in the *SDG2050All* scenario and further in the *SDG2050LowBio* scenario, highlighting the importance of achieving universal energy access and reducing reliance on biomass as a cooking fuel option. Previous studies have raised concerns about the sustainability of continued biomass reliance (Sterman et al. 2022; Cherubini et al. 2011; Haberl 2013), making it imperative to focus on alternative energy sources.

Figure 3.11 shows the cumulative regional fossil fuel and biogenic CO₂ emissions under SSP 3 by fuel and emission scenario between 2020 and 2050. Notably, the lowest CO₂ emissions occur in the *SDG2050LowBio* scenario, which assumes that biomass is not an option for cooking fuel. Emission reductions are also evident in the *SDG2050All* scenario, where countries achieve SDG target 7.1 with 100% Tier 1+ access by 2030 and 100% Tier 3+ access in 2050. The decline in emissions is due to the increased use of renewable sources for meeting higher consumption households' electricity needs and BECCS displacing emissions from LPG use for cooking.

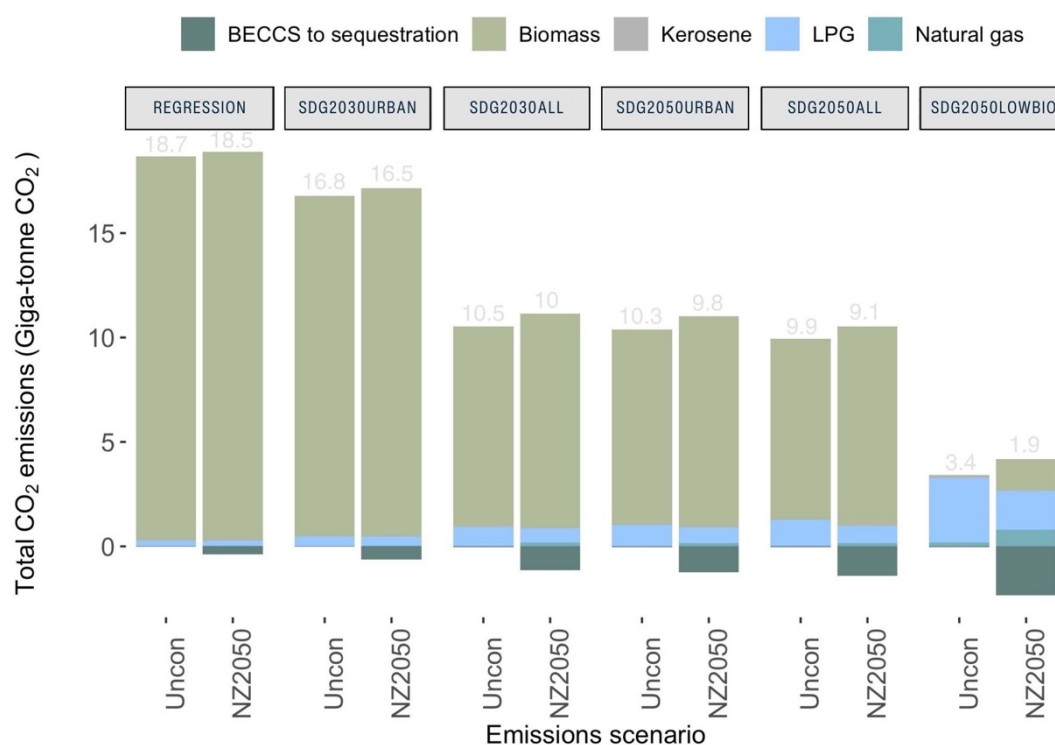


Figure 3.11 Cumulative CO₂ emissions between 2020 and 2050 including biogenic emissions from biomass under linear reduction to net zero by 2050 constraints.

In the unconstrained scenario, the absence of biomass as a cooking fuel choice in the *SDG2050LowBio* scenario reduces cumulative emissions by 67% decrease from 9.9 Gt CO₂ emissions (*SDG2050All*) to 3.4 Gt CO₂ emissions. Between *SDG2050LowBio* and *SDG2050All* replacing 8.6 Gt CO₂ emissions from biogenic emissions, results in an increase of 2.08 Gt CO₂ fossil-based emissions.

Under the *NZ2050* scenario, which assumes that countries align universal energy access goals with climate mitigation targets to avoid overshoot, 1.66 Gt CO₂ of fossil-based emissions replace the 8.6 Gt CO₂ biogenic emissions. While in the unconstrained *Uncon* scenario 1 tCO₂ of fossil-based emissions displaces 4.1 tCO₂ from biogenic emissions, in the *NZ2050* 1 tCO₂ of fossil-based emissions replaces 5.2 tCO₂ biogenic emissions. Less efficient biomass technologies lead to higher emissions when biogenic emissions are unchecked, and continued reliance on biomass increases CO₂ concentrations without certainty regarding regrowth or the time when growth will equate to harvesting (Sterman et al. 2022).

Total costs by 2050

The cost of providing universal access to electricity and clean cooking energy in rural and urban households while meeting climate mitigation goals varies across the study's scenarios.

In the *Regression* scenario, the total cost ranges from \$222 billion (SSP 1) to \$334 billion (SSP 3) between 2020 and 2050. Under the SSP 3 assumptions total costs increase from \$334 billion to \$451 billion in SDG2030All, and \$455 in SDG2050All scenarios. The average annual cost to meet Tier 1+ access by 2030 is \$15 billion (SDG2030All), and in \$15.2 billion (SDG2050All). Current international financing for electricity access totals 8 USD\$ billion in East Africa, which is insufficient to close the access gap (Hafner et al. 2019; Li, Bae, and Rishi 2023; Mbanda et al. 2021). Our results are consistent with research that shows domestic capital alone will not suffice to meet the SDG 7.1 target in addition to supporting climate action in East Africa (Szabó et al. 2016).

According to Figure 3.12 shows the total discounted net present value (NPV) cost to meet household energy demand in SSP 3 between 2020 and 2050 under the six access scenarios. The x-axis represents the emissions scenarios, while the y-axis represents the total cost in billions of 2020 US dollars. The color-coded bars show the breakdown of discounted costs as capital, fixed non-fuel operation and maintenance, and variable fuel costs. The grey figures represent the percentage growth in total costs between the *Regression* and alternative access scenarios, with a lower range in unconstrained (*Uncon*) and an upper range in the linear reduction to net zero (*NZ2050*) emission scenarios.

The percentage increase in the total cost of achieving 100% Tier 1+ access in both rural and urban households (*SDG2030All*) relative to the *Regression* scenario is 35% to 36%, which is significantly higher than the 7.5% percent increase in cost to meet 100% Tier 1+ access in urban households (*SDG2030urban*). This result indicates larger financing gap in achieving universal electricity access in rural areas, where policymakers will need more concentrated efforts to meet 2030 targets. The capital costs, primarily incurred as upfront costs, represent the larger share of total costs between 2020 and 2050 across all the scenarios. This result implies a need for more concentrated effort to invest in rural areas as higher costs to facilitate move away from biomass dependency may disproportionately affect the poorest in rural areas if left unaddressed by targeted policies.

After achieving 100% Tier 1+ access, the incremental cost of providing energy access at 100% Tier 3+ level in 2050 is relatively low compared to meeting minimum access targets

(Tier 1+) in rural and urban households. The difference in incremental costs relative to the Regression access scenario between the *Uncon* and *NZ2050* scenarios increases in the more ambitious scenarios from 1% in the *SDG2030All* to 2% in the *SDG2050All* and 8% in the *SDG2050LowBio* scenarios. Traditional biomass is almost free in the model, so costs will be higher in scenarios where the cheaper fuel option is no longer available. The cost difference between the emissions scenarios highlights the reduced cost of tapping into mitigation potential at the household level as “low hanging fruit” compared to the significant investments and targeted policies required to achieve universal energy access, particularly in rural areas.



Figure 3.12 Total cost to meet household energy demand in SSP3 between 2020 and 2050

Table 3.6 presents the abatement cost results for the *SDG2050*- scenarios where we consider only the fossil-fuel based emissions in the cost of abatement calculation. We observe that the absence of biomass as a cooking fuel choice in the *SDG2050LowBio* scenario increases the cost of abatement relative to the *SDG2050All*. Specifically, the cost of abatement increases from approximately 60 \$/tCO₂ in the *SDG2050All* scenario to 74 \$/tCO₂ in *SDG2050LowBio* scenario under SSP 3 assumptions. However, policymakers

must ensure that policies designed to penalize biogenic emissions and reduce reliance on biomass do not impose a disproportionate cost burden on lower access and rural households. These households may not have the financial resources to adopt cleaner and more efficient technologies without adequate support from policymakers. Therefore, it is essential to develop policies that ensure the transition to sustainable energy sources is equitable and accessible to all, especially the poorest in rural areas who are most vulnerable to the impacts of climate change.

Table 3.6 Cost of abatement between SDG2050- scenarios under 100% Tier 3+ access assumptions in SSP3

	SDG2050All	SDG2050LowBio
Cost of abatement, \$/tCO ₂ (Fossil fuel only)	60 \$/tCO ₂	74 \$/tCO ₂
Cost of abatement = Difference in total costs in Million US\$ / Difference in cumulative emissions in Mt-CO ₂		

Our study provides valuable insights into technology-specific policies that promote efficient water heaters and electric technologies such as LED bulbs and electric cookers to achieve universal energy access targets. It is important to acknowledge the limitations of our study, such as the assumptions made in the scenarios and the uncertainties associated with the data. It is essential to understand the input assumptions in each scenario when interpreting the results with the scope of analysis. Furthermore, our analysis does not account for fuel stacking or the cost of time spent collecting fuel wood due to data limitations on a granular level. Additionally, the modeling assumptions do not reflect any supply limits, reliability issues, and competing forces from higher electricity consumption and hard-to-decarbonize sectors like the transport sector. Our findings are still relevant and provide quantitative evidence to advocate for clearer universal energy access targets beyond 2030. The results can serve as a valuable reference for stakeholders in the development of sustainable energy access policies. Additionally, further research is needed to explore low-carbon pathways for the entire energy system in these countries. Overall, our study contributes to ongoing efforts to address the challenge of sustainable energy access while mitigating the impacts of climate change.

3.6 Conclusion

To improve comprehension of the implications of meeting universal energy access targets, there is a need for clarity on the target population (tier access level), subgroup location (rural or urban), and the time frame (mid- to long-term). This chapter provides clearly demonstrates the importance of refining universal energy access targets for equitable energy transitions in East Africa. The study demonstrates that limited energy access definitions influence analytical results, emphasizing the need for clarity in developing universal energy access targets that account for disparities within countries. Beyond 2030, the study also provides a benchmark for refining SDG 7.1 targets by tier access level to achieve universal access in rural and urban areas.

The study also emphasizes the need for affordable electricity and reducing upfront costs for selected technologies in rural areas to prevent widening inequalities. While climate mitigation targets in the form of carbon dioxide emission constraints, have minimal impact on the technology mix in access deficit populations, there is a trade-off between meeting universal energy access with biomass and fossil fuels, and achieving emission reductions in higher tier access levels with increasing energy consumption levels. The shift from traditional biomass to LPG to meet growing cooking needs signals a need for long term strategies to facilitate electricity use to avoid negative environmental implications in the long term.

One unanticipated result is the low magnitude of fossil fuel emissions, which displace substantial levels of biogenic emissions. The study's results also suggest a need to internalize the externalities of biomass use and induce demand for alternative modern energy options to address increasing biogenic CO₂ emissions. With growing population under the different socioeconomic pathways (SSPs), without intervention, traditional biomass dependency persists with increasing biogenic CO₂ emissions, these findings are consistent with previous studies (Rammelt et al. 2022; Stoner et al. 2021).

The findings also demonstrate that decentralized generation technologies play a crucial role in rural electrification, as the majority of the access deficit population in rural areas. Cost to achieve universal access in rural areas are notably higher than closing the access gap in urban households, and scenarios exploring universal energy access pathways in rural areas point to the need for strategies to reduce upfront costs for selected technologies such as solar PV water heaters, LED bulbs, and more efficient cooking technologies.

Policymakers can encourage consumer behavior towards more efficient technologies by implementing electric cooking tariffs and modern energy subsidies.

The international community must recognize the substantial financing required for East African countries to equitably achieve universal energy in rural and urban East Africa beyond minimum access. The Temoa East Africa model demonstrates the potential for open-source country-level datasets with the representation of diverse groups within countries to improve the relevance of modeling applications in exploring energy transition pathways in Africa.

4 LAND-BASED PASSENGER AND FREIGHT TRANSPORTATION SERVICES DEMAND MODEL¹

According to the United Nations (UN), the global population will continue to grow exponentially in the coming decades (United Nations Department of Economic and Social Affairs 2022b). Therefore, there is a need to consider how to sustainably meet the growing population's demand for goods and services while improving living standards and simultaneously mitigating and avoiding CO₂ emissions. Due to surging economic activity, past population growth has led to increased demand for transportation services, a pattern that is likely to continue in the coming decades. Indeed, the International Transportation Forum (ITF) (2019; 2021a) reports that global passenger demand for

¹A version of this chapter has been published in peer reviewed journal, titled: J. Nkiriki, P. Jaramillo, N. Williams, A. Davis, and D. E. Armanios (2022). Estimating Global Demand for Land-based Transportation Services Using the Shared Socioeconomic Pathways Scenario Framework. *Environmental Research: Infrastructure and Sustainability*, <https://doi.org/10.1088/2634-4505/ac823b>.

transportation will triple by 2050, while demand for freight transportation will increase two-fold in the same time frame. The ITF projects that this growth in transportation demand will primarily occur in developing countries in East Asia, South Asia, West Asia, Latin America, and the Caribbean (ITF 2021a). This increased need for transportation services will likely lead to an increasing demand for energy used in the transportation sector, which could in turn increase global CO₂ emissions. Simultaneously meeting the growing demand for transportation services and reaching a net-zero CO₂ emissions target will require the deployment of mitigation strategies in both developed and developing countries. While there has been significant research about the barriers and opportunities for low-carbon transport systems in developed countries, there is still a need for a more detailed analysis of low-carbon transportation pathways in developing countries (Daggash and Mac Dowell 2021).

The CO₂ emissions from the transportation sector are a function of travel demand, travel mode, transport technology, CO₂ intensity of fuels, and energy efficiency. Thus, a crucial input to the models used for evaluating low-carbon transportation pathways is the future demand for transportation services. Statistical models use historical data to predict the relationship between socioeconomic variables and aggregate demand for services (Ajanovic, Dahl, and Schipper 2012; Tsekeris and Tsekeris 2011). Researchers have used regression models to estimate the demand for residential electricity, industrial energy consumption, and energy use in agriculture (Karkacier, Goktolga, and Cicek 2006; Peña-Guzmán and Rey 2020; Wolde-Rufael 2004). Statistical models have also been used to estimate service and energy demand in the transportation sector (Ajanovic, Dahl, and Schipper 2012; Daldoul, Jarboui, and Dakhlaoui 2016; Limanond, Jomnonkwao, and Srikaew 2011; Singh 2006; Varjan, Rovňaníková, and Gnáp 2017). For developing countries, the sporadic nature of transport data and lack of statistics across countries constitute a barrier to estimating future energy demand for the transportation sector (Ajanovic, Dahl, and Schipper 2012). Using data on transportation-related energy consumption in developed countries to estimate energy use in the transportation sector in developing countries creates internal validity concerns. For example, the efficiency of vehicle fleets in developing countries is likely different from that in developed countries, which affects the validity of the energy demand estimated using data from developed countries. Thus, Ajanovic *et al.* (2012) suggested an alternative approach that models demand for transport services (in passenger-kilometers and tonne-kilometers) instead of the vehicle and fuel demand.

4.1 Literature review: Transport demand models

In selecting the variables for the transportation model, we consider two factors: elasticity of transport demand and data availability. The income elasticity of demand represents the relationship between changes in income and demand for given services, whereas the price elasticity of demand describes the relationship between prices and demand. Empirically, in a study spanning over 100 years, Fouquet (2012) reported income elasticities of transport demand between 0.8 and 3.1. In comparison, Fouquet (2012) estimates that price elasticities of transport demand are negative but have a lower magnitude ranging between -1.5 and -0.6 across the study period. In line with these findings, Dunkerley *et al.* (2014) suggest that income elasticities are more substantial than price elasticities when evaluating long-term trends in demand for transport services. Similarly, Dunkerley *et al.* (2014) suggest higher and positive income elasticities of freight demand (Dunkerley, Rohr, and Daly 2014). Specifically, Dunkerley *et al.* (2014) find income elasticity ranges between 0.5 and 1.4 for passenger transport and between 0.5 and 1.5 for freight transport. Van Dender and Clever (2013) argue that the price elasticity of transport demand does not sufficiently influence to be included in transport vehicle demand forecasting. According to the literature, income elasticity is not only positively correlated to demand but is also 1.5 to 3 times higher than price elasticity (Litman 2013).

4.2 Research goals and modeling framework

In this study, we rely on a two-stage process to develop scenarios for land-based passenger and freight transport demand worldwide up to 2050. Figure 4.1 describes the modeling approach for combining a double log model with a scenario-based framework. At the baseline, we trained the log-log model on historical land-based transportation consumption trends. We then added the SSP projections to fit the land-based passenger and freight transport demand from 2020 to 2050 at the global, regional, and national levels.

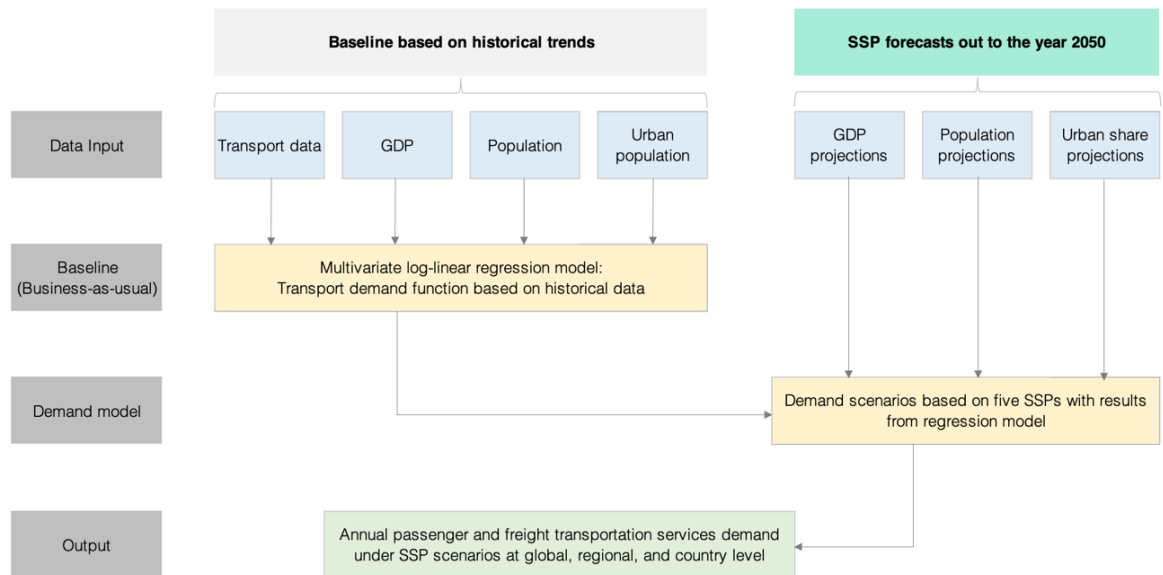


Figure 4.1 Summary of country-level log-log model for estimating transportation demand. The grey side boxes show title labels, and the blue boxes represent inputs to the model while the green is the output. The yellow boxes are representative of the model.

4.3 Regression model

This study builds on prior work by presenting a regression-based model for estimating country-level passenger and freight transport demand. The log-log linear model relies on publicly available historical land-based transportation demand and socioeconomic data. We then use the model to develop future scenario-based demand projections for transportation services in 179 countries. We rely on socio-economic data from the Shared Socioeconomic Pathways (SSPs) previously developed by the international community to support climate-related research, ensuring that we develop internally consistent demand scenarios that other researchers can use to evaluate low-carbon transportation futures.

In this study, we aim to estimate possible scenarios for transport demand in countries around the world. Given the unavailability of long-term, region-specific prices for transport services and the higher values of income elasticity of demand, we rely on income data to build a regression-based model. Note that this study does not aim to identify the causal relationship between the dependent and independent variables. Instead, we aim to develop a model that accounts for the correlation among the variables to create future demand scenarios. The future demand scenarios are not predictions of the future but instead provide a range of possible futures. Specifically, we rely on the gross domestic

product (GDP) as a proxy for income (Cuaresma 2017; The World Bank Group 2022). The model also included total population as an explanatory variable. Dunkerley *et al.* (2014) also suggest that urbanization rates affect demand for transport services, so we include an urbanization variable in the model. The choice to use population and urbanization as model variables was also a result of this cross-validation to evaluate the log-log model performance that we trained on historical data. The cross-validation exercise, shown in more detail in the Appendix, confirms that GDP, population, and urbanization improve model performance compared to the GDP per capita measure.

We relied on the World Bank's historical income and population data for 217 countries and territories from the World Development Indicators dataset (The World Bank Group 2022). We use country-level GDP, population, and urban population data, where the absolute urban population is a product of the urban share and total population. We calculated population density by dividing the total population by land area. We use GDP and GDP per capita (purchasing power parity (PPP) data from the World Bank databank in constant 2017 USD. We also collect demand data for the land-based passenger (in passenger-kilometers (km)) for 38 countries and freight transport (in tonne-km) for 43 countries between 1990 and 2018 from the Transport Outlook of the International Transport Forum (ITF 2019; 2021b). These demand data for passenger and freight transport are an aggregate for each country across all modes (cars, buses, trucks, rail, etc.). Thus, the model in this paper doesn't disaggregate transportation demand by mode. Furthermore, an implicit assumption in this study is that historical correlations between population, GDP, and transportation demand for sample countries represent the future relationship between the variables in all countries. A wide range of economic development levels is represented across the historic panel data which includes 911 observations for passenger transport and 1,351 data points for freight transport. Across the countries and years in the historical dataset, many of the countries transitioned through different income groups; therefore, the data points covered a range of broader economic development levels delivering a reliable benchmark to develop consistent scenarios for global analysis.

This analysis used a log-log linear regression model. This statistical analysis technique uses multiple regression analysis to explore the relationship between the log-transformed dependent and independent variables (Limanond, Jomnonkwao, and Srikaew 2011). The coefficients of a log-transformed data can be interpreted as their elasticity. These models

are common in transportation modeling (Wadud 2011). For example, Souche (2010) used a log-log linear model to estimate daily car and transport trips in urban areas, noting the logarithmic model reduces the risk of heteroskedasticity. Recently, log-transformed models have also been used to estimate demand for freight and passenger air transport (Carmona-Benítez, Nieto, and Miranda 2017; Tabassum and Khan 2020). Finally, log-transformed models have been used to estimate demand for public transit (Alam, Nixon, and Zhang 2018), micro-mobility (Lee, Park, and Kilic 2021) and active transport (Singhvi et al. 2015).

Based on the exploratory analysis, the independent and dependent variables have skewed distributions, therefore, we log-transformed the variables and addressed outliers while checking for any zero values. Equation (7) describes the general form of the log-log linear model, which was trained on historical transport demand data. The independent variables in Equation (7) are GDP in 2020 US dollars (US\$), population (POP), and urbanization (URB) as a share of the population in urban areas. We used the United States Bureau of Labor Statistics Consumer Price Index (CPI) conversion rates to compute the 2020 US\$ equivalent for all economic data (U.S. Bureau of Labor Statistics 2021). We estimate the demand for passenger and freight services in passenger-km traveled (pkm) and tonne-km (tkm) traveled, respectively. β_n is the estimated corresponding coefficient and ε is the error term.

$$\ln(pkm \text{ or } tkm) = \beta_0 + \beta_1 \ln(GDP) + \beta_2 \ln(POP) + \beta_3 \ln(URB) + \varepsilon \quad (7)$$

The regression in Equation (7) shows a statistically significant relationship between population, GDP, and transportation demand. The adjusted coefficient of determination (R^2) measures how well the regression line fits the data and how well the predictions are correlate with the actual values. The results in Table 4.1 indicate that the log-log linear model fits the data well.

Based on the coefficients, land-based passenger and freight transport demands are positively correlated with income and population levels. The scatterplot for passenger travel against urbanization, as well as freight transport demand against urbanization, show a strong positive correlation (Figure 4.2). Urbanization is positively related to demand in Figure 4.2, which could indicate an increase in short trips. However, urbanization also suggests a dense population for which less travel may be required to perform economic activities (Dunkerley, Rohr, and Daly 2014). The sign change for urbanization

coefficients in the regression may be attributed to the severe correlation between population and urbanization, resulting in a suppressor variable effect (Ludlow and Klein 2014). Therefore, taking into account income and population, a one percent increase in urbanization results in reductions of 2.3% in passenger-km and 0.04% in freight tonne-km for land-based transport demand.

It is worth noting, however, that multicollinearity poses challenges to the interpretation of the coefficients. The correlation matrix in shows the correlation coefficients among the independent and dependent variables which were indeed high. Kutner et al. (2004) suggests that the need to correct the model to address multicollinearity depends on the purpose of the study noting that *“the fact that some or all predictor variables are correlated among themselves does not, in general, inhibit our ability to obtain a good fit nor does it tend to affect inferences about mean responses or predictions of new observations.”* The purpose of this study is not to evaluate the causal relationship between the independent and dependent variables. Instead, the goal is to develop scenario-based estimates of future transportation demand consistent with the narratives of the Shared Socioeconomic Pathways so they can be used in future transportation, energy, and climate systems research.

Table 4.1 Model fit results in comparison with selected model highlighted yellow. The asterisks relate to the statistical significance (*p<0.05, **p<0.01, *p<0.001), while the bracketed ranges represent 95% confidence intervals clustered at the country level.**

(a) Passenger	Model 1	Model 2	Model 3	Model 4
Intercept, β_0	-3.69*** (-4.45 – -2.93)	0.31 (-0.34 – 0.95)	-0.45 (-1.18 – 0.27)	7.52*** (6.85 – 8.19)
log(GDP), β_1	1.06*** (1.00 – 1.12)	0.68*** (0.64 – 0.72)	0.97*** (0.94 – 0.99)	
log(POP), β_2	2.29*** (2.05 – 2.53)	0.41*** (0.36 – 0.46)		1.07*** (1.03 – 1.11)
log(URB), β_3	-2.30*** (-2.59 – -2.01)			
Observations	911	911	911	929
Countries	40	40	40	40
Years	28	28	28	28
Adjusted R ²	0.90	0.88	0.84	0.75

(b) Freight	Model 1	Model 2	Model 3	Model 4
Intercept, β_0	-0.01 (-0.70 – 0.68)	0.07 (-0.48 – 0.62)	-0.86** (-1.48 – -0.24)	7.62*** (7.09 – 8.15)
$\log(\text{GDP}), \beta_1$	0.66*** (0.61 – 0.71)	0.65*** (0.62 – 0.69)	0.95*** (0.92 – 0.97)	
$\log(\text{POP}), \beta_2$	0.45*** (0.25 – 0.66)	0.42*** (0.38 – 0.46)		1.00*** (0.97 – 1.03)
$\log(\text{URB}), \beta_3$	-0.04*** (-0.28 – 0.20)			
Observations	1351	1351	1351	1390
Countries	50	50	50	50
Years	28	28	28	28
Adjusted R ²	0.87	0.86	0.82	0.73

Additionally, we explore the model performance by examining the residuals of the predicted values against the actual historical data in the models by varying the variables included (as shown in Figure 4.2). The model underestimates the demand for outliers, such as India, with a high income and population at the country level. Meanwhile, in the historical data, Montenegro has the lowest income and population across all years, and the model overestimates the demand in similar countries such as Albania and Armenia. These observations speak to the limitations of the linear regression model and global averages, which cannot accurately capture patterns for countries that deviate from the historical demand trends. Using a broad scenario analysis framework helps to address reduce some of the uncertainties in forecasting demand using the historical dataset.

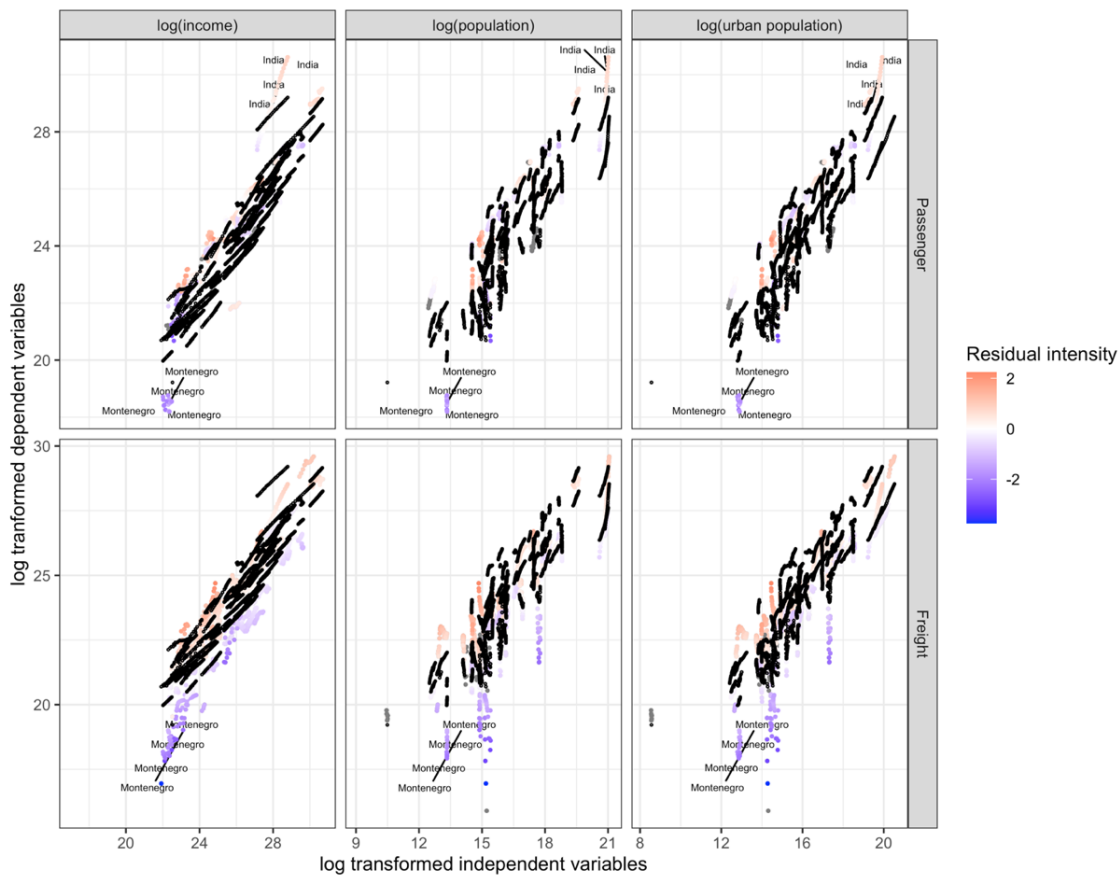


Figure 4.2 Log-linear model performance by the predictor. The residual scatter plots above show the dependent variables against the regressors in each facet (income, population, and urban population). The predicted values are shaded black, while the continuous intensity scale from blue to red shows actual data (colored) values. We use a blue, white, and orange color range to represent under-, exact, and over-estimates.

Cross-validation is a statistical method used to evaluate model performance by dividing data into training and testing sub-sets, with the latter being used to validate the model (Refaeilzadeh, Tang, and Lui 2009). Cross-validation can be done in a couple of ways: 1) leave observations out randomly, 2) leave whole countries out randomly, and 3) leave future time periods out. We use 5-fold cross-validation, grouping the panel data by country guided by the ‘groupdata2’ R package (Olsen 2021). The method is used to estimate the generalization performance (ability to handle out-of-sample data) and compare the performance of two or more models to support selection. The log-log model we select in this analysis, has the lowest average errors for passenger travel, and this study concluded it is the best predictor of out-of-sample data compared to the other models.

4.4 Demand scenarios

Using the demand function for transportation described in equation (7), we developed five scenarios for future transportation demand in countries around the world. For this purpose, we rely on data from the Shared Socioeconomic Pathways (SSPs) database maintained by the International Institute for Applied System Analysis (IIASA). These SSPs characterize future socioeconomic scenarios that would influence greenhouse gas emissions and drive climate impacts (Bauer et al. 2017; Ebi et al. 2014; IPCC 2021; Riahi et al. 2017). Researchers have used these SSP scenarios to explore challenges and response options to climate change (Bauer et al. 2017; Chen et al. 2020; Dellink et al. 2017; O'Neill et al. 2017; Rao et al. 2019).

The SSP storylines are not predictions of the future but exploratory narratives of plausible 'what ifs,' though they cannot cover every possible scenario. Table 4.2 presents the characteristics of the five SSPs. The IIASA provides quantitative GDP, population, and urbanization data for each SSP for 232 countries and territories via the SSP database (IIASA, 2018). Figure 4.3 shows the global estimates of the primary key drivers used in this regression model until 2050 for each SSP.

Table 4.2 Summary of SSP storylines and assumptions for quantification of the socioeconomic projections with ratios provided by IIASA (Cuaresma 2017; Jiang and O'Neill 2017; KC and Lutz 2017; Riahi et al. 2017).

Scenario	Name (Description)	GDP [Ratio 2050:2010]	Population [Ratio 2050:2010]	Urban share [Ratio 2050:2010]
SSP1	Sustainable – Taking the Green Road (Inclusive and sustainable pathway)	High economic growth [4.3]	Low population growth [1.2]	High urbanization [1.9]
SSP2	Middle-of-the-road (Path follows historical patterns)	Uneven income growth [3.4]	Moderate population growth [1.3]	Moderate urbanization [1.8]
SSP3	Regional rivalry – A Rocky Road (Low international priority and regional cooperation)	Slow income growth [2.7]	High population growth in developing countries [1.5]	Low urbanization [1.6]
SSP4	Inequality – A Road Divided (Increasing regional inequalities within and across countries)	Unequal GDP growth [3.3]	Moderate population growth [1.3]	High urbanization [2.0]
SSP5	Fossil-fueled development	High economic growth [5.4]	Global population peaks [1.3]	High urbanization [1.9]

(Rapid global technological
progress and heavy fossil-
fuel deployment)

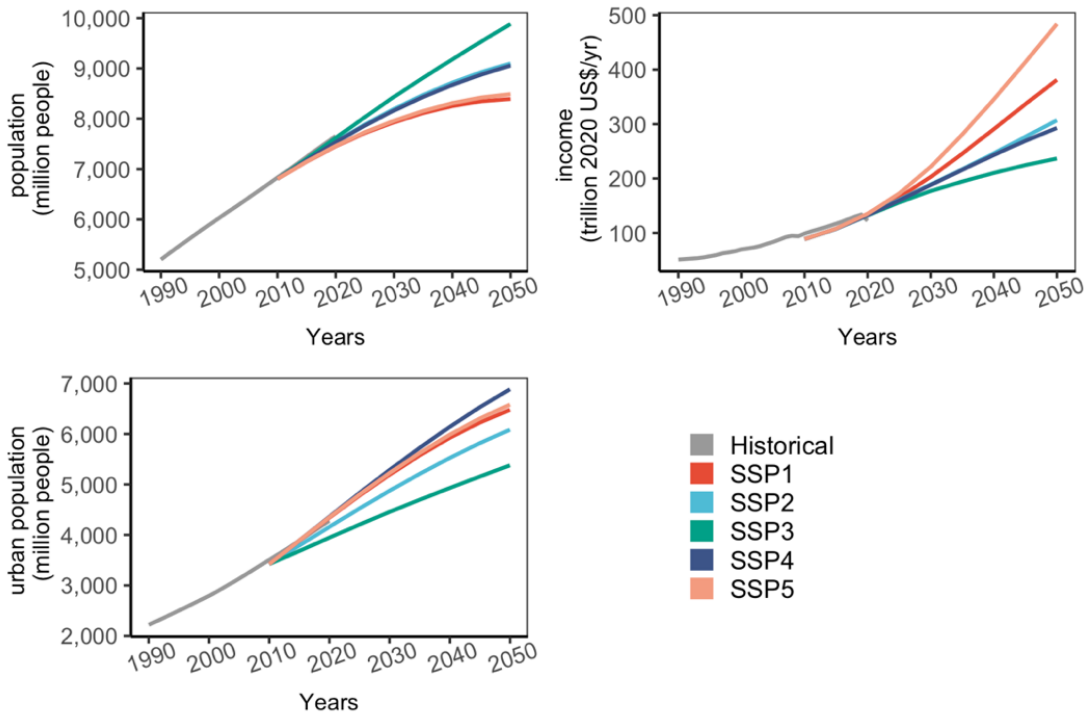


Figure 4.3 Global historical trends (grey: World Bank data) and SSP projections (colored: IIASA dataset) of transport demand drivers. The historical income trend shows degrowth in the years corresponding to the great recession (~2008) and the COVID-19 pandemic (~2020).

In addition to reporting absolute values for the demand for passenger and freight transport in each SSP, we estimated passenger and freight demand growth rates across each country and scenario for these demands between 2020 and 2050 using the nonparametric smoothing spline regression from the R “growthrates” package in equation (8) (Petzoldt 2020). β_0 is the estimated coefficient corresponding to the intercept, β_1 is the coefficient of the exponential time component on demand, K is the demand saturation level to ensure that forecasts are within the plausible range of the forecasted population, ε represents the error term, and t is the time in years from 2020 to 2050 in 5-year increments.

$$\text{annual growth rate} = \frac{K \times \beta_0}{\beta_0 + (K - \beta_0)e^{\beta_1 t}} + \varepsilon \quad (8)$$

4.5 Results and Discussion

Global demand for land-based passenger freight transportation services

Using the log-log linear model and SSP projections, we projected the demand for land-based passenger and freight services over a 30-year time horizon from 2020 to 2050 under different socioeconomic scenarios. These results are *not* predictions of the future but exploratory narratives of possible futures, and throughout the rest of the paper, we refer to them as scenario-based projections. Figure 4.4 shows the scenario-based projections of global demand for land-based passenger and freight transport. We also compared the results with the existing global projections from the ITF to validate the model. ITF (2019) uses more complex independent variables, which makes for a useful validation with the representative model. The global results were within the range of the ITF model shown in Appendix. The results go a step further to provide more granular data on a country-level data for 179 countries.

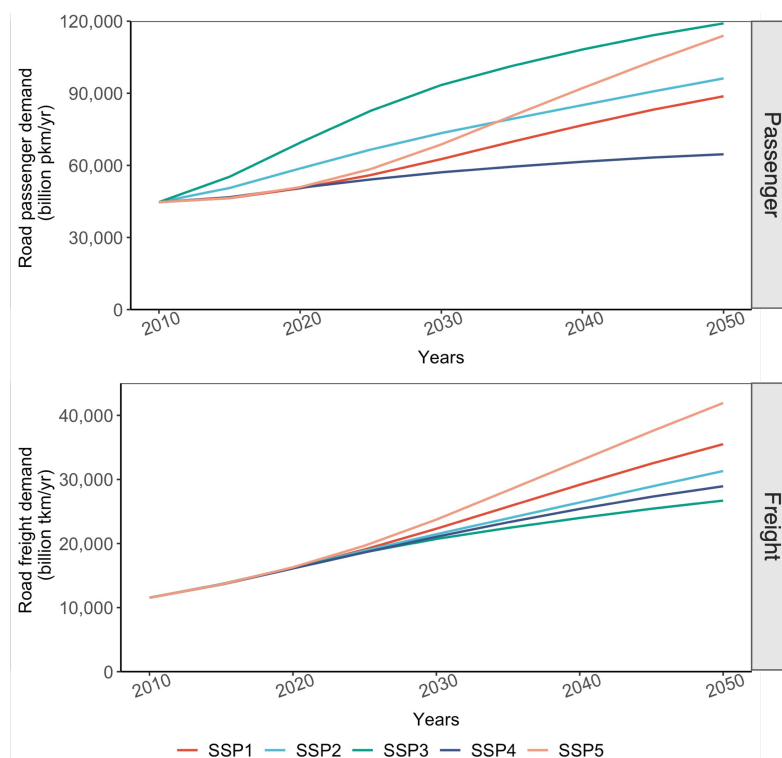


Figure 4.4 Passenger and freight demand projections by SSP scenarios. These results for the annual land-based passenger demand in billion pkm and land-based freight transport in billion tkm on a global scale aggregated across each of the SSP scenarios.

The results show that the global land-based passenger services demand will increase by a ratio of between 1.2 and 2.2 across all the SSPs, while global land-based freight transport demand will increase by a factor of between 1.6 and 2.6 over the 30-year period (Figure 4.4). The reference demand in 2020 across all scenarios ranges from 50,500 billion passenger-km (pkm) in SSP1 to 69,400 billion pkm in SSP3, and between 16,100 billion tonne-km (tkm) in SSP4 and 16,300 billion tkm in SSP5. Across the scenarios, the absolute passenger mobility by 2050 ranges between 64,600 billion pkm (SSP4) and 119,000 billion pkm (SSP3). The global demand for land-based freight transportation is expected to double from 26,700 billion tkm (SSP3) to 41,900 billion tkm by 2050 (SSP5). The global annual growth rate of land-based passenger demand scenario-based projections ranges between 1.3% (in SSP4) and 4.1% (in SSP3), whereas the annual growth rate for land-based freight demand ranges between 3.1% (in SSP3) and 3.6% (in SSP5) over the 30 years.

The global land-based passenger demand is the highest in SSP3, with the highest population growth across all scenarios. Land-based freight transport is the highest in the SSP5 scenario, with the highest economic activity growth. The rapid growth in passenger transport demand in SSP5 after 2030 (Figure 4.4) may be attributed to the sharp growth in income around 2030 in SSP income projections.

Demand for land-based passenger and freight transportation services by region and income group

Comparing income group and region classifications, as described in Appendix, the results suggest that lower and upper-middle-income countries and countries in Asia will have the highest global demand for land-based transportation services by 2050. Figure 4.5 plots the results across each SSP and shows the transport demand by region and income groups. We used the income levels and regional classifications from the World Bank (Serajuddin and Hamadeh 2020; The World Bank 2021). The results suggest that low-income countries, particularly in sub-Saharan Africa, have the lowest absolute values of total demand for land-based transportation services across the period of analysis, whereas Asian and middle-income countries have the highest overall demand for land-based transportation services by 2050. Accounting for 24% of the global population and 34% of global GDP, the East Asia and Pacific region passenger demand projections are approximately 36% of the global passenger transport service demand. The freight

transport demand of the region ranges from 8,600 billion tkm (SSP3) to 14,000 billion tkm (SSP5) by 2050, accounting for approximately 33% of the global freight transport share.

Figure 4.6 shows country-level estimates of land-based transport demand. The detailed data underlying this figure are available via a Zenodo repository. Figure 4.6 highlights that India, China, the United States, and Indonesia will remain the countries with the highest absolute demand for land-based passenger and freight transportation. However, the share of global land-based transport demand start shifting after 2030, as shown in Figure 4.7. This figure shows that after 2030, the share of global demand from East Asia, North America, Europe, and Central Asia starts decreasing while South Asia and sub-Saharan Africa account for an increasing share of global land-based transport demand. By 2050, the demand in South Asia and sub-Saharan Africa is projected to reach near parity with demand from East Asia. Figure 4.7 also shows that middle-income countries account for between 50% and 70% of the global land-based passenger and freight transportation demand across all the scenarios and all periods.

Demand growth rates

The scenario-based analysis shows that low-income countries and countries in sub-Saharan Africa could experience the highest percentage changes in transportation demand over the 30 years. Figure 4.8 quantifies the annual growth rate of the land-based transportation demand between 2020 and 2050 by country. The rapid growth in the demand for transportation services could translate into a rise in transportation-related demand for energy and greenhouse gas emissions in these countries unless cost-effective mitigation options are deployed. These trends suggest that while countries with higher initial baselines will continue to see high absolute demand, developing countries currently experiencing rapid population growth and economic activity are likely to experience accelerated growth.

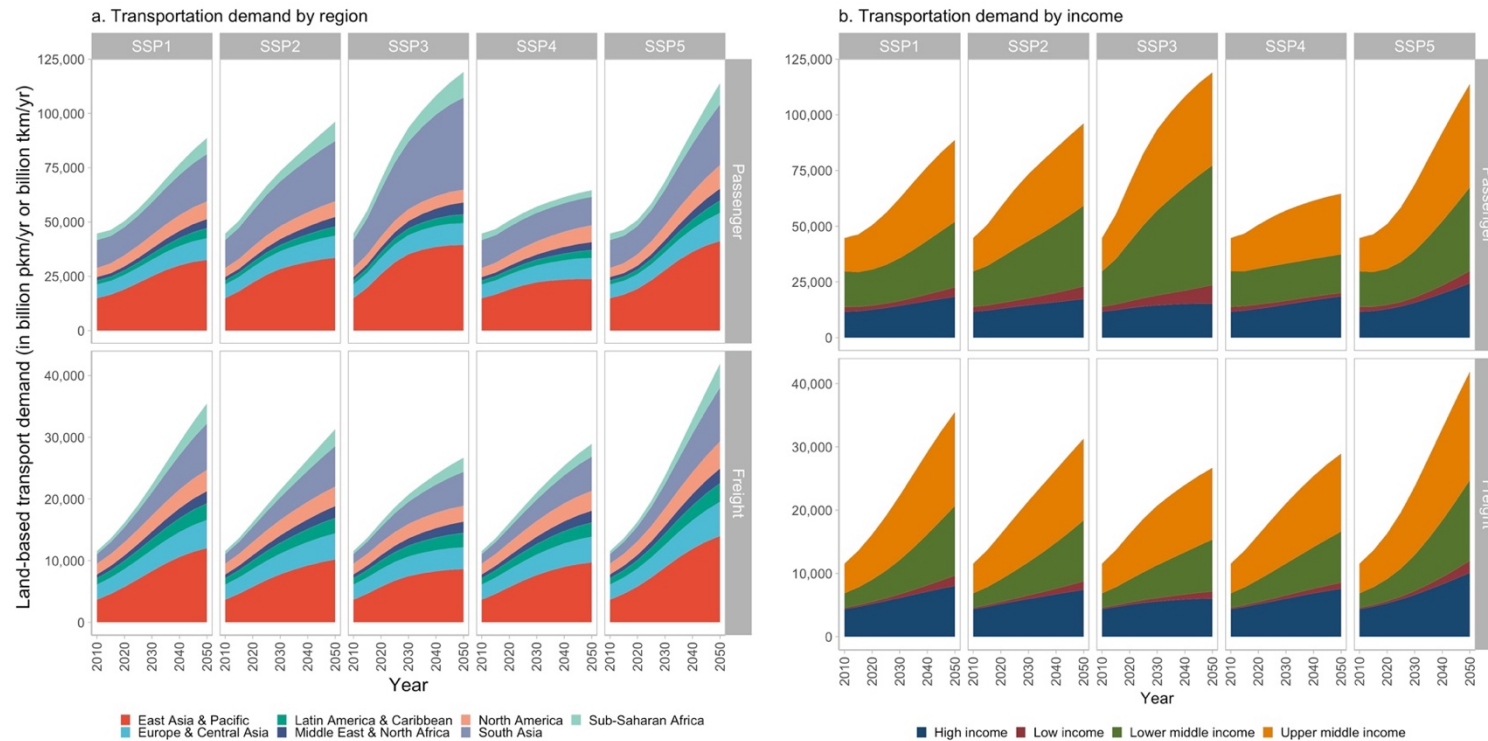


Figure 4.5 Scenario-based projections of land-based transportation demand by (a) region and (b) income group. The panels in each plot represent the land-based passenger scenario-based projections on the right and the land-based freight demand on the left, with the results grouped and faceted by scenario. Panel (a) shows that South Asia and East Asia & Pacific regions will have the highest absolute demand for land-based transport by 2050. In panel (b), we observe the most considerable demand in middle-income countries.

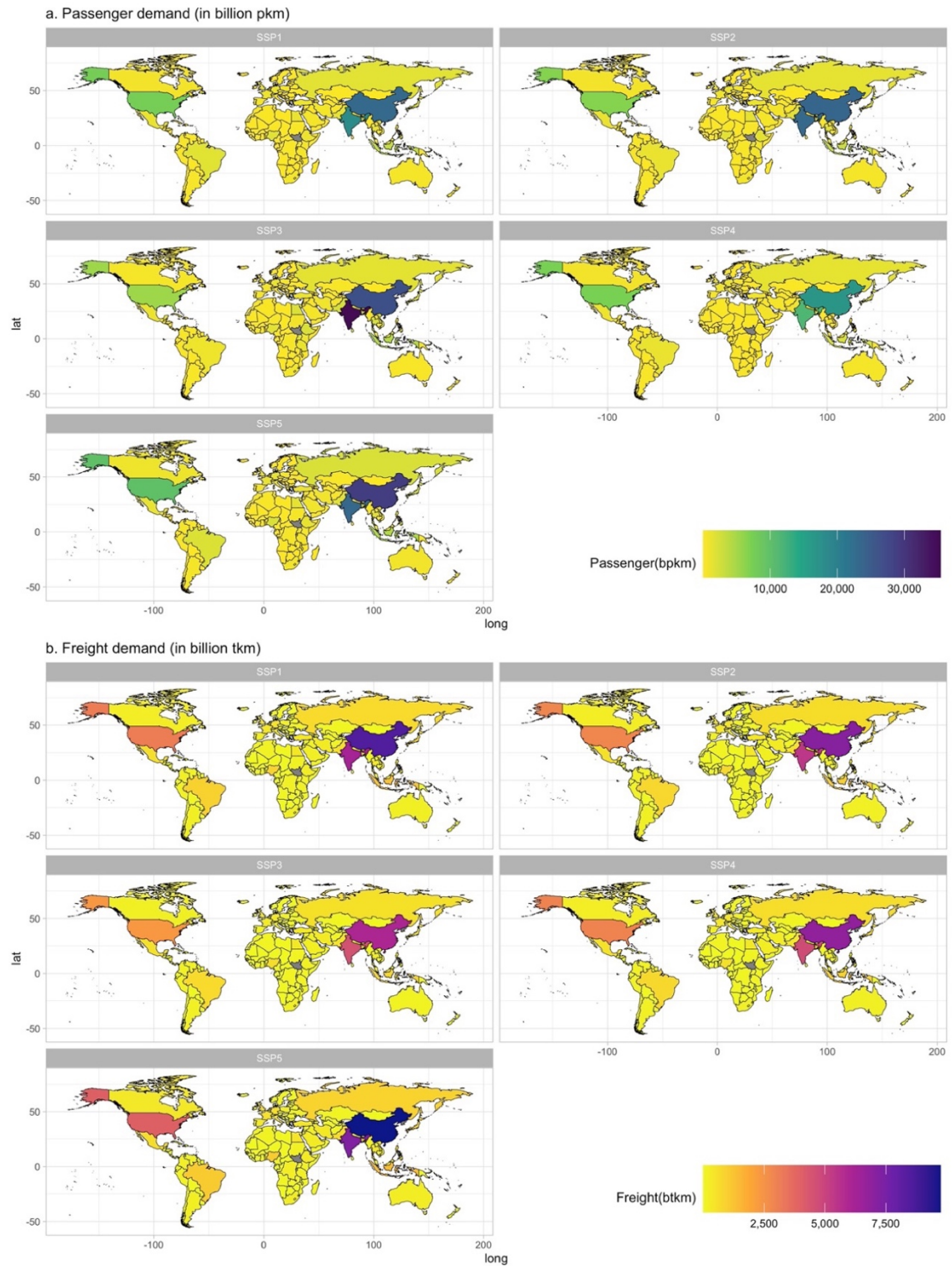


Figure 4.6 Land-based Passenger (a) and Freight (b) transport demand by country in 2050 across each scenario. The coloring represents absolute demand in 2050 for each country. Countries with missing data are shaded gray.

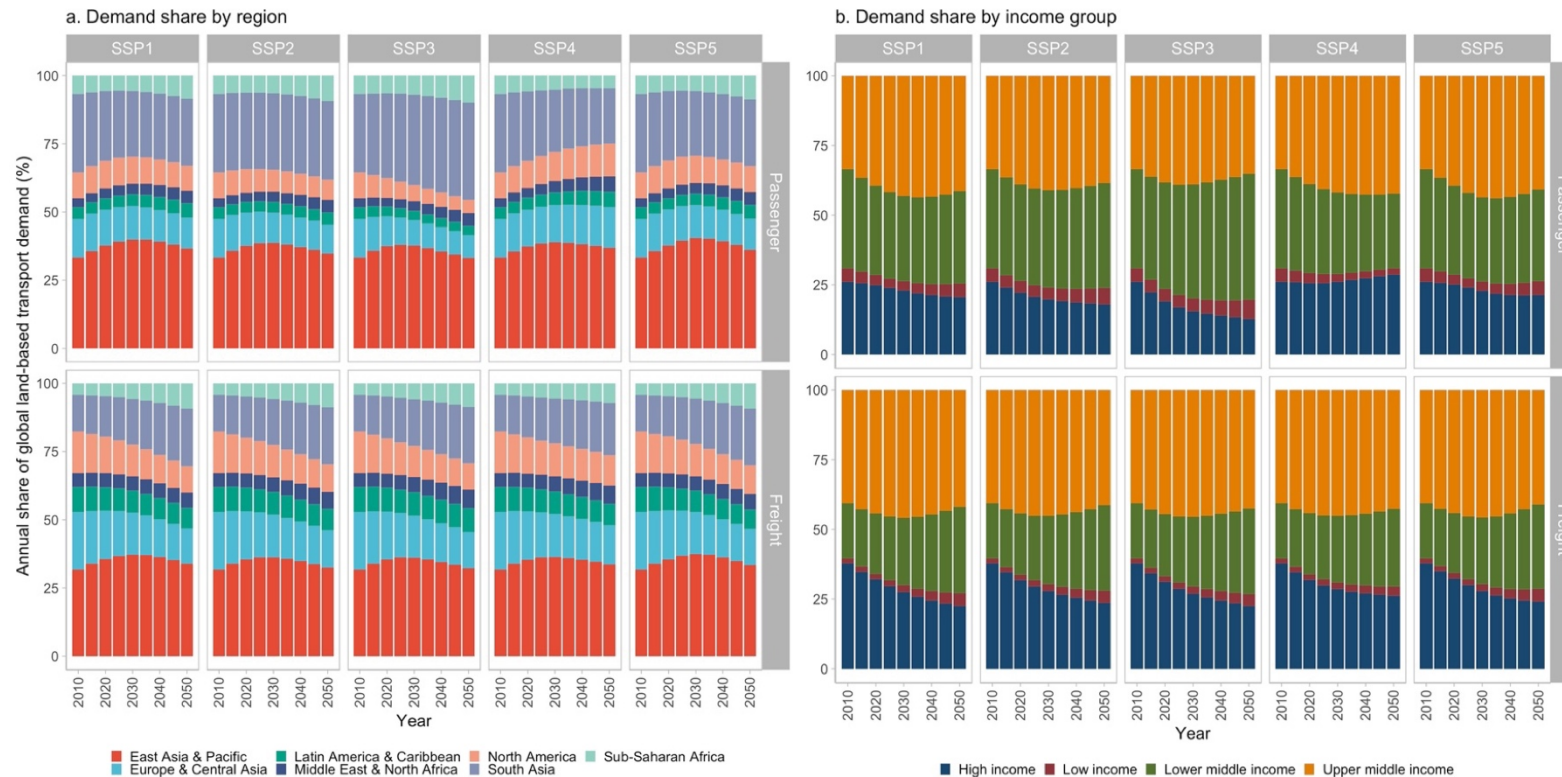


Figure 4.7 Share of global land-based transportation demand by region and income group. The panels in each plot represent the share of the scenario-based projections on global land-based passenger and freight transport by region on the right and by income-group on the left, with the results grouped and faceted by scenario. Panel (a) shows that the share of transport demand from East Asia & Pacific, North America, and Europe & Central Asia start decreasing by 2030, while the share from South Asia and sub-Saharan Africa start increasing. In panel (b), the lower and upper-middle-income countries account for over 60% of global transport demand.

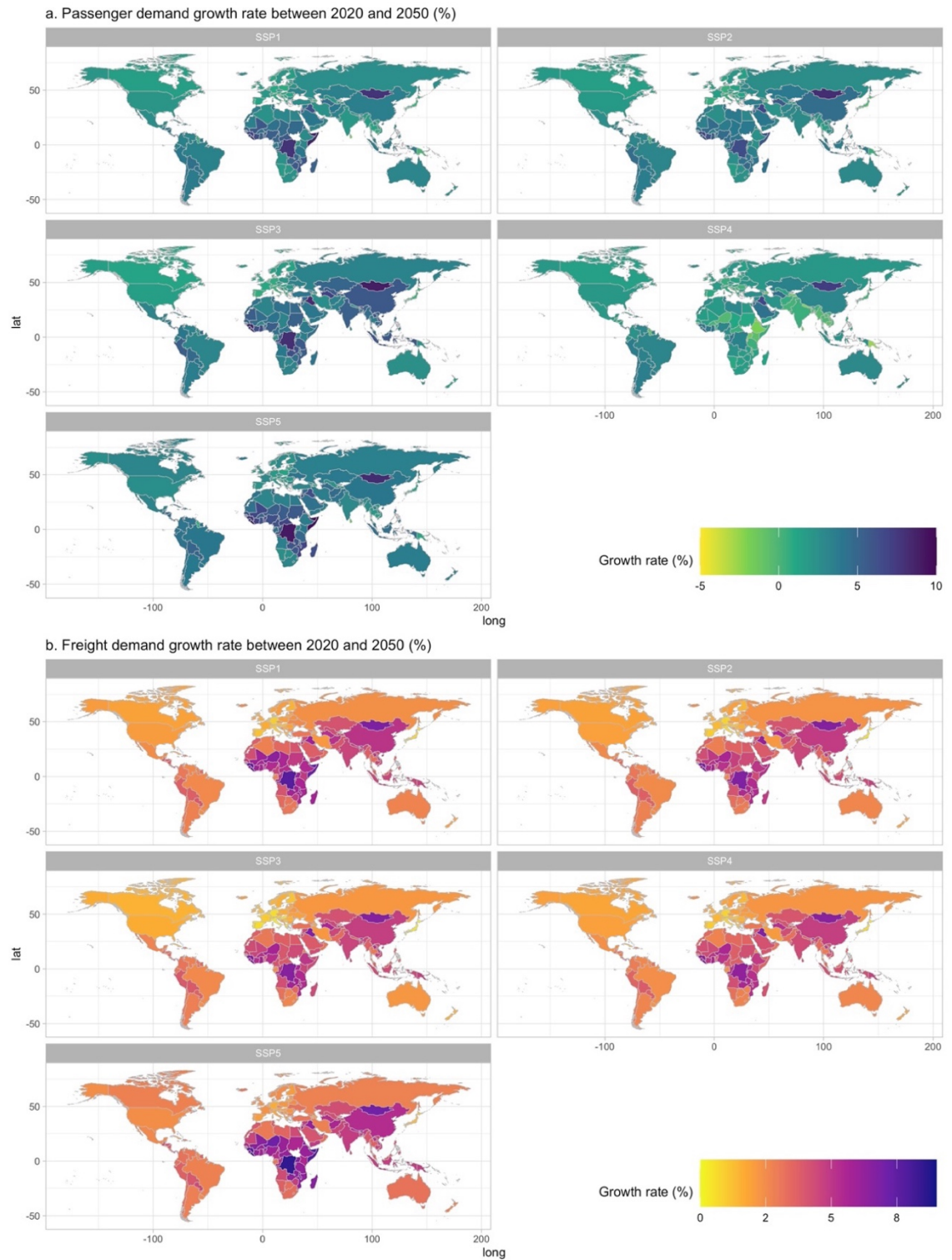


Figure 4.8 Annual growth rates in land-based Passenger (a) and Freight (b) transport demand by country. The color sharing of the countries represents the annual growth rate over the 30 years between 2020 and 2050.

4.6 Conclusion

We develop a statistical model to evaluate the potential demand for land-based transportation services worldwide. The log-log linear model relies on publicly available historical land-based transportation demand and socioeconomic data. We then use the model to develop country-specific future demand scenario-based projections for land-based transportation services worldwide by relying on different SSPs. Using a model that depends only on three variables, we can include estimates for low-income countries for which detailed empirical data is not readily available and advanced demand estimation methods are difficult to deploy. This study identified internally consistent demand estimates for land-based transportation services across 179 countries under five socioeconomic SSP scenarios. Thus, this study is forecasting anchored in five agreed-upon and oft-used scenario-based narratives of energy transitions and decarbonization efforts throughout the world without probability assignment.

Globally, the results indicate that passenger mobility and freight transportation are expected to double by 2050. The results of this study suggest that the highest transportation demand occurs in East Asia and the Pacific. On the other hand, low-income developing countries or sub-Saharan African nations would experience the highest growth rates in demand for transportation services. These two trends come together at an inflection point around 2030. East Asia and the Pacific account for the greatest demand share before 2030. However, after 2030, there is an increase in demand for South Asia and sub-Saharan Africa, reaching near parity with East Asia by 2050.

Much attention has been paid to decarbonizing the transport sector in Europe, North America, and East Asia, as these regions have the largest absolute demand values, resulting in larger energy use and emissions from the transportation sector. However, the larger growth rates in land-based transport demand in developing countries in South Asia and sub-Saharan Africa could pose challenges for their climate mitigation and adaptation targets. Countries in these regions have relatively lower greenhouse gas emissions but they still have developed emission reduction targets under the Paris Agreement. The rapid growth in the demand for land-based transportation services could limit the ability of these countries to reduce emissions, as detailed by their National Determined Contributions to the Paris Agreement. Indeed, the growing demand for land-based transportation services could lead to growing GHG emissions from these regions unless

appropriate strategies are deployed to decouple demand and emissions. For example, low-income African countries constitute a 40% share of used vehicle sales globally, and the lack of regulation leads to an influx of highly polluting vehicles (UN Environment Programme 2020). The results indicate higher demand growth rates for transportation services in low-income countries that are already the most vulnerable to climate change impacts. Continuing to meet this rising demand with high-polluting second-hand vehicles with low fuel-economy ratings is concerning. Some countries such as Rwanda, have imposed restrictions on the import of used vehicles (Ayetor et al. 2021). These countries need to consider policies to expand the availability of more efficient low-carbon transport options.

Rapid growth in demand for land-based transportation services has implications on the need for new infrastructure. Road infrastructure in emerging economies is inadequate and characterized by traffic congestion, which would worsen with increased demand if population growth outpaces the rate of infrastructure expansion (Meijer et al. 2018). For example, the results indicate the Democratic Republic of Congo (DRC) could have one of the highest growth rates in land-based transport demand up to 2050, but the country does not have sufficient road infrastructure for its current population. According to Meijer *et al.*, (2018), the road network in the DRC will need to increase by 81% by 2050 under the SSP scenarios. Furthermore, an increase in land-based freight demand could impose additional requirements on the quality of roads that need to be constructed. Similarly, meeting the growing demand for transportation services in places where the electricity infrastructure is still underdeveloped offers opportunities for co-optimizing power and transport infrastructure to support low-carbon technologies. Consumer behavior is a key driver of useful energy demand. Encouraging consumers to purchase more efficient and low carbon alternatives must go hand in hand with meeting human needs at a low cost.

While the use of personal vehicles will likely increase in developing countries where demand for transportation services is expected to increase rapidly, these countries also have the opportunity to design and invest in low-carbon public transit (e.g., rapid bus systems) and active transport (e.g., protected bike lanes) infrastructure. Investing in such infrastructure, while transport demand is still relatively low, could help establish more sustainable behavioral patterns. While more people will continue to “move around,” providing affordable public transportation can reduce the energy intensity and resulting emissions from meeting the passenger transport demand globally. With the rising rate of urbanization, city planners have the opportunity to leverage urban form strategies to

reduce the emissions implications of increased mobility (Creutzig et al. 2015). Similarly, governments will need to provide affordable and less energy-intensive modal options in sparsely populated rural areas where residents may travel longer distances. Finally, the expansion of rail transportation infrastructure could reduce the burden of road freight and minimize the use of carbon-intensive passenger vehicles. The demand estimates reported in this study can serve as inputs for future analyses of these infrastructure options.

An underlying assumption in the model is that all countries follow the same trends that link GDP and population to transportation demand. While the data points used in the regression include a large range of GDP, population, and urbanization values that cover some of the characteristics of developing countries unpredictable systemic changes can affect the relationships between these variables. For instance, in 2020, the COVID-19 pandemic decreased passenger mobility as people worked from home (ITF 2021a). Although we do not explore these effects, additional improvements in digitalization could further decouple GDP and travel demand (ITF 2019; IEA 2020b). Similarly, other systemic changes like shared and smart mobility, teleworking, dematerialization of the economy, vehicle automation, and supply chain management could affect the relationship between the dependent and independent variables used in this paper (Mouratidis, Peters, and van Wee 2021). These impacts are highly uncertain, and there isn't sufficient empirical data to include them in a statistical model (Jaramillo et al. 2022). Despite this caveat, this study sheds light on important data gaps and the results provide a starting point for understanding demand patterns in developing countries. For example, the results could serve as inputs for macro-energy system models to evaluate opportunities for providing low-carbon energy to the transportation sector.

In conclusion, the global transition to low-carbon infrastructure systems is crucial for simultaneously meeting the growing demand for transport services identified in this study and climate stabilization goals. Thus, there is a growing need to evaluate the barriers to and opportunities for the deployment of these low-carbon systems. Although significant efforts have been made to model decarbonization pathways in developed countries using energy system models, these evaluations are lacking for developing countries. In the context of the least developed countries, current energy modeling research has traditionally focused on providing minimum energy access. However, the results of this study suggest that such models should also include the energy needs of the transportation

sector, which will be driven by the demand for transportation services analyzed in this chapter.

5 CONCLUSION AND FUTURE WORK

This chapter concludes the dissertation by highlighting the key takeaways of the thesis and their policy implications. The objective of this study was to tackle three core areas in progressing the movement towards achieving universal energy access while adhering to climate goals in six East African countries. First, the need to develop a well parameterized household model for the six countries, Burundi, the Democratic Republic of Congo, Kenya, Rwanda, Tanzania, and Uganda. Second, the need to better understand the level of access these countries are trying to achieve and explore the implication of different definitions of energy access with access tier considerations. Lastly, considering climate mitigation efforts, a forward-looking approach to encourage expansion of analysis to include the hard-to-decarbonize transport sector, the need to extend the knowledge of how demand for land-based transportation services will grow by developing long-term scenario-based logistic regression demand model. To satisfy these objectives:

- First, this thesis presents a high-resolution dataset and open access household model to help us better understand where demand is. Therefore, this dissertation introduces the Temoa East Africa version 1 model using an established modeling framework with resource and technology parameters for rural and urban households in six East African countries.
- Second, this thesis develops open-source datasets with long term household energy demand for energy services by access levels out to 2050 under different scenarios and uses the Temoa East Africa model to assess impacts of different access definitions on least-cost technology choice, emissions, and costs in East Africa
- Lastly, the thesis extends knowledge in a hard-to-decarbonize sector with a focus on the transportation sector to support future efforts to expand modeling scope by presenting an inclusive demand model for land-based passenger and freight transport on a country level.

5.1 Revisiting the dissertation research goals

- (i) To improve open-source modeling tools to explore energy transition pathways at the country level in rural and urban East Africa
 - a. This dissertation achieved this goal in Chapter 2 by introducing the first version of an open-source high resolution model that details the parameterization of the Temoa East Africa model that uses the Temoa modeling framework. Building the Temoa East Africa database was a significant accomplishment, as it offers a detailed depiction of supply, electricity, building, and transport sector technologies, and includes carbon removal and hydrogen production pathways.
 - b. By providing an open-source model, this modeling contribution seeks to encourage further collaboration and sustainable energy planning in East Africa. Future work will focus on improving the database by incorporating more detailed transportation service demand data, electric vehicle charging station characterizations, and extending technology characterizations to the industrial and agricultural sectors.
- (ii) To examine impacts of hypothetical pathways towards achieving universal energy access goals across different access levels in rural and urban East African households
 - a. Using the Temoa East Africa model from Chapter 2, Chapter 3 provided insights into impacts of different energy access definitions, under different socioeconomic and emissions scenarios, on final energy consumption, least-cost technology mix, emissions, and costs by access tier group. The findings suggest that policymakers must identify obstacles and opportunities for each access tier to progress towards equitable energy access and low carbon pathways. This requires the internalization of externalities from biogenic CO₂ emissions and policies that promote access to modern, efficient, and cleaner energy sources, particularly in rural households.
 - b. The shift towards natural gas and electricity in Tier 3 – 5 access levels can facilitate progress towards a low carbon energy system, dependent on renewable energy and carbon removal technologies. Policymakers can

encourage the adoption of electric technologies in lower energy consumption households (Tier 1 – 2) by offering innovative financing models to reduce high up-front costs for efficient technologies such as electric cookers, solar water heaters and LED bulbs. Decentralized technologies such as off-grid solar PV will play pivotal role in increasing electricity access to the access deficit households in sparsely populated rural areas, to provide access where centralized grids may not be feasible. The results also emphasize that policymakers must ensure policies designed to reduce reliance on biomass do not impose a disproportionate cost burden on lower access and rural households.

- (iii) To provide long-term land-based passenger and freight transport demand estimates on a country-level to support open modeling community efforts in developing countries including those in East Africa
 - a. In Chapter 4, we developed a long-term land-based passenger and freight transport demand model to support open-modeling community efforts in developing countries, including East Africa. Using the log-log- linear model and SSP projections, this chapter presented the long-term land-based passenger and freight transportation service demand over a 30-year time horizon from 2020 to 2050 under different socioeconomic scenarios. The global average annual land-based passenger demand growth rate ranges between 1.3% and 4.1%, while the annual growth rate for land-based freight demand ranges between 3.1% and 3.6% across the 30 years between 2020 and 2050. Middle-income countries in Asia such as India and China, show the highest expected transport demand across all scenarios.
 - b. Meanwhile, the results suggest that low-income countries in the sub-Saharan African region are likely to experience the largest growth in demand for passenger and freight transport services. These two trends come together at an inflection point around the year 2030. Prior to 2030, the transport demand was the highest in East Asia. After 2030, there is an ascendancy in transport demand in South Asia and sub-Saharan Africa, whereby the cumulative demand share of these two regions reaches near parity with that of East Asia by 2050. Sustainably meeting this growing

demand will require the adoption of data-driven transport planning tools and leveraging cross-linkages across other energy sectors such as electricity.

5.2 Limitations

The Temoa East Africa version 1 model is a useful tool for examining sector specific pathways. However, there are limitations to be noted before using the tools.

- The demand characterization assumptions use socioeconomic projections from the SSPs, the broad range of scenarios can only capture some of the uncertainties, however future demand projections remain largely uncertain. Therefore, the results of this dissertation must be interpreted within the scope of the work with an understanding of the underlying assumptions. Demand is defined exogenously therefore there is no feedback between technology choice and demand drivers.
- The fuel prices are not country specific given data limitations. For instances, countries like Tanzania with large natural gas proven reserves will have lower gas prices compared to Burundi that does not discovered reserves at present and lacks pipeline infrastructure. The current fuel prices do not reflect fuel cost curves therefore the model will not model increasing cost of fossil fuels as reserves are depleted.
- The electricity demand in the tool is determined endogenously however, the current version of the model does not account for demand from other sectors such as the industrial and agricultural sector. Accounting for electricity demand from these sectors will change the fuel mix and resulting emissions and cost results.
- The current iteration of the Temoa East Africa model does not allow for cross-border power exchange or regional power pooling.
- The model provides results for technology choice (what technology), emissions and costs (how much), but with current parameterization cannot determine where technologies will be deployed geospatially beyond the rural-urban split by access level.
- In the end-use sectors there are not existing technology stocks. For the building sector over long term analysis these may be negligible as many of the technologies have short lifespans. However, future modeling to including the transport sector in the analysis would benefit from extending the database to represent vehicle stock.

- Uncertainty from the data inputs for instance, the global discount rate of 5% may not depict the cost of capital from high-risk perceptions that investors may have in financing projects in the region (Ameli et al. 2021).

Understanding the underlying assumptions that underlie the modeling results and its applicability go hand in hand, for instance:

- The modeling approach assumed a single rational decision maker, but in reality, decisions are frequently reached in collaboration with a number of stakeholders who have multiple objectives. As a result, given the existing system representation, the results are not designed to replace a decision maker, but rather to provide quantitative support for technology deployment and the impacts of explicitly specified energy access and mitigation targets over the long term.
- The demand trajectories are speculative and do not represent predictions of the future. The scenario-based demand trajectories can be changed by modelers and decision-makers to create and explore other long-term pathways to evaluate implications under various plausible scenarios. The SDG2050- scenarios in Chapter 3 make target-based hypothetical assumptions such as the improbable 100% Tier 3+ access, because there is no empirical data of how population distribution evolves across the tiers. Therefore, decision-makers can improve the scenarios as research efforts advance and aim to better comprehend these household demand consumption patterns in the long-term.
- The modeling approach's emissions and costs trade-offs do not account for more nuanced geopolitical or cultural behavior. The Temoa modeling tool is unable to take into account non-monetary objectives such as cultural preferences, effective science communication, or corruption, which are crucial factors in the selection of technology. This indicates that each output from a model is only reliable within the parameters of the modeling assumptions and level of aggregation.

5.3 Future work

Future iterations will strive to improve the database by disaggregating transportation service demand based on access level across rural-urban divisions and incorporating electric vehicle charging station characterizations. In addition, the forthcoming Temoa East Africa version will encompass the industrial and agricultural sectors, boasting

extensive technology characterizations. The long-term objective is to continuously expand the modeling scope to encompass end-use technologies in all major energy system sectors, ultimately publishing the complete model database in a publicly accessible GitHub repository. By providing an open-source model, we hope to encourage further research and collaboration on sustainable energy planning in East Africa.

In summary, this dissertation focuses on addressing three core areas in progressing the East African region's movement towards achieving universal energy access while adhering to climate goals, we develop the Temoa East Africa model using established modeling framework, we use the open-source modeling tool with different access tiers to account for different access scenarios and the impacts of varying degrees of household energy access definitions, and lastly, our key contribution in the transportation sector is to provide a demand model for land-based passenger and freight transport on a country level including the countries in East Africa.

6 REFERENCES

- Adwek, George, Shen Boxiong, Paul O. Ndolo, Zachary O. Siagi, Chebet Chepsaigutt, Cicilia M. Kemunto, Moses Arowo, John Shimmon, Patrobers Simiyu, and Abel C. Yabo. 2020. *The Solar Energy Access in Kenya: A Review Focusing on Pay-As-You-Go Solar Home System. Environment, Development and Sustainability*. Vol. 22. Springer Netherlands. <https://doi.org/10.1007/s10668-019-00372-x>.
- Ajanovic, Amela, Carol Dahl, and Lee Schipper. 2012. “Modelling Transport (Energy) Demand and Policies - an Introduction.” *Energy Policy* 41: iii–xiv. <https://doi.org/10.1016/j.enpol.2011.12.033>.
- Alam, Bhuiyan Monwar, Hilary Nixon, and Qiong Zhang. 2018. “Factors Affecting Travel Demand by Bus: An Empirical Analysis at U.S. Metropolitan Statistical Area Level.” *Transportation Research Record* 2672 (8): 817–26. <https://doi.org/10.1177/0361198118798714>.
- Allington, Lucy, Carla Cannone, Ioannis Pappis, Karla Cervantes Barron, Will Usher, Steve Pye, Edward Brown, et al. 2022. “Selected ‘Starter Kit’ Energy System Modelling Data for Selected Countries in Africa, East Asia, and South America (#CCG, 2021).” *Data in Brief* 42: 108021. <https://doi.org/10.1016/j.dib.2022.108021>.
- Amanam, Usua U. 2017. “Natural Gas in East Africa : Domestic and Regional Use.” https://ngi.stanford.edu/sites/default/files/NGI_EAfrica_LitReview%284-17%29.pdf.
- Ameli, Nadia, Olivier Dessens, Matthew Winning, Jennifer Cronin, Hugues Chenet, Paul Drummond, Alvaro Calzadilla, Gabriel Anandarajah, and Michael Grubb. 2021. “Higher Cost of Finance Exacerbates a Climate Investment Trap in Developing Economies.” *Nature Communications* 12 (1): 1–12. <https://doi.org/10.1038/s41467-021-24305-3>.
- Ayeter, G. K., Innocent Mbonigaba, M. N. Sackey, and P. Y. Andoh. 2021. “Vehicle Regulations in Africa: Impact on Used Vehicle Import and New Vehicle Sales.”

- Transportation Research Interdisciplinary Perspectives* 10 (May): 100384. <https://doi.org/10.1016/j.trip.2021.100384>.
- Bär, Roger, Andreas Heinimann, and Albrecht Ehrensperger. 2017. “Assessing the Potential Supply of Biomass Cooking Fuels in Kilimanjaro Region Using Land Use Units and Spatial Bayesian Networks.” *Energy for Sustainable Development* 40: 112–25. <https://doi.org/10.1016/j.esd.2017.05.007>.
- Bauer, Nico, Katherine Calvin, Johannes Emmerling, Oliver Fricko, Shinichiro Fujimori, Jérôme Hilaire, Jiyong Eom, et al. 2017. “Shared Socio-Economic Pathways of the Energy Sector – Quantifying the Narratives.” *Global Environmental Change* 42: 316–30. <https://doi.org/10.1016/j.gloenvcha.2016.07.006>.
- Bhatia, Mikul, and Nicolina Angelou. 2015. “Beyond Connections Energy Access Redefined.” <https://openknowledge.worldbank.org/bitstream/handle/10986/24368/Beyond0connect0d000technical0report.pdf?sequence=1&isAllowed=y>.
- Bhattacharyya, Subhes C., and Govinda R. Timilsina. 2010a. “A Review of Energy System Models.” *International Journal of Energy Sector Management* 4 (4): 494–518. <https://doi.org/10.1108/17506221011092742>.
- . 2010b. “Modelling Energy Demand of Developing Countries: Are the Specific Features Adequately Captured?” *Energy Policy* 38 (4): 1979–90. <https://doi.org/10.1016/j.enpol.2009.11.079>.
- Calvin, Katherine, Pralit Patel, Leon Clarke, Ghassem Asrar, Ben Bond-Lamberty, Ryna Yiyun Cui, Alan Di Vittorio, et al. 2019. “GCAM v5.1: Representing the Linkages between Energy, Water, Land, Climate, and Economic Systems.” *Geoscientific Model Development* 12 (2): 677–98. <https://doi.org/10.5194/gmd-12-677-2019>.
- Capros, P, D van Regemorter, L Paroussos, P Karkatsoulis, C Fragkiadakis, S Tsani, I Charalampidis, and T Revesz. 2013. *GEM-E3 Model Documentation. JRC Technical Reports*. <https://doi.org/10.2788/47872>.
- Capros, Pantelis, Maria Kannavou, Stavroula Evangelopoulou, Apostolos Petropoulos, Pelopidas Siskos, Nikolaos Tasios, Georgios Zazias, and Alessia DeVita. 2018. “Outlook of the EU Energy System up to 2050: The Case of Scenarios Prepared for European Commission’s ‘Clean Energy for All Europeans’ Package Using the PRIMES Model.” *Energy Strategy Reviews* 22 (October): 255–63.

<https://doi.org/10.1016/j.esr.2018.06.009>.

Carmona-Benítez, Rafael Bernardo, Maria Rosa Nieto, and Danya Miranda. 2017. “An Econometric Dynamic Model to Estimate Passenger Demand for Air Transport Industry.” *Transportation Research Procedia* 25: 17–29.

<https://doi.org/10.1016/j.trpro.2017.05.191>.

Carvallo, Juan-Pablo, Brittany J Shaw, Nkiruka I Avila, and Daniel M Kammen. 2017. “Sustainable Low-Carbon Expansion for the Power Sector of an Emerging Economy: The Case of Kenya.” *Environmental Science and Technology* 51: 10232–42. <https://doi.org/10.1021/acs.est.7b00345>.

Chen, He, Keisuke Matsushashi, Kiyoshi Takahashi, Shinichiro Fujimori, Keita Honjo, and Kei Gomi. 2020. “Adapting Global Shared Socio-Economic Pathways for National Scenarios in Japan.” *Sustainability Science* 15 (3): 985–1000. <https://doi.org/10.1007/s11625-019-00780-y>.

Cherubini, Francesco, Glen P. Peters, Terje Berntsen, Anders H. Strømman, and Edgar Hertwich. 2011. “CO₂ Emissions from Biomass Combustion for Bioenergy: Atmospheric Decay and Contribution to Global Warming.” *GCB Bioenergy* 3 (5): 413–26. <https://doi.org/10.1111/j.1757-1707.2011.01102.x>.

Clemens, Harry, Rob Bailis, Anne Nyambane, and Victoria Ndung'u. 2018. “Africa Biogas Partnership Program: A Review of Clean Cooking Implementation through Market Development in East Africa.” *Energy for Sustainable Development* 46: 23–31. <https://doi.org/10.1016/j.esd.2018.05.012>.

Creutzig, Felix, Giovanni Baiocchi, Robert Bierkandt, Peter Paul Pichler, and Karen C. Seto. 2015. “Global Typology of Urban Energy Use and Potentials for an Urbanization Mitigation Wedge.” *Proceedings of the National Academy of Sciences of the United States of America* 112 (20): 6283–88. <https://doi.org/10.1073/pnas.1315545112>.

Cuaresma, Jesus Crespo. 2017. “Income Projections for Climate Change Research: A Framework Based on Human Capital Dynamics.” *Global Environmental Change* 42: 226–36. <https://doi.org/10.1016/j.gloenvcha.2015.02.012>.

Daggash, Habiba Ahut, and Niall Mac Dowell. 2021. “Delivering Low-Carbon Electricity Systems in Sub-Saharan Africa: Insights from Nigeria.” *Energy and Environmental*

- Science* 14 (7): 4018–37. <https://doi.org/10.1039/d1ee00746g>.
- Dagnachew, Anteneh G., Andries F. Hof, Paul L. Lucas, and Detlef P. van Vuuren. 2020. “Scenario Analysis for Promoting Clean Cooking in Sub-Saharan Africa: Costs and Benefits.” *Energy* 192: 116641. <https://doi.org/10.1016/j.energy.2019.116641>.
- Dagnachew, Anteneh G., Paul L. Lucas, Andries F. Hof, David E.H.J. Gernaat, Harmen Sytze de Boer, and Detlef P. van Vuuren. 2017. “The Role of Decentralized Systems in Providing Universal Electricity Access in Sub-Saharan Africa – A Model-Based Approach.” *Energy* 139: 184–95. <https://doi.org/10.1016/j.energy.2017.07.144>.
- Dagnachew, Anteneh G., Miguel Poblete-Cazenave, Shonali Pachauri, Andries F. Hof, Bas Van Ruijven, and Detlef P. Van Vuuren. 2019. “Integrating Energy Access, Efficiency and Renewable Energy Policies in Sub-Saharan Africa: A Model-Based Analysis.” *Environmental Research Letters* 15 (12). <https://doi.org/10.1088/1748-9326/abcbb9>.
- Daiglou, Vassilis, Bas J. van Ruijven, and Detlef P. van Vuuren. 2012. “Model Projections for Household Energy Use in Developing Countries.” *Energy* 37 (1): 601–15. <https://doi.org/10.1016/j.energy.2011.10.044>.
- Daldoul, Manel, Sami Jarboui, and Ahlem Dakhlaoui. 2016. “Public Transport Demand: Dynamic Panel Model Analysis.” *Transportation* 43 (3): 491–505. <https://doi.org/10.1007/s11116-015-9586-1>.
- DeCarolis, J. F., S. Babae, B. Li, and S. Kanungo. 2016. “Modelling to Generate Alternatives with an Energy System Optimization Model.” *Environmental Modelling and Software* 79 (May): 300–310. <https://doi.org/10.1016/j.envsoft.2015.11.019>.
- Decarolis, Joseph, Kevin Hunter, and Sarat Sreepathi. 2020. “The TEMOA Project : Tools for Energy Model Optimization and Analysis.” *Energy*, no. June: 1–18.
- Dellink, Rob, Jean Chateau, Elisa Lanzi, and Bertrand Magné. 2017. “Long-Term Economic Growth Projections in the Shared Socioeconomic Pathways.” *Global Environmental Change* 42 (January): 200–214. <https://doi.org/10.1016/J.GLOENVCHA.2015.06.004>.
- Dender, Kurt Van, and Martin Clever. 2013. “Recent Trends in Car Usage in Advanced Economies: Slower Growth Ahead?” *International Transport Forum*. <http://dx.doi.org/10.1787/5k46n4121k5d-en>.

- DHS Program - USAID. 2022. “Demographic and Health Surveys.” USAID. 2022. <https://dhsprogram.com/methodology/survey/survey-display-485.cfm>.
- Diawuo, Felix Amankwah, Marriette Sakah, André Pina, Patricia C. Baptista, and Carlos A. Silva. 2019. “Disaggregation and Characterization of Residential Electricity Use: Analysis for Ghana.” *Sustainable Cities and Society* 48 (March): 101586. <https://doi.org/10.1016/j.scs.2019.101586>.
- Diawuo, Felix Amankwah, Ian J Scott, Patricia C Baptista, and Carlos A Silva. 2020. “Assessing the Costs of Contributing to Climate Change Targets in Sub-Saharan Africa: The Case of the Ghanaian Electricity System” 57: 32–47. <https://doi.org/10.1016/j.esd.2020.05.001>.
- Dioha, Michael O., and Nnaemeka Vincent Emodi. 2019. “Investigating the Impacts of Energy Access Scenarios in the Nigerian Household Sector by 2030.” *Resources* 8 (3). <https://doi.org/10.3390/resources8030127>.
- Dioha, Michael O., and Atul Kumar. 2020a. “Exploring Sustainable Energy Transitions in Sub-Saharan Africa Residential Sector: The Case of Nigeria.” *Renewable and Sustainable Energy Reviews* 117 (October 2019): 109510. <https://doi.org/10.1016/j.rser.2019.109510>.
- . 2020b. “Exploring the Energy System Impacts of Nigeria’s Nationally Determined Contributions and Low-Carbon Transition to Mid-Century.” *Energy Policy* 144 (April): 111703. <https://doi.org/10.1016/j.enpol.2020.111703>.
- Dioha MO. 2017. “Energy System Models for Sub-Saharan African Countries - A Systematic Review.” *Journal of Sustainable Energy* 8 (4): 159–68.
- Dunkerley, Fay, Charlene Rohr, and Andrew Daly. 2014. “Road Traffic Demand Elasticities: A Rapid Evidence Assessment.” *Road Traffic Demand Elasticities: A Rapid Evidence Assessment*. <https://doi.org/10.7249/rr888>.
- E3Mlab/NTUA. 2016. “PRIMES Model Version 6 2016-2017,” 50. [http://www.e3mlab.ntua.gr/e3mlab/PRIMES Manual/The PRIMES MODEL 2016-7.pdf](http://www.e3mlab.ntua.gr/e3mlab/PRIMES%20Manual/The%20PRIMES%20MODEL%202016-2017.pdf).
- Ebi, Kristie L., Stephane Hallegatte, Tom Kram, Nigel W. Arnell, Timothy R. Carter, Jae Edmonds, Elmar Kriegler, et al. 2014. “A New Scenario Framework for Climate Change Research: Background, Process, and Future Directions.” *Climatic Change*

- 122 (3): 363–72. <https://doi.org/10.1007/s10584-013-0912-3>.
- EIA. 2016a. “Kenya.” 2016.
- . 2016b. “Uganda.” 2016.
- Energy Information Administration (EIA). 2023. “Updated Building Sector Appliances and Equipment Costs and Efficiencies.” U.S. EIA. 2023.
- ERA. 2021. “Energy Minister Launches Reviewed Electricity Tariff Structure.”
- Eshraghi, Hadi, Anderson Rodrigo De Queiroz, and Joseph F. DeCarolus. 2018. “US Energy-Related Greenhouse Gas Emissions in the Absence of Federal Climate Policy.” Research-article. *Environmental Science and Technology* 52 (17): 9595–9604. <https://doi.org/10.1021/acs.est.8b01586>.
- Falchetta, Giacomo, Manfred Hafner, and Simone Tagliapietra. 2020. “Pathways to 100% Electrification in East Africa by 2030.” *The Energy Journal* 41 (3). <https://doi.org/10.5547/01956574.41.3.gfal>.
- Falchetta, Giacomo, Shonali Pachauri, Edward Byers, Olha Danylo, and Simon C. Parkinson. 2020. “Satellite Observations Reveal Inequalities in the Progress and Effectiveness of Recent Electrification in Sub-Saharan Africa.” *One Earth* 2 (4): 364–79. <https://doi.org/10.1016/j.oneear.2020.03.007>.
- Fouquet, Roger. 2012. “Trends in Income and Price Elasticities of Transport Demand (1850-2010).” *Energy Policy* 50: 62–71. <https://doi.org/10.1016/j.enpol.2012.03.001>.
- Gambhir, Ajay, Isabela Butnar, Pei Hao Li, Pete Smith, and Neil Strachan. 2019. “A Review of Criticisms of Integrated Assessment Models and Proposed Approaches to Address These, through the Lens of BECCs.” *Energies* 12 (9): 1–21. <https://doi.org/10.3390/en12091747>.
- Groissböck, Markus. 2019. “Are Open Source Energy System Optimization Tools Mature Enough for Serious Use?” *Renewable and Sustainable Energy Reviews*. Elsevier Ltd. <https://doi.org/10.1016/j.rser.2018.11.020>.
- Haberl, Helmut. 2013. “Net Land-Atmosphere Flows of Biogenic Carbon Related to Bioenergy: Towards an Understanding of Systemic Feedbacks.” *GCB Bioenergy* 5 (4): 351–57. <https://doi.org/10.1111/gcbb.12071>.
- Hafner, Manfred, Simone Tagliapietra, Giacomo Falchetta, and Giovanni Occhiali. 2019.

Renewables for Energy Access and Sustainable Development in East Africa.
SpringerBriefs in Energy.

Howells, Mark, Holger Rogner, Neil Strachan, Charles Heaps, Hillard Huntington, Socrates Kypreos, Alison Hughes, et al. 2011. "OSeMOSYS: The Open Source Energy Modeling System. An Introduction to Its Ethos, Structure and Development." *Energy Policy* 39 (10): 5850–70. <https://doi.org/10.1016/j.enpol.2011.06.033>.

Hunter, Kevin, Sarat Sreepathi, and Joseph F. DeCarolis. 2013. "Modeling for Insight Using Tools for Energy Model Optimization and Analysis (Temoa)." *Energy Economics* 40 (November): 339–49. <https://doi.org/10.1016/j.eneco.2013.07.014>.

IEA. 2017. "Energy Technology Perspectives 2017 - Catalysing Energy Technology Transformations." https://doi.org/10.1787/energy_tech-2017-en.

———. 2019a. "Africa Energy Outlook 2019." *World Energy Outlook Special Report*, 288. <https://www.iea.org/reports/africa-energy-outlook-2019#energy-access%0Ahttps://www.iea.org/reports/africa-energy-outlook-2019%23africa-case>.

———. 2019b. "The Future of Hydrogen." *The Future of Hydrogen*, no. June. <https://doi.org/10.1787/1e0514c4-en>.

———. 2019c. "World Energy Outlook 2019." *World Energy Outlook 2019*, 1. <https://www.iea.org/reports/world-energy-outlook-2019>.

———. 2020a. "Defining Energy Access: 2020 Methodology." 2020. <https://www.iea.org/articles/defining-energy-access-2020-methodology>.

———. 2020b. "Energy Technology Perspectives 2020." *Energy Technology Perspectives 2020*. <https://doi.org/10.1787/ab43a9a5-en>.

———. 2022a. "CO₂ Emissions in 2022." *Global Energy* 62 (10): 20–21. <https://www.iea.org/news/global-co2-emissions-rebounded-to-their-highest-level-in-history-in-2021>.

———. 2022b. "World Energy Outlook Special Report Africa Energy Outlook 2022." *International Energy Agency*, 249. www.iea.org/t&c/.

IEA, and IRENA. 2017. "Perspectives for Energy Transition: Investment Needs for a Low-Carbon Energy System." *IEA Publications*.

- https://www.irena.org/DocumentDownloads/Publications/Perspectives_for_the_Energy_Transition_2017.pdf.
- IEA, IRENA, UNDSO, World Bank, and WHO. 2022. “Tracking SDG 7: The Energy Progress Report.” *World Bank, Washington DC*.
- IEA, IRENA, UNDSO, World Bank, and WHO. 2018. “Tracking SDG7: The Energy Progress Report 2018.” *Executive Summary*, 29.
- IIASA. 2018. “SSP Database (Shared Socioeconomic Pathways) - Version 2.0.” 2018. <https://tntcat.iiasa.ac.at/SspDb>.
- International Monetary Fund African Department. 2022. “Democratic Republic of the Congo: Selected Issues.” <https://doi.org/https://doi.org/10.5089/9798400214288.00>.
- International Institute for Applied Systems Analysis (IIASA). 2018. “Integrated Assessment Modeling of SSPs.”
- IPCC. 2018. “Special Report Global Warming of 1.5C: Glossary.” <https://www.ipcc.ch/sr15/chapter/glossary/>.
- . 2019. “2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2 Energy: Chapter 2.” <https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol2.html>.
- . 2021. “Assessment Report 6 Climate Change 2021: The Physical Science Basis.” <https://www.ipcc.ch/report/ar6/wg1/>.
- . 2022a. *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegr. Intergovernmental Panel on Climate Change*. Cambridge University Press. Cambridge University Press, Cambridge, UK and New York, NY, USA., <https://doi.org/10.1017/9781009325844>.
- . 2022b. “EFDB Emission Factor Database.” 2022.
- . 2023. “AR6 Synthesis Report: Climate Change 2023.” *Cambridge University Press*. <https://www.ipcc.ch/report/ar6/syr/>.
- IRENA. 2012. “Prospects for the African Power Sector.” *Scenarios and Strategies for Africa Project*, 60 pp.

- . 2021. *Planning and Prospects for Renewable Power: Eastern and Southern Africa*. Publication forthcoming.
- . 2023. “IRENASTAT Online Data Query Tool.” IRENA. 2023.
- ITF. 2019. *ITF Transport Outlook 2019*. *ITF Transport Outlook 2019*. <https://doi.org/10.1787/9789282103937-en>.
- . 2021a. *International Transport Forum (ITF) Transport Outlook 2021*. <https://doi.org/10.1787/16826a30-en>.
- . 2021b. “ITF Transport Statistics.” OECD-ILibrary. 2021. https://www.oecd-ilibrary.org/transport/data/itf-transport-statistics_trsptr-data-en.
- Jaramillo, P., S Kahn Ribeiro, P. Newman, S. Dhar, O.E. Diemuodeke, T. Kajino, D.S. Lee, et al. 2022. “2022: Transport. In IPCC, 2022: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [P.R. Shukla, J. Skea, R. Slade, A. Al Khourdajie, R. Van.” <https://doi.org/10.1017/9781009157926.012>.
- Jiang, Leiwen, and Brian C. O'Neill. 2017. “Global Urbanization Projections for the Shared Socioeconomic Pathways.” *Global Environmental Change* 42 (January): 193–99. <https://doi.org/10.1016/J.GLOENVCHA.2015.03.008>.
- Joint Global Change Research Institute (JGCRI). 2017. “GCAM Model Release.” <http://www.globalchange.umd.edu/gcam/download/>.
- Karkacier, Osman, Z Goktolga, and Adnan Cicek. 2006. “A Regression Analysis of the Effect of Energy Use in Agriculture.” *Energy Policy* 34: 3796–3800. <https://doi.org/10.1016/j.enpol.2005.09.001>.
- KC, Samir, and Wolfgang Lutz. 2017. “The Human Core of the Shared Socioeconomic Pathways: Population Scenarios by Age, Sex and Level of Education for All Countries to 2100.” *Global Environmental Change* 42: 181–92. <https://doi.org/10.1016/j.gloenvcha.2014.06.004>.
- Keith, David W., Geoffrey Holmes, David St. Angelo, and Kenton Heidel. 2018. “A Process for Capturing CO₂ from the Atmosphere.” *Joule* 2 (8): 1573–94. <https://doi.org/10.1016/j.joule.2018.05.006>.
- Khalili, Siavash, Eetu Rantanen, Dmitrii Bogdanov, and Christian Breyer. 2019. “Global

- Transportation Demand Development with Impacts on the Energy Demand and Greenhouse Gas Emissions in a Climate-Constrained World.” *Energies* 12 (20). <https://doi.org/10.3390/en12203870>.
- Kutner, Michael H., John Neter, Christopher J. Nachtsheim, and William Li. 2004. *Applied Linear Statistical Models*. McGraw-Hill Education.
- Lamb, William F., Thomas Wiedmann, Julia Pongratz, Robbie Andrew, Monica Crippa, Jos G.J. Olivier, Dominik Wiedenhofer, et al. 2021. “A Review of Trends and Drivers of Greenhouse Gas Emissions by Sector from 1990 to 2018.” *Environmental Research Letters* 16 (7). <https://doi.org/10.1088/1748-9326/abee4e>.
- Lecaros, Santiago Sinclair, Winnie Musivo, Dimitris Mentis, and Benson Ireri. 2021. “Integrated Planning Helps Kenya Close Its Energy Access Gap.” World Resources Institute (WRI). 2021. <https://www.wri.org/insights/integrated-energy-access-plans-kenya>.
- Lee, Sunkee, Jisoo Park, and Arman Kilic. 2021. “Bridging Two Worlds: Application of Organizational Theory to Cardiac Surgery.” *The Annals of Thoracic Surgery*. <https://doi.org/10.1016/j.athoracsur.2021.01.060>.
- Levesque, Antoine, Robert C. Pietzcker, Lavinia Baumstark, Simon De Stercke, Arnulf Grübler, and Gunnar Luderer. 2018. “How Much Energy Will Buildings Consume in 2100? A Global Perspective within a Scenario Framework.” *Energy* 148: 514–27. <https://doi.org/10.1016/j.energy.2018.01.139>.
- Li, Dmitriy, Jeong Hwan Bae, and Meenakshi Rishi. 2023. “Sustainable Development and SDG-7 in Sub-Saharan Africa: Balancing Energy Access, Economic Growth, and Carbon Emissions.” *European Journal of Development Research* 35 (1): 112–37. <https://doi.org/10.1057/s41287-021-00502-0>.
- Lighting Global. 2013. “Minimum Quality Standards and Recommended Performance Targets for Lighting Global, Lighting Africa, and Lighting Asia” 16 (2007). https://www.lightingglobal.org/wp-content/uploads/2013/06/109_LG-QA-StdsTargets_Memo-June2013.pdf.
- Limanond, Thirayoot, Sajjakaj Jomnonkwao, and Artit Srikaew. 2011. “Projection of Future Transport Energy Demand of Thailand.” *Energy Policy* 39 (5): 2754–63. <https://doi.org/10.1016/j.enpol.2011.02.045>.
- Litman, Todd Alexander. 2013. “Understanding Transport Demands and Elasticities:

- How Prices and Other Factors Affect Travel Behavior.” *Victoria Transport Policy Institute*, 1–76. <http://www.vtpi.org/elasticities.pdf>.
- Lopion, Peter, Peter Markewitz, Martin Robinius, and Detlef Stolten. 2018. “A Review of Current Challenges and Trends in Energy Systems Modeling.” *Renewable and Sustainable Energy Reviews*. Elsevier Ltd. <https://doi.org/10.1016/j.rser.2018.07.045>.
- Ludlow, Larry, and Kelsey Klein. 2014. “Suppressor Variables: The Difference between ‘is’ versus ‘Acting as.’” *Journal of Statistics Education* 22 (2). <https://doi.org/10.1080/10691898.2014.11889703>.
- Maigné, Vanessa, Patrick Brockmann, Pep Canadell, Corinne Le Quéré, Phillippe Peylin, Robert Andres, Anna Peregon, Pascal Evano, and Phillippe Clais. 2021. “Global Carbon Atlas.” <http://globalcarbonatlas.org/>.
- Manirambona, Egide, Stephen M. Talai, and Stephen K. Kimutai. 2022. “A Review of Sustainable Planning of Burundian Energy Sector in East Africa.” *Energy Strategy Reviews* 43 (April): 100927. <https://doi.org/10.1016/j.esr.2022.100927>.
- Mbanda, Njoke L., Patrobers Simiyu, Zhongqun Wu, and Innocent E. Davidson. 2021. “Tracking the Progress of Universal Access to Electricity in Africa: Status Quo, Challenges and Opportunities.” *2021 IEEE PES/IAS PowerAfrica, PowerAfrica 2021*. <https://doi.org/10.1109/PowerAfrica52236.2021.9543355>.
- McCollum, David L., Luis Gomez Echeverri, Sebastian Busch, Shonali Pachauri, Simon Parkinson, Joeri Rogelj, Volker Krey, et al. 2018. “Connecting the Sustainable Development Goals by Their Energy Inter-Linkages.” *Environmental Research Letters* 13 (3). <https://doi.org/10.1088/1748-9326/aaafe3>.
- Meijer, Johan R., Mark A.J. Huijbregts, Kees C.G.J. Schotten, and Aafke M. Schipper. 2018. “Global Patterns of Current and Future Road Infrastructure.” *Environmental Research Letters* 13 (6). <https://doi.org/10.1088/1748-9326/aabd42>.
- Mentis, Dimitrios, Mark Howells, Holger Rogner, Alexandros Korkovelos, Christopher Arderne, Eduardo Zepeda, Shahid Siyal, et al. 2017. “Lighting the World: The First Application of an Open Source, Spatial Electrification Tool (OnSSET) on Sub-Saharan Africa.” *Environmental Research Letters* 12 (8). <https://doi.org/10.1088/1748-9326/aa7b29>.

- Ministry for Energy and Minerals - The United Republic of Tanzania. 2015. “Tanzania’s SE4All Action Agenda.”
- Ministry of Energy and Mineral Development - Republic of Uganda. 2015. “Uganda’s Sustainable Energy For All (SE4All) Initiative Action Agenda.” *Uganda’s Sustainable Energy for All Initiative - Action Agenda*, no. June: 1–78. www.energyandminerals.go.ug.
- Ministry of Finance. 2007. “Uganda Vision 2040.” *Government of Uganda* 12 (3): 1–7. <https://doi.org/10.1007/s11947-009-0181-3>.
- Ministry of Finance - The United Republic of Tanzania. 2011. “The Tanzania Development Vision 2025.” *The United Republic of Tanzania*. <https://mof.go.tz/mofdocs/overarch/vision2025.htm>.
- Ministry of Planning and Communal Development - Burundi. 2011. “Vision Burundi 2025.” https://www.undp.org/content/dam/burundi/docs/publications/UNDP-bi-vision-burundi-2025_complete_EN.pdf.
- Morrissey, James. 2019. “Achieving Universal Electricity Access at the Lowest Cost: A Comparison of Published Model Results.” *Energy for Sustainable Development* 53: 81–96. <https://doi.org/10.1016/j.esd.2019.09.005>.
- Mouratidis, Kostas, Sebastian Peters, and Bert van Wee. 2021. “Transportation Technologies, Sharing Economy, and Teleactivities: Implications for Built Environment and Travel.” *Transportation Research Part D: Transport and Environment* 92 (February): 102716. <https://doi.org/10.1016/j.trd.2021.102716>.
- Muchunku, Charles, Kirsten Ulsrud, Debajit Palit, and Wim Jonker-Klunne. 2018. “Diffusion of Solar PV in East Africa: What Can Be Learned from Private Sector Delivery Models?” *Wiley Interdisciplinary Reviews: Energy and Environment* 7 (3): 1–15. <https://doi.org/10.1002/wene.282>.
- Munyaneza, Janvier, Muhammad Wakeel, and Bin Chen. 2016. “Overview of Rwanda Energy Sector: From Energy Shortage to Sufficiency.” *Energy Procedia* 104 (December): 215–20. <https://doi.org/10.1016/j.egypro.2016.12.037>.
- Nandi Moksnes, Alexandros Korkovelos, Dimitrios Mentis, and Mark Howells. 2017. “Electrification Pathways for Kenya - Linking Spatial Electrification Analysis and Medium to Long Term Energy Planning.” *Environmental Research Letters* 095008. <https://doi.org/10.1088/1748-9326/aa7e18>.

- Nicoli, Matteo, Francesco Gracceva, Daniele Lerede, and Laura Savoldi. 2022. “Can We Rely on Open-Source Energy System Optimization Models? The TEMOA-Italy Case Study.” *Energies* 15 (18). <https://doi.org/10.3390/en15186505>.
- O’Neill, Brian C., Elmar Kriegler, Kristie L. Ebi, Eric Kemp-Benedict, Keywan Riahi, Dale S. Rothman, Bas J. van Ruijven, et al. 2017. “The Roads Ahead: Narratives for Shared Socioeconomic Pathways Describing World Futures in the 21st Century.” *Global Environmental Change* 42 (January): 169–80. <https://doi.org/10.1016/j.gloenvcha.2015.01.004>.
- Olsen, Ludvig Renbo. 2021. “Cross-Validation with Groupdata2.” 2021. https://cran.r-project.org/web/packages/groupdata2/vignettes/cross-validation_with_groupdata2.html.
- Ouedraogo, Nadia S. 2017a. “Africa Energy Future: Alternative Scenarios and Their Implications for Sustainable Development Strategies.” *Energy Policy* 106 (July): 457–71. <https://doi.org/10.1016/j.enpol.2017.03.021>.
- Ouedraogo, Nadia S. 2017b. *WIDER Working Paper 2017/56 Energy Futures Modelling for African Countries: LEAP Model Application*. <https://www.wider.unu.edu/sites/default/files/wp2017-56.pdf>.
- Pachauri, Shonali, Miguel Poblete-Cazenave, Arda Aktas, and Matthew J. Gidden. 2021. “Access to Clean Cooking Services in Energy and Emission Scenarios after COVID-19.” *Nature Energy* 6 (11): 1067–76. <https://doi.org/10.1038/s41560-021-00911-9>.
- Pappis, Ioannis, Vignesh Sridharan, Mark Howells, Hrvoje Medarac, Ioannis Kougias, Rocío González Sánchez, Abhishek Shivakumar, and Will Usher. 2022. “The Effects of Climate Change Mitigation Strategies on the Energy System of Africa and Its Associated Water Footprint.” *Environmental Research Letters* 17 (4). <https://doi.org/10.1088/1748-9326/ac5ede>.
- Parzen, Maximilian, Hazem Abdel-Khalek, Ekaterina Fedorova, Matin Mahmood, Martha Maria Frysztacki, Johannes Hampp, Lukas Franken, et al. 2022. “PyPSA-Earth. A New Global Open Energy System Optimization Model Demonstrated in Africa.” <http://arxiv.org/abs/2209.04663>.
- Patankar, Neha, Anderson Rodrigo de Queiroz, Joseph F. DeCarolus, Morgan D. Bazilian, and Debabrata Chattopadhyay. 2019. “Building Conflict Uncertainty into Electricity

- Planning: A South Sudan Case Study.” *Energy for Sustainable Development* 49: 53–64. <https://doi.org/10.1016/j.esd.2019.01.003>.
- Peña-Guzmán, Carlos, and Juliana Rey. 2020. “Forecasting Residential Electric Power Consumption for Bogotá Colombia Using Regression Models.” *Energy Reports* 6: 561–66. <https://doi.org/10.1016/j.egyr.2019.09.026>.
- Petzoldt, Thomas. 2020. “R Package Growthrates.” 2020. <https://www.rdocumentation.org/packages/growthrates/versions/0.8.2>.
- Pfenninger, Stefan, Adam Hawkes, and James Keirstead. 2014. “Energy Systems Modeling for Twenty-First Century Energy Challenges.” *Renewable and Sustainable Energy Reviews* 33: 74–86. <https://doi.org/10.1016/j.rser.2014.02.003>.
- Poblete-Cazenave, Miguel, Shonali Pachauri, Edward Byers, Alessio Mastrucci, and Bas van Ruijven. 2021. “Global Scenarios of Household Access to Modern Energy Services under Climate Mitigation Policy.” *Nature Energy* 6 (8): 824–33. <https://doi.org/10.1038/s41560-021-00871-0>.
- Prina, Matteo Giacomo, Giampaolo Manzolini, David Moser, Benedetto Nastasi, and Wolfram Sparber. 2020. “Classification and Challenges of Bottom-up Energy System Models - A Review.” *Renewable and Sustainable Energy Reviews*. Elsevier Ltd. <https://doi.org/10.1016/j.rser.2020.109917>.
- Rammelt, Crelis F., Joyeeta Gupta, Diana Liverman, Joeri Scholtens, Daniel Ciobanu, Jesse F. Abrams, Xuemei Bai, et al. 2022. “Impacts of Meeting Minimum Access on Critical Earth Systems amidst the Great Inequality.” *Nature Sustainability*, no. 2. <https://doi.org/10.1038/s41893-022-00995-5>.
- Rao, Narasimha D., Petra Sauer, Matthew Gidden, and Keywan Riahi. 2019. “Income Inequality Projections for the Shared Socioeconomic Pathways (SSPs).” *Futures* 105 (August 2018): 27–39. <https://doi.org/10.1016/j.futures.2018.07.001>.
- Refaeilzadeh, Payam, Lei Tang, and Huan Lui. 2009. “Cross-Validation.” In *Encyclopedia of Database Systems*, 532–38. Springer. <https://doi.org/10.1201/b17767-13>.
- Remy, Tom, and Debabrata Chattopadhyay. 2020. “Promoting Better Economics, Renewables and CO2 Reduction through Trade: A Case Study for the Eastern Africa Power Pool.” *Energy for Sustainable Development* 57: 81–97. <https://doi.org/10.1016/j.esd.2020.05.006>.

- Renewable Energy Agency, International, KfW Development Bank, and Deutsche Gesellschaft für Internationale Zusammenarbeit. n.d. “The Renewable Energy Transition in Africa Powering Access, Resilience and Prosperity On Behalf of The.”
- Republic of Kenya. 2018. “Updated Least Cost Power Development Plan 2017 - 2037,” no. June. <http://gak.co.ke/wp-content/uploads/2019/02/Updated-Least-Cost-Power-Development-Plan-2017-2022-min.pdf>.
- Republic of Kenya - Ministry of Energy. 2021. “Least Cost Power Development Plan.”
- Riahi, Keywan, Detlef P. van Vuuren, Elmar Kriegler, Jae Edmonds, Brian C. O’Neill, Shinichiro Fujimori, Nico Bauer, et al. 2017. “The Shared Socioeconomic Pathways and Their Energy, Land Use, and Greenhouse Gas Emissions Implications: An Overview.” *Global Environmental Change* 42: 153–68. <https://doi.org/10.1016/j.gloenvcha.2016.05.009>.
- Rubin, Edward S., John E. Davison, and Howard J. Herzog. 2015. “The Cost of CO₂ Capture and Storage.” *International Journal of Greenhouse Gas Control* 40: 378–400. <https://doi.org/10.1016/j.ijggc.2015.05.018>.
- Ruijven, Bas van, Frauke Urban, René M.J. Benders, Henri C. Moll, Jeroen P. van der Sluijs, Bert de Vries, and Detlef P. van Vuuren. 2008. “Modeling Energy and Development: An Evaluation of Models and Concepts.” *World Development* 36 (12): 2801–21. <https://doi.org/10.1016/j.worlddev.2008.01.011>.
- Rwanda Energy Group. 2019. “Rwanda Least Cost Power Development Plan (LCPDP) 2019-2040.” *Rwanda Energy Group*, no. June: 37.
- Rwanda Energy Group (REG). 2022. “Rwanda: Least Cost Power Development Plan(LCPDP) 2022 – 2040,” no. June: 2022–40.
- Safari, Bonfils. 2010. “A Review of Energy in Rwanda.” *Renewable and Sustainable Energy Reviews* 14 (1): 524–29. <https://doi.org/10.1016/j.rser.2009.07.009>.
- Schandl, Heinz, Yingying Lu, Nhu Che, David Newth, James West, Stefan Frank, Michael Obersteiner, Andrew Rendall, and Steve Hatfield-Dodds. 2020. “Shared Socio-Economic Pathways and Their Implications for Global Materials Use.” *Resources, Conservation and Recycling* 160 (September): 104866. <https://doi.org/10.1016/J.RESCONREC.2020.104866>.

- Serajuddin, Umar, and Nada Hamadeh. 2020. “New World Bank Country Classifications by Income Level: 2020 - 2021.” World Bank Blogs. 2020.
- Singh, S.K. 2006. “The Demand for Road-Based Passenger Mobility in India: 1950-2030 and Relevance for Developing and Developed Countries.” *European Journal of Transport and Infrastructure Research* 6 (3): 247–74.
- Singhvi, Divya, Somya Singhvi, Peter I. Frazier, Shane G. Henderson, Eoin O’Mahony, David B. Shmoys, and Dawn B. Woodard. 2015. “Predicting Bike Usage for New York City’s Bike Sharing System.” *AAAI Workshop - Technical Report WS-15-06*: 110–14.
- Soergel, Bjoern, Elmar Kriegler, Benjamin Leon Bodirsky, Nico Bauer, Marian Leimbach, and Alexander Popp. 2021. “Combining Ambitious Climate Policies with Efforts to Eradicate Poverty.” *Nature Communications* 12 (1): 1–12. <https://doi.org/10.1038/s41467-021-22315-9>.
- Souche, Stéphanie. 2010. “Measuring the Structural Determinants of Urban Travel Demand.” *Transport Policy* 17 (3): 127–34. <https://doi.org/10.1016/j.tranpol.2009.12.003>.
- Sridharan, Vignesh, Oliver Broad, Abhishek Shivakumar, Mark Howells, Brent Boehlert, David G. Groves, H. Holger Rogner, et al. 2019. “Resilience of the Eastern African Electricity Sector to Climate Driven Changes in Hydropower Generation.” *Nature Communications* 10 (1): 1–9. <https://doi.org/10.1038/s41467-018-08275-7>.
- Sterman, John, William Moomaw, Juliette N. Rooney-Varga, and Lori Siegel. 2022. “Does Wood Bioenergy Help or Harm the Climate?” *Bulletin of the Atomic Scientists* 78 (3): 128–38. <https://doi.org/10.1080/00963402.2022.2062933>.
- Stoner, Oliver, Jessica Lewis, Itzel Lucio Martínez, Sophie Gumy, Theo Economou, and Heather Adair-Rohani. 2021. “Household Cooking Fuel Estimates at Global and Country Level for 1990 to 2030.” *Nature Communications* 12 (1): 1–8. <https://doi.org/10.1038/s41467-021-26036-x>.
- Szabó, Sándor, Magda Moner-Girona, Ioannis Kougias, Rob Bailis, and Katalin Bódis. 2016. “Identification of Advantageous Electricity Generation Options in Sub-Saharan Africa Integrating Existing Resources.” *Nature Energy* 1 (10): 1–8. <https://doi.org/10.1038/nenergy.2016.140>.
- Tabassum, Nafisa, and A F S Ahad Rahman Khan. 2020. “Forecasting Air Freight

Transportation Demand Using Log-Linear Model for Socio-Economic Factors,” no. February: 1–6.

Taliotis, Constantinos, Asami Miketa, Mark Howells, Sebastian Hermann, Manuel Welsch, Oliver Broad, Holger Rogner, Morgan Bazilian, and Dolf Gielen. 2014. “An Indicative Assessment of Investment Opportunities in the African Electricity Supply Sector.” *Journal of Energy in Southern Africa* 25 (1): 2–12. <https://doi.org/10.17159/2413-3051/2014/v25i1a2681>.

Taliotis, Constantinos, Abhishek Shivakumar, Eunice Ramos, Mark Howells, Dimitris Mentis, Vignesh Sridharan, Oliver Broad, and Linus Mofor. 2016. “An Indicative Analysis of Investment Opportunities in the African Electricity Supply Sector - Using TEMBA (The Electricity Model Base for Africa).” *Energy for Sustainable Development* 31: 50–66. <https://doi.org/10.1016/j.esd.2015.12.001>.

TEMOA Developers. 2023. “TEMOA Project GitHub Repository.” 2023. <https://github.com/TemoaProject/oeo>.

The World Bank. 2021. “World Bank Country and Lending Groups.” 2021. <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>.

The World Bank et al. 2020. *The State of Access to Modern Energy Cooking Services. The State of Access to Modern Energy Cooking Services*. <https://journals.openedition.org/factsreports/6131>.

The World Bank Group. 2022. “Data Bank: World Development Indicators.” 2022. <https://databank.worldbank.org/source/world-development-indicators#>.

Tiba, Sofien, and Fateh Belaid. 2021. “Modeling the Nexus Between Sustainable Development and Renewable Energy: The African Perspectives.” *Journal of Economic Surveys* 35 (1): 307–29. <https://doi.org/10.1111/joes.12401>.

Tinungki, G. M. 2019. “The Analysis of Partial Autocorrelation Function in Predicting Maximum Wind Speed.” *IOP Conference Series: Earth and Environmental Science* 235 (1). <https://doi.org/10.1088/1755-1315/235/1/012097>.

Trotter, Philipp A., Nathaniel J. Cooper, and Peter R. Wilson. 2019. “A Multi-Criteria, Long-Term Energy Planning Optimisation Model with Integrated on-Grid and off-Grid Electrification – The Case of Uganda.” *Applied Energy* 243 (June): 288–312.

- <https://doi.org/10.1016/J.APENERGY.2019.03.178>.
- Tsekeris, Theodore, and Charalambos Tsekeris. 2011. "Demand Forecasting in Transport: Overview and Modeling Advances." *Economic Research* 24 (1): 82–94. <https://doi.org/10.1080/1331677X.2011.11517446>.
- U.S. Bureau of Labor Statistics. 2021. "CPI Inflation Calculator." 2021. https://www.bls.gov/data/inflation_calculator.htm.
- UN Environment Programme. 2020. "Used Vehicles and The Environment: A Global Overview of Used Light Duty Vehicles."
- United Nations. 2019. "The Sustainable Development Goals Report." New York.
- United Nations. 2015. "Paris Agreement." Vol. 45. https://unfccc.int/sites/default/files/english_paris_agreement.pdf.
- . 2019. "The Sustainable Development Goals Report 2019." *United Nations Publication Issued by the Department of Economic and Social Affairs*, 64. <https://undocs.org/E/2019/68>.
- United Nations Climate Change (UNCC). 2023. "Nationally Determined Contributions Registry." United Nations. 2023.
- United Nations Department of Economic and Social Affairs. 2022a. "World Population Prospects 2022: Summary Results." *United Nations Publication*. www.un.org/development/desa/pd/.
- . 2022b. "World Population Prospects 2022." Washington, DC: United Nations. <https://population.un.org/wpp/>.
- Varjan, Pavol, Dominika Rovňaníková, and Jozef Gnap. 2017. "Examining Changes in GDP on the Demand for Road Freight Transport." *Procedia Engineering* 192: 911–16. <https://doi.org/10.1016/j.proeng.2017.06.157>.
- Venkatesh, Aranya, Aditya Sinha, and Katherine Jordan. 2022. "GitHub-TemoaProject/Oeo: Open Energy Outlook for the United States." 2022.
- Wadud, Zia. 2011. "Modeling and Forecasting Passenger Demand for a New Domestic Airport with Limited Data." *Transportation Research Record*, no. 2214: 59–68. <https://doi.org/10.3141/2214-08>.
- WHO, and SEI. 2021. *Household Energy Policy Repository*. World Health Organization.

- Wolde-Rufael, Yemane. 2004. “Disaggregated Industrial Energy Consumption and GDP: The Case of Shanghai, 1952-1999.” *Energy Economics* 26 (1): 69–75. [https://doi.org/10.1016/S0140-9883\(03\)00032-X](https://doi.org/10.1016/S0140-9883(03)00032-X).
- World Bank. 2020a. “Increasing Access to Electricity in the Democratic Republic of Congo.” *The World Bank*. <https://doi.org/10.1596/33593>.
- . 2020b. “RISE: Regulatory Indicators for Sustainable Energy.” 2020. <https://rise.esmap.org/>.
- World Economic Council. 2011. “Global Transport Scenarios 2050.” *World Energy Council*. <https://www.worldenergy.org/publications/entry/world-energy-scenarios-global-transport-scenarios-2050>.
- Yoro, Kelvin O., Michael O. Daramola, Patrick T. Sekoai, Uwemedimo N. Wilson, and Orevaoghene Eterigho-Ikelegbe. 2021. “Update on Current Approaches, Challenges, and Prospects of Modeling and Simulation in Renewable and Sustainable Energy Systems.” *Renewable and Sustainable Energy Reviews*. Elsevier Ltd. <https://doi.org/10.1016/j.rser.2021.111506>.
- Zwaan, Bob van der, Tom Kober, Francesco Dalla Longa, Anouk van der Laan, and Gert Jan Kramer. 2018. “An Integrated Assessment of Pathways for Low-Carbon Development in Africa.” *Energy Policy* 117 (June): 387–95. <https://doi.org/10.1016/J.ENPOL.2018.03.017>.

7 APPENDICES

APPENDIX 1 TEMOA EAST AFRICA	116
APPENDIX 2 IMPACT ASSESSMENT UNDER DIFFERENT SCENARIOS	121
APPENDIX 3 LAND-BASED TRANSPORTATION DEMAND MODEL	129

APPENDIX 1 TEMOA EAST AFRICA

SDG 7.1 in East Africa

Burundi, the Democratic Republic of Congo, Kenya, Tanzania, and Uganda were home to about 159 million people who lacked access to electricity and 183 million who lacked access to clean cooking fuels in 2020 (IEA et al. 2022). Burundi and the Democratic Republic of Congo are fragile and conflict-prone countries making planning for universal energy access devastatingly difficult. On the other hand, East Africa is also home to two of the countries with fastest electricity access growth rate globally, Kenya and Uganda, growing at an annual rate that was 2 percentage points higher than the global average between 1990 and 2020 (IEA et al. 2022). Kenya and Rwanda have adopted electricity access policies credited with increasing off-grid technology penetration and higher access growth rates in the region (Li, Bae, and Rishi 2023; Mbanda et al. 2021). The six countries have higher progress in electricity access shares compared to clean cooking as most, aside from Burundi and Democratic Republic of Congo, have national electrification plans. Kenya and Rwanda have gone a step further developing least cost development plans to inform power capacity expansion using geospatial and system modeling tools (Republic of Kenya 2018; Republic of Kenya - Ministry of Energy 2021; Rwanda Energy Group 2019).

The Temoa modeling framework

The Tools for Energy Modeling Optimization and Analysis (Temoa) tool is an open source, bottom-up energy system optimization model that employs linear programming optimization. Developers from North Carolina (NC) State University developed the modeling framework to allow for ease of third party verification with transparent model results (Hunter, Sreepathi, and DeCarolis 2013). The interface uses open-source Python programming language. Temoa is capable of system-wide energy system analysis from supply sector with primary resources, energy conversion processes to secondary energy forms such as electricity, to end-use sectoral representation. Therefore, the modeling interface is suitable for household energy model with rich technology detail and disaggregated representation of different access tiers in East Africa. The objective function minimizes present value costs by optimizing technology deployment and

utilization over a given time horizon. Deployment of least-cost technologies is restricted by user-defined constraints such as maximum primary energy potential. Previous studies have used Temoa for a wide range of applications nationally (Nicoli et al. 2022; Patankar et al. 2019; Eshraghi, De Queiroz, and DeCarolis 2018). For example, DeCarolis et al (2016) used TEMOA, an energy optimization model, to generate alternative energy pathways for the U.S. Energy database inputs and technology parameterization (transformation representation) are the backbone of models that use Temoa. Demand is a key input exogenously defined by the modeler.

In Temoa, modelers can map an energy system network with resources, conversion process, end-use services, and resulting emissions. The key components in the network are commodities and technologies. Commodities represent forms of energy. The technologies are processes that convert primary or secondary energy commodities into consumable energy commodities to meet demand. The Temoa model includes flexible time slicing (time-of-day and seasons), variable time periods, vintaging technology options, technology specific lifetimes, discount rates (global or technology specific), stochastic optimization capabilities, and design features to allow for technology rich parameterization (Decarolis, Hunter, and Sreepathi 2020). The modeling framework offers mathematical model formulation, open-source code, environment tools, and boasts of an online community to support open modeling efforts. The GitHub repository and Temoa website provide practical steps on how to install software elements and run Temoa (TEMOA Developers 2023)

Table 7.1 Fuel reserve assumptions on supply-side (data source: (Allington et al. 2022)).

Country	Total Recovery coal (mil. Short. tons in 2017)	Crude oil proven reserves (billion barrels in 2019)	Natural gas proven reserves
Burundi	-	-	-
Congo, Dem. Rep.	97	0.18	0.04
Kenya	-	-	-
Rwanda	-	-	2
Tanzania	297	-	0.23
Uganda	-	2.5	0.5

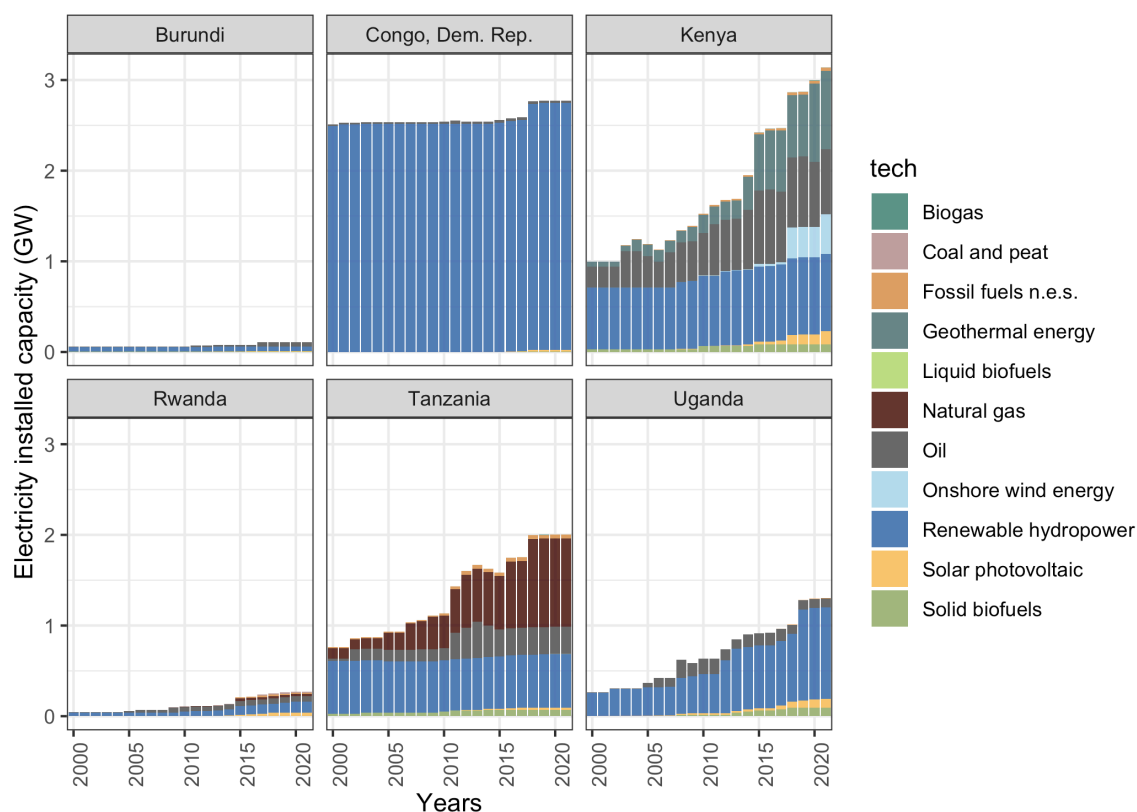


Figure 7.1 Total installed electricity capacity (GW) in six countries between 2000 and 2020

Table 7.2 Powerplant renewable energy potential by country (data source: (Allington et al. 2022; IRENA 2021))

Country	Solar PV (TWh/yr)	Concentrating Solar Power (TWh/yr)	Wind (CF 20%, TWh/yr)	Wind (CF 30%, TWh/yr)	Hydropower (>10MW, MW)	Small hydropower (<10MW, MW)	Geothermal (MW)
Burundi	888	786	-	-	1700	61	-
Congo, Dem. Rep.	22,862	12,439	2,173	41	22,573	101	-
Kenya	23,046	15,399	22,746	4,446	6,000	3,000	10,000
Rwanda	892	789	-	-	500	25	700
Tanzania	38,804	31,482	18,456	2,295	3,800	480	650
Uganda	9,470	8,582	815	100	4,500	200	450

Table 7.3 Powerplant technology parameter inputs and assumptions

Technology	Capital cost			Fixed O&M costs	Efficiency	Capacity factor	Lifetime (years)
	(Million USD\$2020/GW)	(Million USD\$2020/GW)	(Million USD\$2020/GW)				
	2020	2030	2050				
Battery storage (4hrs)	1,349	778	585	16.9 – 33.4	0.85		15
Biomass	2,500	2,500	2,500	75	0.35	0.5	30
Coal with CCS	5,000	4,500	3,500	123	0.39	0.85	35
Coal with IGCC	2,500	3,850	3,300	123	0.39	0.85	35
Subcritical coal steam turbine	2,500	2,500	2,500	78	0.35	0.85	35
CSP with storage	5,797	3,763	3,660	58	1	0.45	30
Diesel	1,200	1,200	1,200	35	0.35	0.8	25
Diesel off-grid generators	750	750	750	23	0.16	0.3	10
Geothermal	4,000	4,000	4,000	120	0.8	0.79	25
Large hydropower (>100MW)	3,000	3,000	3,000	90	1	varies	50
Medium hydropower (10 – 100MW)	2,500	2,500	2,500	75	1	varies	50
Small hydropower mini-grid (<10MW)	3,000	3,000	3,000	90	1	varies	50
Natural gas combined cycle gas turbine	1,200	1,200	1,200	35	0.48	0.85	30
Natural gas single cycle gas turbine	700	700	700	20	0.3	0.85	25
Natural gas combined cycle turbine with CCS	2,800	2,500	2,000	35	0.58	0.85	30
Heavy fuel oil	1,450	1,450	1,450	45	0.35	0.8	25
Solar PV	1,378	886	723	18	1	varies	25
Stand-alone solar PV	4,320	2,700	2,090	86	1	varies	24
Nuclear	6,137	6,137	6,137	184	0.33	0.9	50
Offshore wind	3,972	2,450	2,100	159	1	varies	25
Onshore wind	1,489	1,087	933	60	1	varies	25
Transmission/ distribution							
(rural)	800	800	800		0.9		
(urban)	365	365	365		0.9		

**Table 7.4 Powerplant fuel price projection assumptions in Million USD\$2020 per PJ
(data source: (Allington et al. 2022; IRENA 2021))**

Fuel commodity	2020	2025	2030	2035	2040	2045	2050
Biomass	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Crude oil imports	12.2	12.8	14.3	14.3	16.9	16.9	19.5
Crude oil domestic	11	11.6	13	13	15	15.4	17.8
Coal imports	5.1	5.3	5.5	5.5	5.9	5.9	5.9
Light fuel oil imports	14.7	15.4	17.3	17.3	20.4	20.4	23.6
Heavy fuel oil imports	8.9	9.3	10.4	10.4	12.3	12.3	14.2

Natural gas imports	8.6	9.5	10.3	10.3	11	11	11
Natural gas domestic	7.1	7.8	8.5	8.5	9.9	9.9	9.9
Uranium	1.4	1.4	1.4	1.4	1.4	1.4	1.4

Table 7.5 Direct fuel use to household price assumptions in Million USD\$2020 per PJ (data sources: (Dioha and Kumar 2020a; Dagnachew et al. 2020))

Fuel commodity	2020	2025	2030	2035	2040	2045	2050
Biogas	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Biomass	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Kerosene	11	11	11	11	11	11	11
LPG	12	12	12	12	12	12	12
Natural gas	12	12	12	12	12	12	12

Table 7.6 Price projections for fuel use in transport sector in Million USD\$2020 per PJ (Khalili et al. 2019)

Fuel commodity	2020	2025	2030	2035	2040	2045	2050
Diesel	14.8	15	17	117.3	20	20.4	24
Gasoline	21.8	22.5	24	24.7	25	25.2	25.2
Natural gas imports	8.6	9.5	10.3	10.3	11	11	11
Synthetic gas	55	50	46	44	42	40	438

[add efficiency tables for end-use technologies and hydrogen pathways]

APPENDIX 2 IMPACT ASSESSMENT UNDER DIFFERENT SCENARIOS

In 2019, more than 85 percent of the global electricity access deficit, or 731 million of the 840 million people without access to electricity, resided in rural areas (United Nations 2019; Mentis et al. 2017). Since the onset of the COVID-19 pandemic, 90 million people in developing countries have rejoined the electricity access deficit, with 77% of the global deficit concentrated in sub-Saharan Africa, totaling 568 million people in 2020 (IEA et al. 2022; Li, Bae, and Rishi 2023). The increase in population in the region has outpaced the increase in electricity access. To achieve SDG indicator 7.1.1 by 2030, the region requires over 80 million annual connections at current (IEA et al. 2022). Decentralized technologies are essential for closing the electricity access gap in rural areas, where sparse population and low consumption patterns make universal electricity access targets more challenging to achieve universal electricity access goals.

Similarly, clean cooking access disparities between rural and urban populations also persist. In 2020, 2.4 billion people worldwide lacked access to clean cooking fuels and technologies; while 86% of urban and 44% of rural populations had access, the share of polluting fuels for cooking increased in sub-Saharan Africa (United Nations 2019; IEA et al. 2022; Stoner et al. 2021). According to the World Health Organization (WHO), electricity, liquified petroleum gas (LPG), solar, biogas, and natural gas are considered clean cooking fuels (WHO and SEI 2021). Sub-Saharan Africa is the only region where the clean cooking deficit has doubled between 1990 and 2020 (IEA et al. 2022; Pachauri et al. 2021). Lack of access to clean cooking fuels results in serious health issues, including respiratory diseases. At the current growth rate of clean cooking access, four out of five people in sub-Saharan Africa will still face the impacts of using polluting fuels for cooking by 2030 (Stoner et al. 2021).

In 2020, the majority of the population in the Eastern Africa countries resided in rural areas (Table 1.1). Burundi, the Democratic Republic of Congo, Kenya, Tanzania and Uganda were among the top 20 energy access deficit countries (IEA et al. 2022). In the same year, all these countries including Rwanda and excluding Kenya were among the top 20 least electrified countries. The Democratic Republic of Congo, Kenya, Tanzania, and Uganda were also among the top 20 countries with largest number of people lacking access to clean cooking fuels (IEA et al. 2022).

To address these challenges, each country has set its own energy access targets. For example, Kenya aims to achieve universal Tier 1+ rural and urban electricity by 2025, while Tanzania aims to achieve over 75% Tier 1+ electricity and clean cooking access rates by 2030. Kenya has made significant progress in electricity access, and its national vision 2030 aims to transform the country into an industrialized middle-income country by 2030. Rwanda, on the other hand has an abundance of natural resources but a low energy development index. However, the country has made significant progress in developing policy and regulatory frameworks for sustainable energy and has committed to cutting emissions by 38% by 2030. Uganda aims to achieve affordable universal energy access to facilitate accelerated industrialization by 2040, and its utility, UMEME, is the only utility in the region with net positive revenue gains.

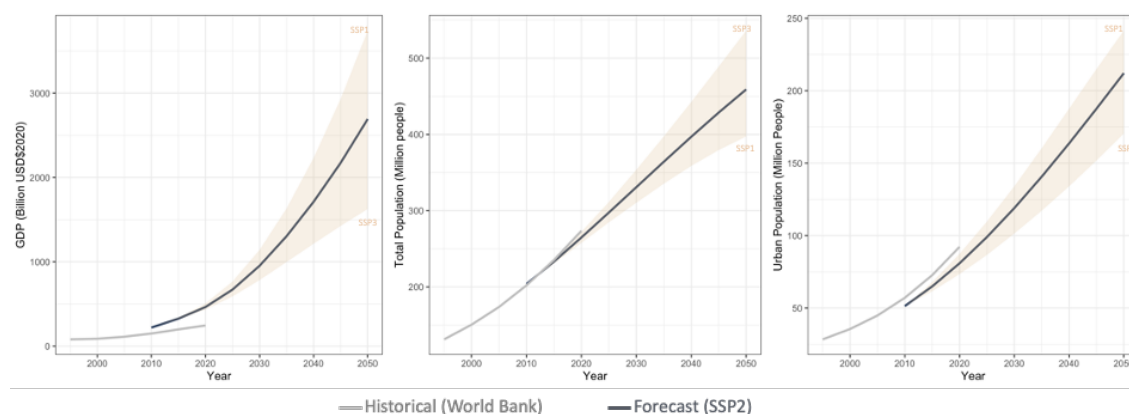


Figure 7.2 Socioeconomic demand drivers for all six countries under 3 SSPs. The grey line is historical demand, the blue line is SSP2 and the lower and upper bounds are labelled, SSP1 and SSP3.

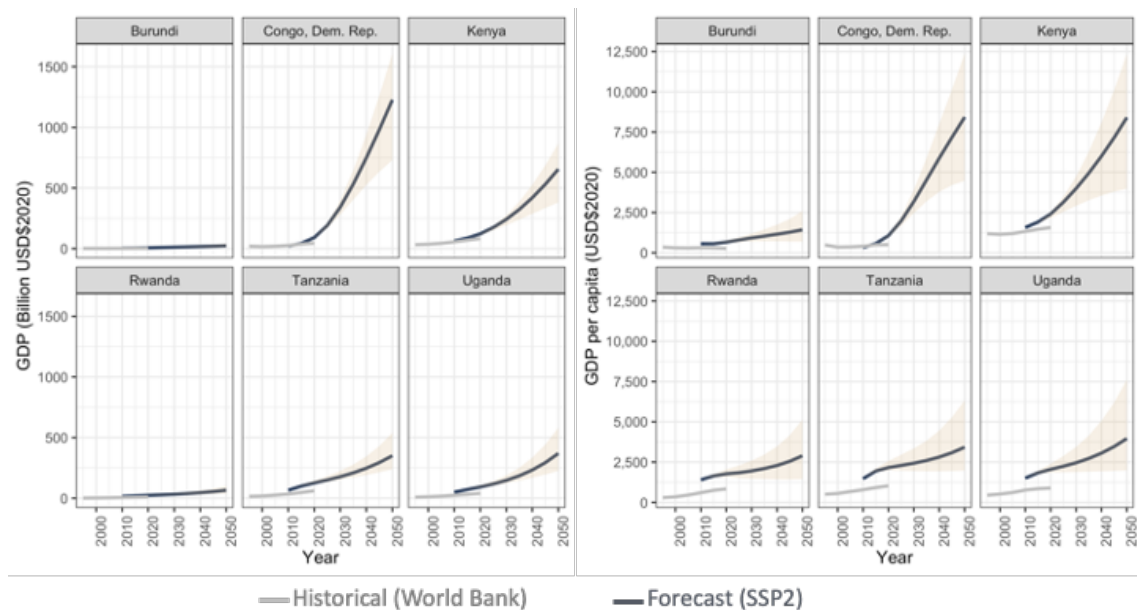


Figure 7.3 GDP projections by countries under the 3 SSPs. The grey line is historical demand, the blue line is SSP2, and the lower and upper bounds are SSP3 and SSP1.

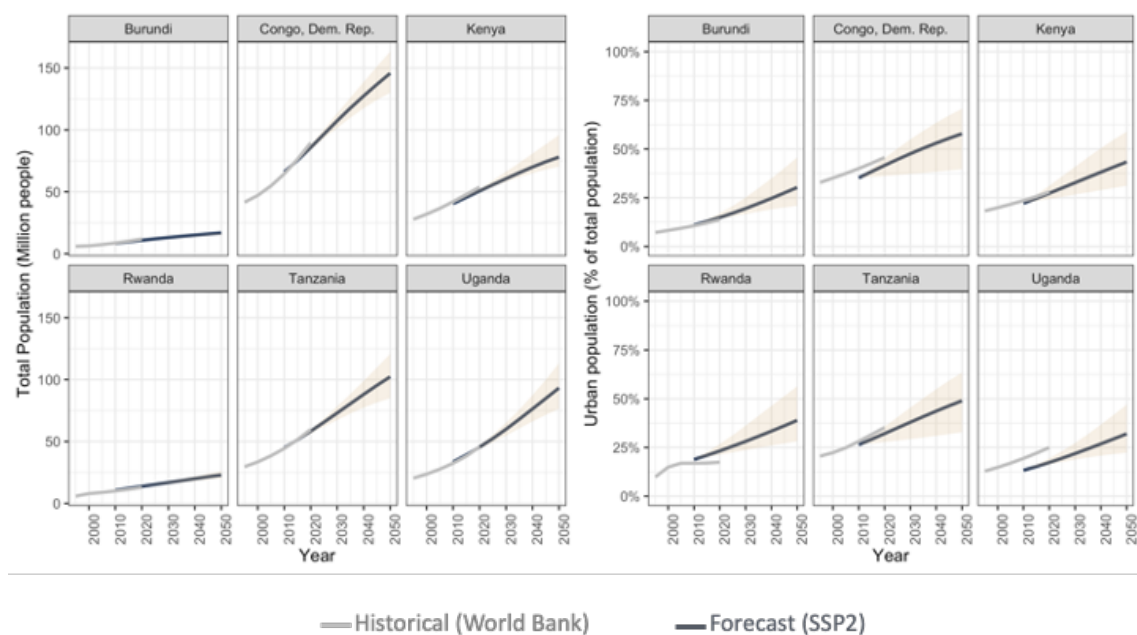


Figure 7.4 Population and urbanization projections by country under the SSPs. The grey line is historical demand, the blue line is SSP2, and the lower and upper bounds are SSP1 and SSP3.



Figure 7.5 The share of population with access to electricity by tier in rural and urban households in 2020 (data source: (Falchetta et al. 2020)).

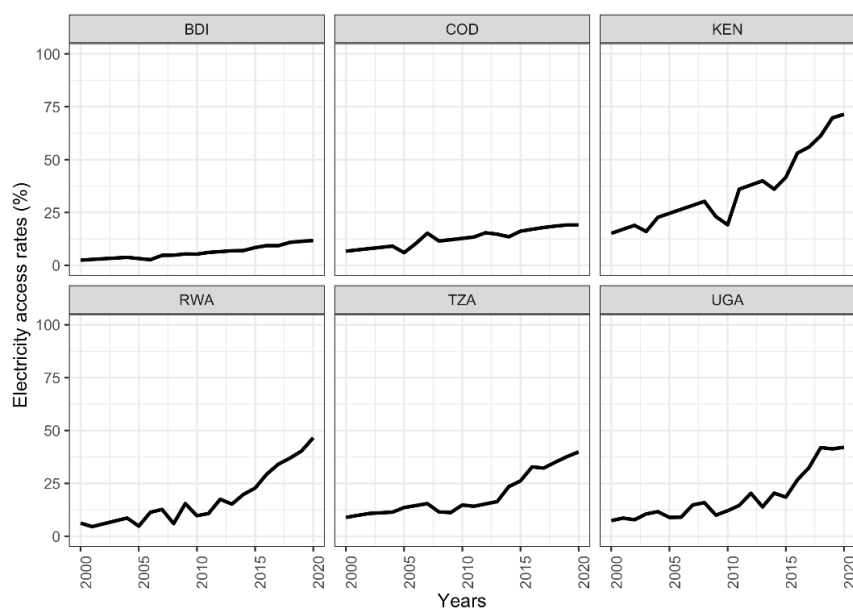


Figure 7.6 Historical electricity access levels by country between 2000 and 2020 (data source: (The World Bank Group 2022)).

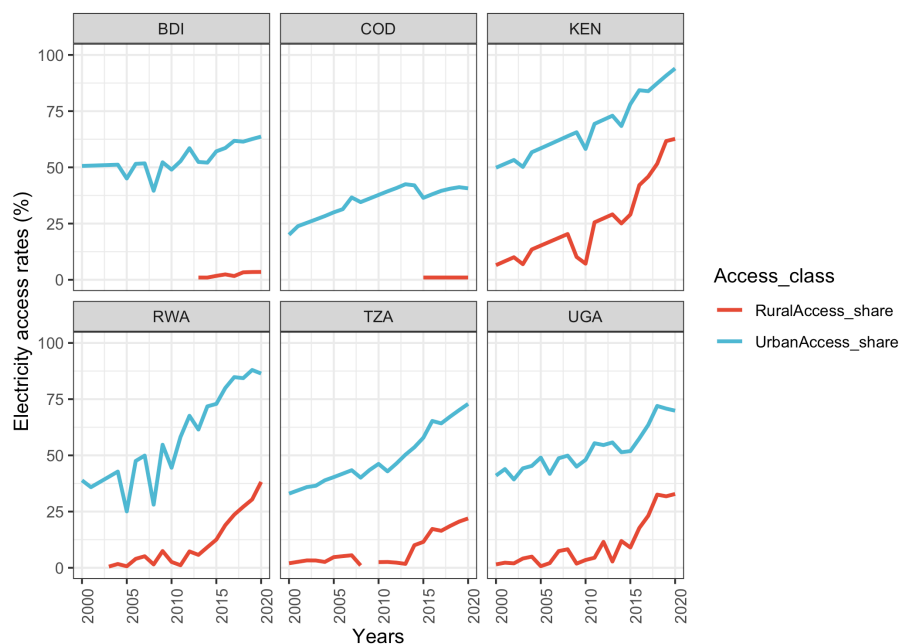


Figure 7.7 Historical electricity access levels by rural and urban share between 2000 and 2020 (data source: (The World Bank Group 2022)).

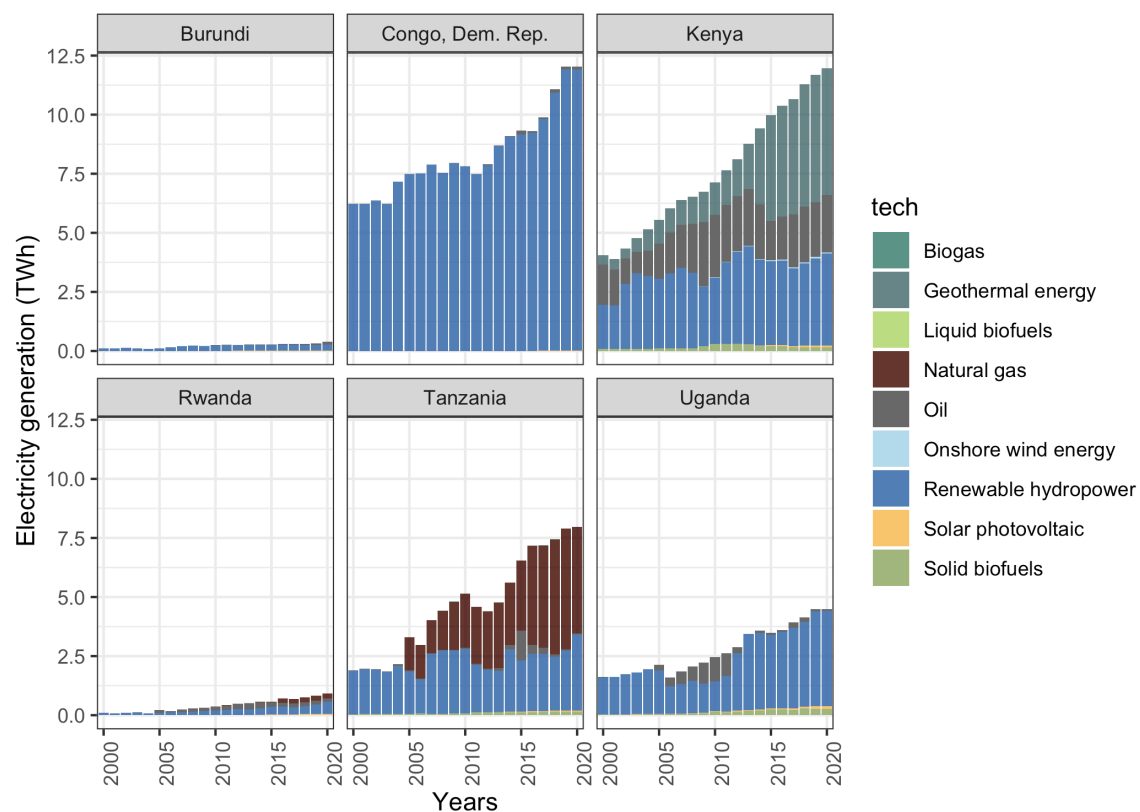


Figure 7.8 Historical total electricity generation by fuel in each country (data source: (IRENA 2023; 2021)).

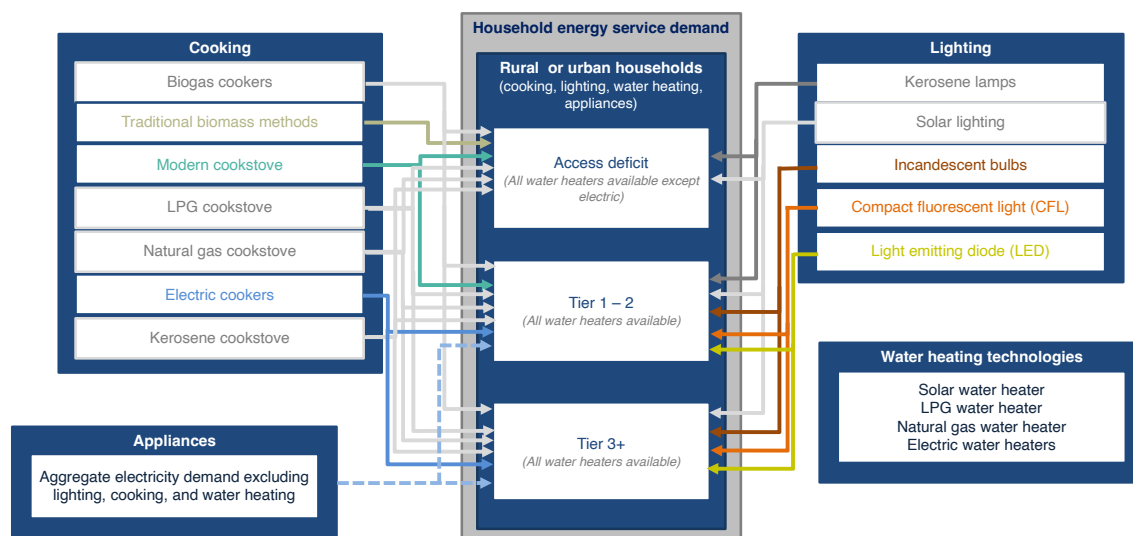


Figure 7.9 Technology choice by access tier. Highlighted pathways are for select technologies available by access level.

Table 7.7 summarizes the total final energy consumption in the six access scenarios under SSP 3 scenarios by 2050. We observe changes in total final energy consumption in households with Tier 3+ access levels.

Table 7.7 Summary of final energy consumption in PJ by access level in 2050 and SSP3

	Access deficit		Tier 1 - 2		Tier 3+	
	<i>Uncon</i>	<i>NZ2050</i>	<i>Uncon</i>	<i>NZ2050</i>	<i>Uncon</i>	<i>NZ2050</i>
Regression	3,822	3,822	640	640	591	589
SDG2030urban	3,085	3,085	841	841	690	686
SDG2030All			1,930	1,930	991	961
SDG2050urban			1,448	1,448	1,267	1,206
SDG2050All			65	65	2,058	1,884
SDG2050LowBio			65	65	2,094	1,901

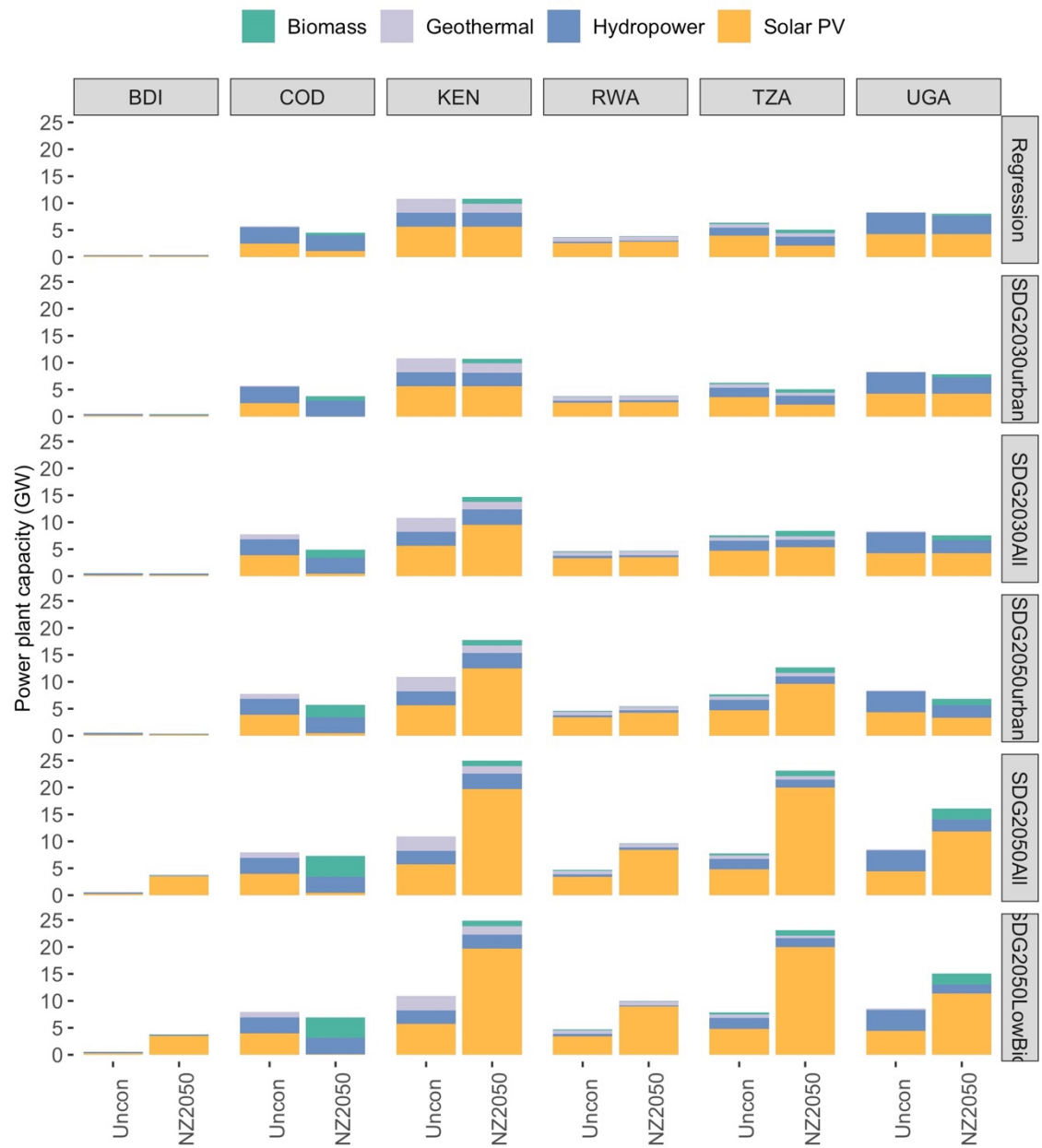


Figure 7.10 Power plant capacity in 2050 by country in SSP3

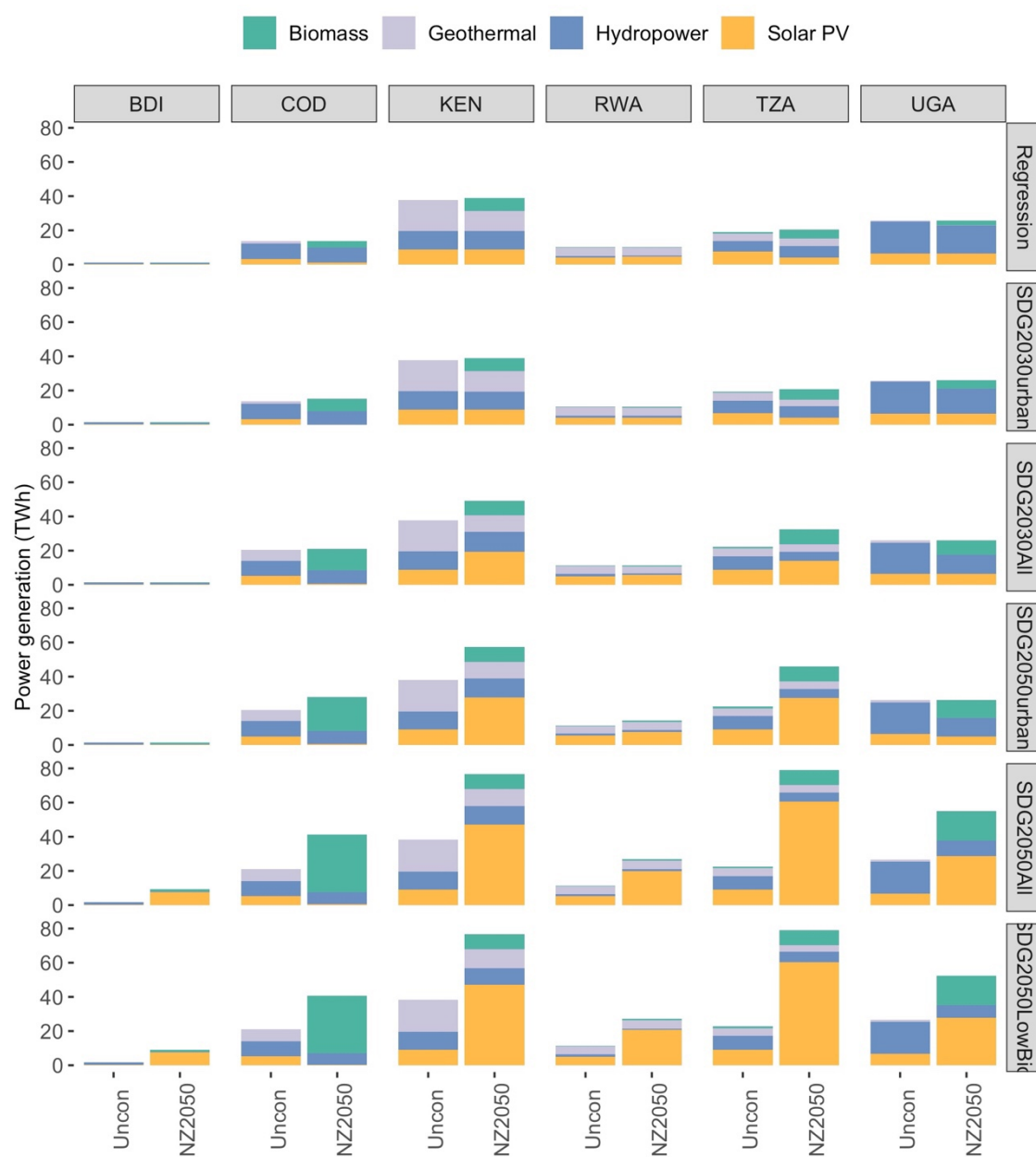


Figure 7.11 Power generation in SSP3 by country

APPENDIX 3 LAND-BASED TRANSPORTATION DEMAND MODEL

Table 7.8 Summary of data sources for log-linear model inputs.

Data	Period in modeling	Countries	Source
Country-level historical transport demand	1990 - 2018	38 (Land-based passenger demand) 43 (Land-based freight)	(ITF 2021a; 2019)
Historical GDP and Population data	1990 - 2018	217 countries and territories	(The World Bank Group 2022)
Country-level GDP, population, and urban share projections	2010 - 2050	179 countries	(Cuaresma 2017; KC and Lutz 2017; Jiang and O'Neill 2017; International Institute for Applied Systems Analysis (IIASA) 2018)

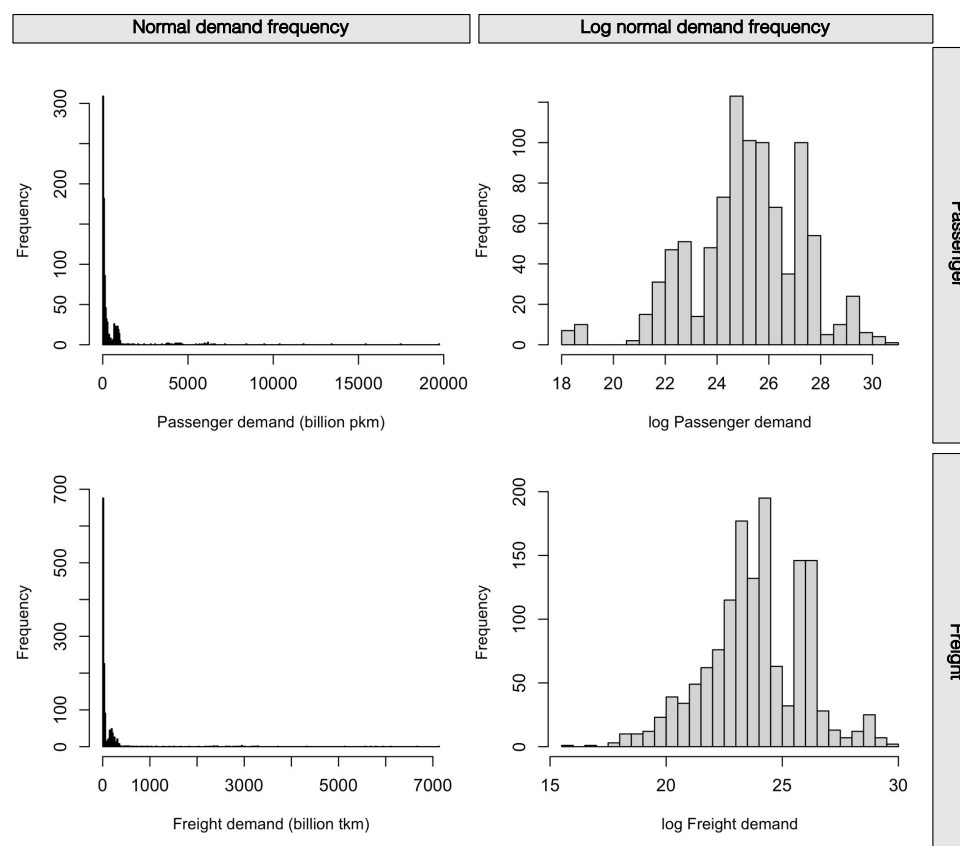


Figure 7.12 Land-based transport demand and log transport demand for all countries in historical ITF dataset.

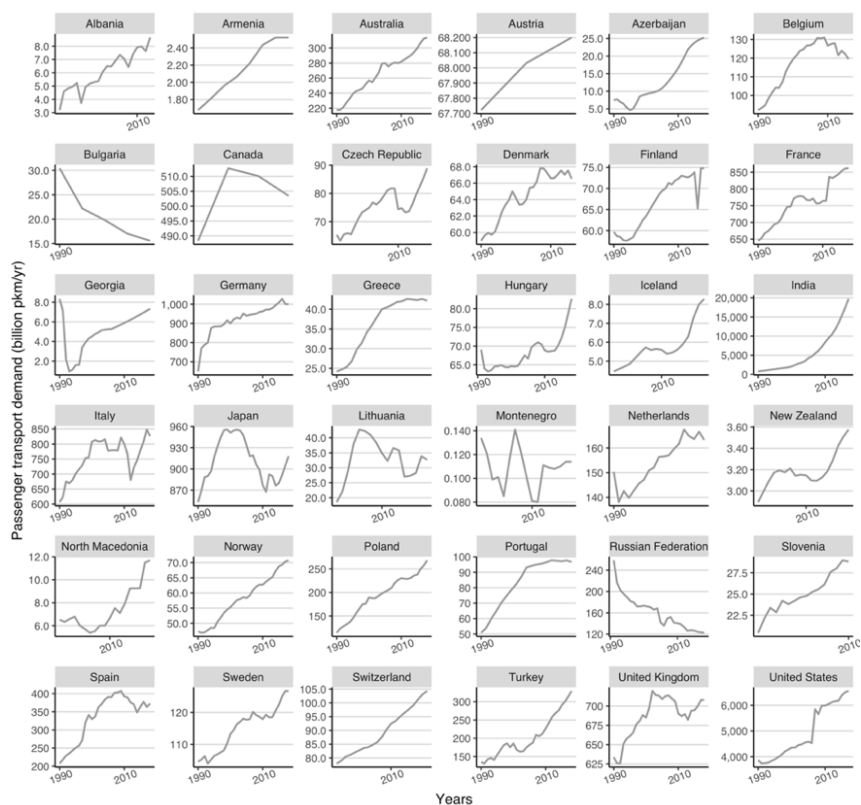


Figure 7.13 Historical Land-based passenger transport demand trends between 1990 and 2018 for the first 36 countries.

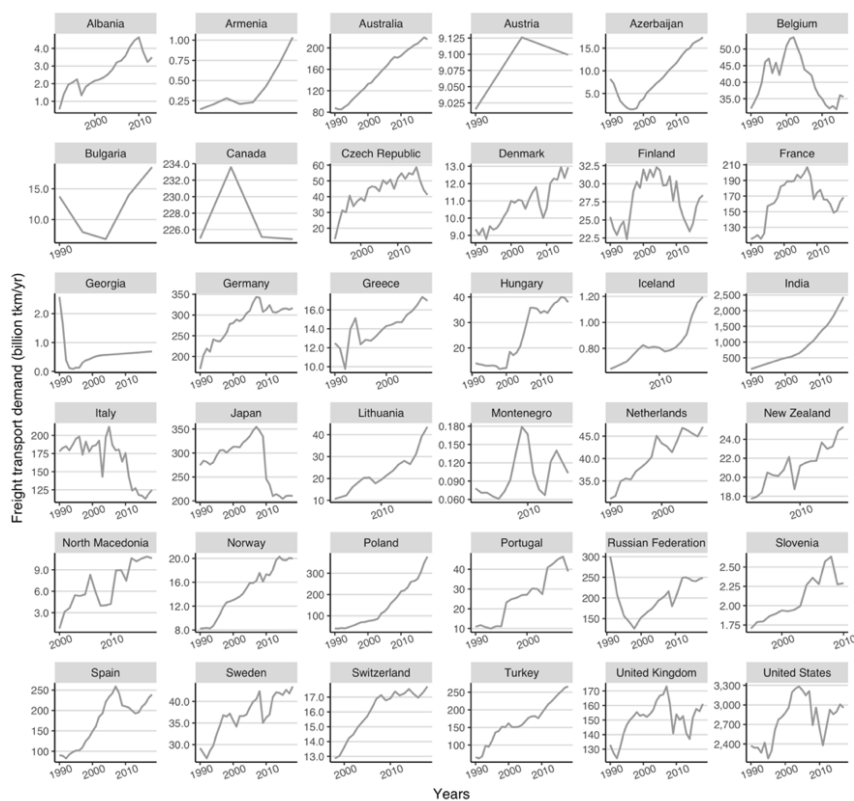


Figure 7.14 Historical land-based freight demand between 1990 and 2018 for 36 out of the 43 available countries in ITF OECD-dataset.

In terms of other explanatory variables, we also tested models with land area and population density; adding those variables did not add extra explanatory power as measured by adj. R-squared. As such, we opted for parsimony, given the additional complexity did not substantially improve our model fit.

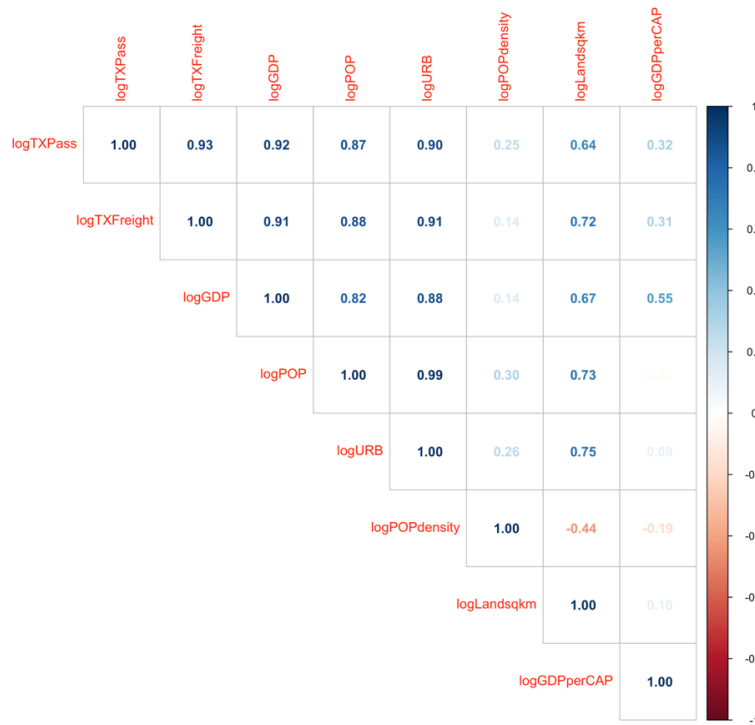


Figure 7.15 Correlation matrix for explanatory variables including GDP per capita, land area, and population density.

The autocorrelation of the errors in the panel model affects the parameter precision, which we account for using the sandwich variance-covariance matrix (Table 7.9). We use the R sandwich variance covariance matrix to allow for autocorrelation between each time series and its lagged observations over time. The variance covariance matrix in Table 7.9 has the variance of each variable on the diagonal and the covariance of that variable with each of the other elements off the diagonal.

Table 7.9 Variance covariance matrix using selected log-log model.

Passenger travel					Freight travel			
	(intercept)	logGDP	logPOP	logURB	(intercept)	logGDP	logPOP	logURB
(intercept)	0.149	-0.010	-0.029	0.037	0.123	-0.007	-0.021	0.026
logGDP	-0.010	0.001	0.003	-0.004	-0.007	0.0006	0.002	-0.002
logPOP	-0.029	0.003	0.015	-0.017	-0.021	0.002	0.011	-0.013
logURB	0.037	-0.004	-0.017	0.021	0.026	-0.002	-0.013	0.015

According to Kutner *et al.*, (2004) “*The fact that some or all predictor variables are correlated among themselves does not, in general, inhibit our ability to obtain a good fit nor does it tend to affect inferences about mean responses or predictions of new observations.*” If we were investigating casual effects there is a need to address multicollinearity, however, this does not affect model performance for purposes of scenario-based demand projections. The correlation matrix in Table 7.10 shows the correlation coefficients among the independent and dependent variables.

Table 7.10 Correlation matrix for selected explanatory variables. Evidence of high correlation between population and urban share.

	logTXPass	logTXFreight	logGDP	logPOP	logURB	logGDPperCAP
logTXPass		0.93***	0.92***	0.86***	0.88***	0.34***
logTXFreight	0.93***		0.91***	0.85***	0.88***	0.25***
logGDP	0.92***	0.91***		0.80***	0.85***	0.48***
logPOP	0.86***	0.85***	0.80***		0.99***	-0.14***
logURB	0.88***	0.88***	0.85***	0.99***		-0.043
logGDPperCAP	0.34***	0.251***	0.48***	-0.14***	-0.043	

Computed correlation used pearson-method with pairwise-deletion.

Note: *= $p < .05$, **= $p < .01$, ***= $p < .001$

TXPass = Passenger-km, TXFreight = Freight tonne-km, GDP = Gross Domestic Product in \$2020, POP = Population, URB = Urbanization, GDPperCAP = GDP per capita in \$2020

When we looked at the correlation matrix, the correlations between variables were indeed high. Therefore, we ran on each variable separately and showed that they are still influential on the models (Table 7.11).

Table 7.11 Function form of double log models in exploratory analysis.

	Model functional form
Model 1	$\ln \ln (pkm \text{ or } tkm) = \beta_0 + \beta_1 \ln(GDP) + \beta_2 \ln(POP) + \beta_3 \ln(URB) + \varepsilon$
Model 2	$\ln \ln (pkm \text{ or } tkm) = \beta_0 + \beta_1 \ln(GDPperCAP) + \beta_3 \ln(URB) + \varepsilon$
Model 3	$\ln \ln (pkm \text{ or } tkm) = \beta_0 + \beta_1 \ln(GDP) + \varepsilon$
Model 4	$\ln \ln (pkm \text{ or } tkm) = \beta_0 + \beta_2 \ln(POP) + \varepsilon$
Model 5	$\ln \ln (pkm \text{ or } tkm) = \beta_0 + \beta_3 \ln(URB) + \varepsilon$
Model 6	$\ln \ln (pkm \text{ or } tkm) = \beta_0 + \beta_1 \ln(GDPperCAP) + \varepsilon$

pkm = passenger-km, tkm = tonne-km

Table 7.12 Log-log passenger and freight models. Fit results in comparison with selected model highlighted yellow. The bracketed ranges represent 95% confidence intervals pooled at the country level accounting for autocorrelation.

Log Passenger (passenger-km)						
Predictors	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
(Intercept)	-3.69 *** (-4.45 – -2.93)	2.64 *** (1.97 – 3.32)	-0.45 (-1.18 – 0.27)	7.52 *** (6.85 – 8.19)	7.36 *** (6.74 – 7.98)	18.99 *** (17.85 – 20.13)
logGDP	1.06 *** (1.00 – 1.12)		0.97 *** (0.94 – 0.99)			
logPOP	2.29 *** (2.05 – 2.53)			1.07 *** (1.03 – 1.11)		
logURB	-2.30 *** (-2.59 – -2.01)	1.08 *** (1.05 – 1.12)			1.10 *** (1.06 – 1.14)	0.62 *** (0.51 – 0.74)
logGDPperCAP		0.50 *** (0.45 – 0.54)				
Observations	911	911	911	929	929	911
R ² / R ² adjusted	0.904 / 0.904	0.847 / 0.847	0.842 / 0.842	0.745 / 0.745	0.776 / 0.776	0.113 / 0.112
Log Freight (tonne-km)						
Predictors	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
(Intercept)	-0.01 (-0.70 – 0.68)	2.13 *** (1.60 – 2.66)	-0.86 ** (-1.48 – -0.24)	7.62 *** (7.09 – 8.15)	7.80 *** (7.33 – 8.27)	19.73 *** (18.84 – 20.62)
logGDP	0.66 *** (0.61 – 0.71)		0.95 *** (0.92 – 0.97)			
logPOP	0.45 *** (0.25 – 0.66)			1.00 *** (0.97 – 1.03)		
logURB	-0.04 (-0.28 – 0.20)	1.06 *** (1.04 – 1.09)			1.01 *** (0.98 – 1.04)	
logGDPperCAP		0.49 *** (0.46 – 0.53)				0.43 *** (0.35 – 0.52)
Observations	1351	1351	1351	1390	1390	1351
R ² / R ² adjusted	0.866 / 0.865	0.857 / 0.857	0.821 / 0.821	0.729 / 0.729	0.767 / 0.767	0.063 / 0.062

$p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

We test for the assumption of autocorrelation using the autocorrelation correlation function (ACF) function in the R ‘stats’ package to plot the estimates of autocorrelation

in the model. The ACF plot in Figure 7.16 shows the correlation (on the y-axis) against the lagged time variable (x-axis). The first vertical bar is always 1 as it represents the correlation of the observation and itself, while the subsequent bars show the correlation between the observation and its lagged values in previous times (Tinungki 2019). Figure 7.16 shows the absence of autocorrelation as the ACF form drops exponentially and remains within the blue dashed lines (limits of the significant levels) and remain close to zero. Therefore, the residuals allow us to conclude the assumption of no correlation.

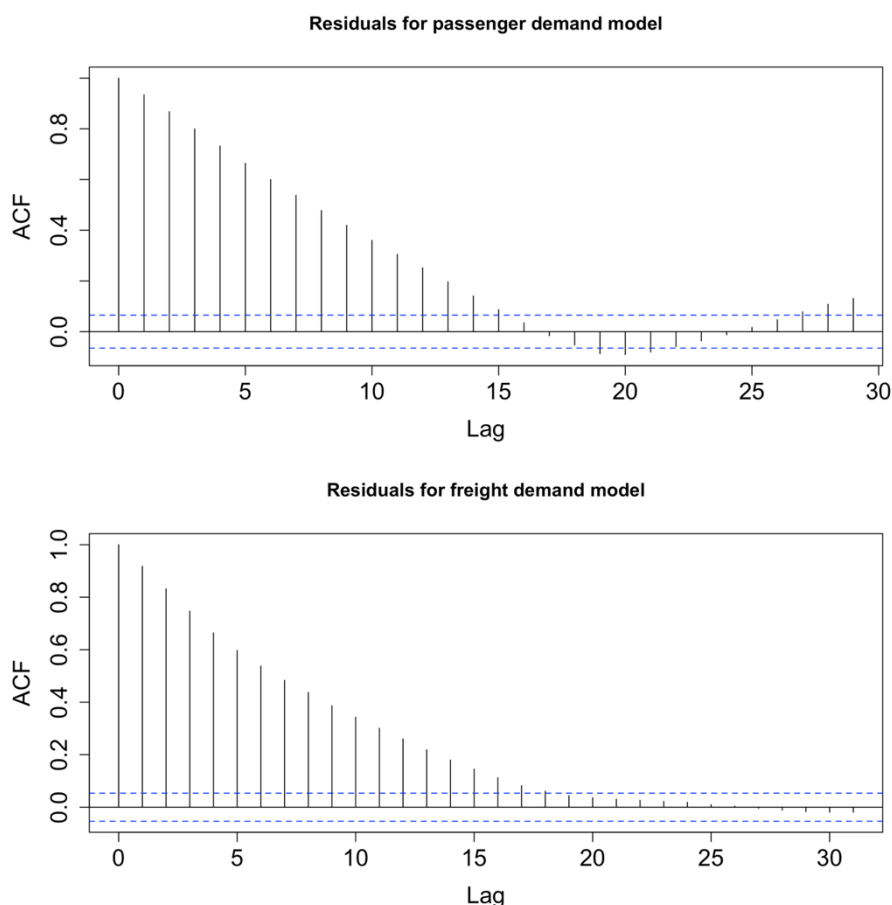


Figure 7.16 Test for autocorrelation among residuals using ACF() function in the (a) passenger and (b) freight double log regression models.

Cross validation is a statistical method used to evaluate model performance by dividing data into training and testing sub-sets with the latter being used to validate the model (Refaeilzadeh, Tang, and Lui 2009). Cross-validation can be done in a couple ways here: 1) leave observations out randomly, 2) leave whole countries out randomly, and 3) leave future time periods out. We use 5-fold cross validation grouping the panel data by country guided by the ‘*groupdata2*’ R package (Olsen 2021). The method is used to estimate the

generalization performance (ability to handle out-of-sample data) and compare performance of two or more models to support selection. We summarize the results of the cross validation in Table 7.13 showing the average of the root mean squared errors (r-MSE) for all the log-log models in the exploratory analysis.

The results from the cross-validation in Table 7.13 show the average root-MSE averaged across the countries. Model 1, the log-log model we select in our analysis, has the lowest average errors for passenger travel, and we assume is the best predictor of out-of-sample data compared to the other models. Although, almost similar in performance to Model 2, we also select Model 1 for freight travel demand model as the model has the lowest average errors. Running multiple iterations for each fold could also improve the diagnostics. We also look at the performance of Model 1 on the test set, find 0.75 average error for the passenger-km model and 0.80 for the freight tonne-km demand model. Therefore, we do not suspect overfitting, and this gives us confidence in our scenario analysis results. The global results were within the range of the ITF model (Figure 7.17 and Figure 7.18).

Table 7.13 The average of the r-MSE for Model 1 is lower than Model 2. Although results are similar, Model 1 has lower average errors and is a better predictor of out-of-sample data compared to Model 2.

	Average root-Mean Squared Error (r-MSE)	
	Passenger	Freight
Model 1 (Y ~ GDP + POP + URB)	0.752	0.802
Model 2 (Y ~ GDPperCAP + URB)	1.046	0.802
Model 3 (Y ~ GDP)	0.930	0.851
Model 4 (Y ~ POP)	1.215	1.095
Model 5 (Y ~ URB)	1.151	1.060
Model 6 (Y ~ GDPperCAP)	2.193	1.882

Table 7.14 Summary of global land-based transport demand results by 2050. We highlight the highest scenario-based projections.

	Population in 2050 (millions)	Urbanization in 2050 (millions)	GDP in 2050 (trillion USD ₂₀₂₀)	Total passenger travel in 2050 (billion pkm)	Total freight demand in 2050 (billion tkm)
SSP1	8,390	6,480	381	88,700	35,500
SSP2	9,100	6,090	307	96,200	31,300
SSP3	9,880	5,380	237	119,000	26,700
SSP4	9,060	6,890	293	64,600	28,900
SSP5	8,490	6,580	484	114,000	41,900

The detailed data for all 179 countries is available via a Zenodo repository.

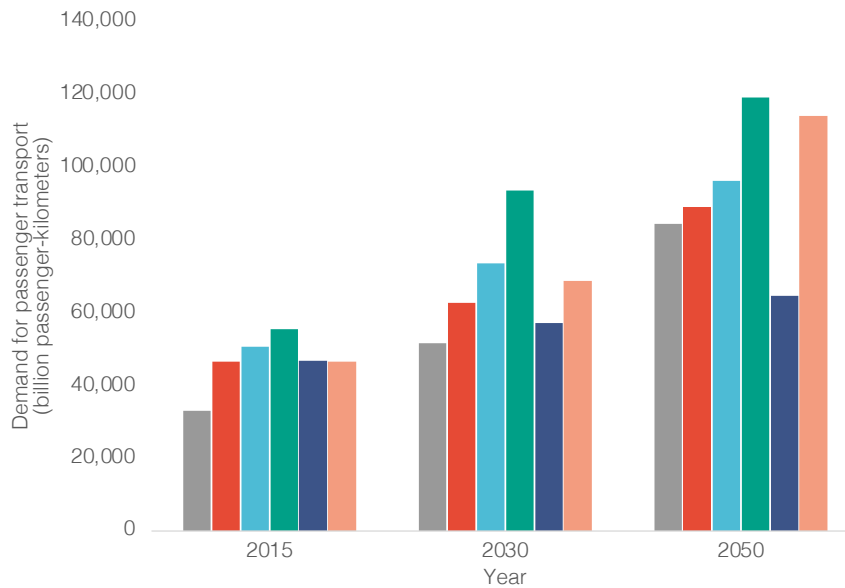


Figure 7.17 Comparison of global aggregate model results with ITF global land-based passenger demand. The ITF Land-based passenger demand by 2050 lies within the scenario-based projections.

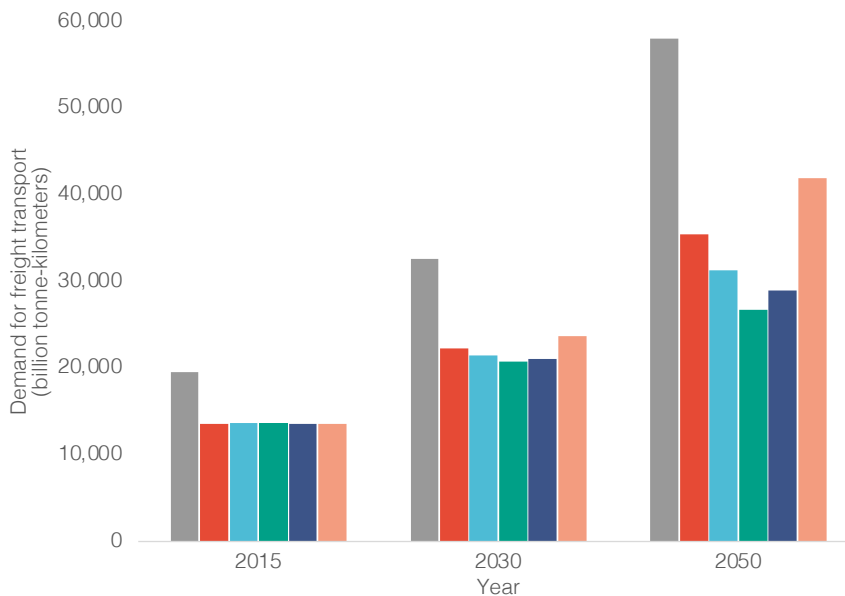


Figure 7.18 Comparison of global aggregate model results with ITF global land-based passenger and freight demand. The ITF has higher 2015 reference/baseline and much higher demand for global land-based freight by 2050.